Exercise list — Topic 4: Moral hazard, credit rationing, asset substitution, debt overhang

Worked solutions

Statements are in `Listas/lista-topic4-moral-hazard-overhang.md` (and its PDF); each exercise is cited by number and printed page and its full text is reproduced right before its solution. Exercises 16–19 are original exercises (no book exercise exists; written with the user's agreement, 2026-11-12), built on Prof. Ricca's Topic 4B slides 37–58; each opens with a one-paragraph data line so that it reads on its own, and the data of exercise 19 are hypothetical. Short answers: `lista-topic4-moral-hazard-overhang-solutions-short.pdf`. Editions are the local copies: Tirole, *The Theory of Corporate Finance* (2005); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013).

Contents

1. DIP financing and existing creditors — Grinblatt & Titman 16.14, p. 593 **2**

2. Asset substitution: project A or B — Grinblatt & Titman 16.1, p. 591 **4**

3. Debt overhang on vacant land — Berk 16.25, pp. 494–495 **6**

4. Short-term debt and incentive problems — Grinblatt & Titman 16.11, p. 593 **8**

5. Random financing — Tirole 3.1, p. 144 **10**

6. Zymase: leverage and strategy choice — Berk 16.26, p. 495 **14**

7. Ajax Manufacturing: debt overhang — Grinblatt & Titman 16.12–16.13, p. 593 **17**

8. Project riskiness and credit rationing — Tirole 3.15, p. 150 **19**

9. Entrepreneurial risk aversion — Tirole Exercise 3.2, p. 144 **22**

10. Maximal incentives principle in the fixed-investment model — Tirole Exercise 3.18, p. 151 **28**

11. Retained-earnings benefit — Tirole Exercise 3.11, p. 148 **34**

12. Renegotiation and debt forgiveness — Tirole Exercise 3.6, p. 146 **38**

13. Loan size and collateral requirements — Tirole Exercise 4.17, p. 195 **43**

14. Dealing with asset substitution — Tirole Exercise 7.3, pp. 323–324 **48**

15. A borrower-friendly bankruptcy court — Tirole Exercise 4.9, pp. 190–191 **53**

16. Default as insurance: the default cutoff — Original exercise **58**

17. Default as insurance: comparative statics, the price of credit and ex ante welfare — Original exercise **61**

18. Karlan–Zinman: which comparison identifies selection and which identifies moral hazard — Original exercise **66**

19. Reading a three-rate experiment: selection, incentives and credit rationing — Original exercise **70**

1. DIP financing and existing creditors — Grinblatt & Titman 16.14, p. 593

Statement.

Source: Grinblatt, Ch. 16, Exercise 16.14, p. 593 (PDF 606) · Answer in book: no

With debtor in possession (DIP) financing, bankrupt firms are able to obtain additional amounts of debt that is senior to the firm's existing debt. Explain how the firm's existing debt holders can benefit from this.

What is asked. A single conceptual question. Chapter 11 lets a bankrupt firm borrow new money that is paid *before* the old debt. On its face this hurts the old creditors, who move down the queue. Explain the mechanism by which they can nevertheless end up better off.

Approach. This is the lecture's *debt overhang* problem (Myers 1977). Write every claim as a state-by-state payoff, debt $= \min\{\text{assets}, F\}$ and equity $= \max\{\text{assets} - F, 0\}$, and compare the old creditors' payoff with and without the new project. Junior money cannot be raised when the old debt is under water, because the old creditors capture the project's value; seniority for the new money removes that obstacle. The old creditors lose priority but gain the project's cash flows, so the comparison is "smaller share of a larger pie" versus "larger share of a smaller pie".

Notation.

- s: state of the world at the payoff date; $V_s \geq 0$: value of the assets in place in state s.
- $F > 0$: face value of the existing debt, protected by a covenant that forbids issuing debt senior to it.
- $I > 0$: cost of a new project that must be carried out inside the firm; $X_s \geq 0$: its payoff in state s; risk-neutral investors, zero riskless rate (as in the book's and the lecture's examples).
- N: face value of the new debtor-in-possession (DIP) debt, senior to F; D, E: market values of the existing debt and of equity.

Step 1 — Why the project is not funded outside bankruptcy. We first show that, without a seniority exception, a good project is passed up; this is the status quo the old creditors compare against. Without DIP financing, new money can only be junior to F. In state s a junior claimant receives at most what is left after the senior debt,

$$\text{junior payoff}_s \leq \max\{V_s + X_s - F,\, 0\}. \tag{1}$$

In the book's Lily Pharmaceuticals example (pp. 564–566), in the unfavourable state the assets in place are worth nothing without further development ($V_s = 0$), development costs $I = 100$ and yields $X_s = 150$, and the senior obligation is $F = 110$. Substituting into (1),

$$\begin{aligned} \text{junior payoff} &\leq \max\{(0) + (150) - (110),\, 0\} \\ &= \max\{40,\, 0\} \\ &= 40 < 100 = I, \end{aligned}$$

so no one lends: junior lenders would pay 100 for a claim worth at most 40. Yet the project is worth doing,

$$\begin{aligned} NPV &= X_s - I \\ &= (150) - (100) \\ &= 50 > 0, \end{aligned}$$

so it is passed up, the firm defaults, and the existing lenders receive $\min\{V_s, F\} = \min\{0, 110\} = 0$ in that state.

Step 2 — DIP debt makes it fundable. We now let the new money jump the queue and recompute the old creditors' payoff. In Chapter 11 the firm may issue debt senior to F. A DIP loan of $I = 100$ at the zero riskless rate has face $N = 100$ and is riskless because $150 \geq 100$. The existing lenders now receive

$$\begin{aligned} \text{existing debt payoff} &= \min\{V_s + X_s - N,\, F\} \\ &= \min\{(0) + (150) - (100),\, (110)\} \\ &= \min\{50,\, 110\} \\ &= 50, \end{aligned} \tag{2}$$

instead of 0: they gain $50 - 0 = 50$, the whole NPV, even though they lost their priority.

Step 3 — The general argument. The example is not special; we show the old creditors never lose when the DIP money funds a project that pays back at least its cost. With a riskless DIP loan ($N = I \leq X_s$ in every state), the existing debt's payoff with the project is $\min\{V_s + X_s - I,\, F\}$, and without it $\min\{V_s,\, F\}$. Since $X_s - I \geq 0$, the first argument of the minimum is larger with the project, $V_s + X_s - I \geq V_s$, and $\min\{\cdot, F\}$ is nondecreasing in its first argument, so

$$\min\{V_s + X_s - I,\, F\} \geq \min\{V_s,\, F\} \qquad \text{for every } s, \tag{3}$$

with strict inequality in every state where the old debt was impaired ($V_s < F$). Seniority is diluted, but the asset pool the old creditors are paid from grows by more than the new claim.

Step 4 — The lecture's numbers. Topic 4B (slides 22–31): assets in place worth 500 (boom) or 200 (slump), each with probability 0.5; $F = 300$; a project costs $I = 60$ and adds 100 in both states. We value both claims without the project, then with it. Without the project,

$$\begin{aligned} D &= (0.5)\min\{500, 300\} + (0.5)\min\{200, 300\} \\ &= (0.5)(300) + (0.5)(200) = 150 + 100 = 250, \\ E &= (0.5)\max\{500 - 300, 0\} + (0.5)\max\{200 - 300, 0\} \\ &= (0.5)(200) + (0.5)(0) = 100. \end{aligned}$$

Financing the 60 with senior DIP debt ($N = 60$, riskless since the slump assets $200 + 100 = 300 \geq 60$), the assets become $500 + 100 = 600$ (boom) and $200 + 100 = 300$ (slump), and the old debt receives, by (2),

$$\begin{aligned} \text{boom:}\ \ &\min\{(600) - (60),\, (300)\} = \min\{540, 300\} = 300, \\ \text{slump:}\ \ &\min\{(300) - (60),\, (300)\} = \min\{240, 300\} = 240, \\ D &= (0.5)(300) + (0.5)(240) \\ &= 150 + 120 = 270, \end{aligned} \tag{4}$$

and equity receives $\max\{600 - 60 - 300, 0\} = 240$ (boom) and $\max\{300 - 60 - 300, 0\} = 0$ (slump), so $E = (0.5)(240) + (0.5)(0) = 120$. Both claims rise by 20 ($270 - 250$ and $120 - 100$), and $20 + 20 = 40 = 100 - 60$ is the project's NPV.

Answer (16.14). The old creditors benefit because, without senior new money, a positive-NPV project cannot be financed (debt overhang) and they are paid out of the assets in place only. With DIP financing the project is done; the DIP lender is repaid first, but the project adds more to the pool than the DIP claim takes out ($X_s \geq I$), so the old creditors' payoff rises in every impaired state (\$0 to \$50 in the Lily example; \$250 to \$270 in the lecture's example).

Step 5 — The limit. (3) needs $X_s \geq I$: a good, safe project. The same provision lets a distressed firm borrow to keep operating when it would be worth more liquidated; then $X_s < I$ in some state and the old creditors lose (Result 16.11).

Intuition. Debt overhang means the old creditors capture part of the value of new investment while the new money bears its full cost, so junior money cannot be raised. Letting new money jump the queue removes the overhang. The old creditors, who would otherwise get little or nothing, take a smaller slice of a larger pie. A covenant that blocks this hurts them, and a diffuse group of bondholders cannot easily agree to lend the money themselves (free-rider problem). Chapter 11 unlocks senior financing only in distress, which keeps most of the protection the covenant gives in normal times.

Common mistakes.

- Comparing the old creditors' payoff with the project under DIP financing against their payoff with the project under *junior* financing. The relevant status quo is no project at all, because junior money cannot be raised.
- Crediting the old creditors with the project's gross payoff X_s and forgetting that the DIP lender is repaid $N = I$ first; their gain is bounded by $X_s - I$, the project's NPV.
- Stating that DIP financing always helps creditors. It helps only when the funded project pays back at least its cost ($X_s \geq I$); financing a value-destroying continuation hurts them.

Ref: Grinblatt & Titman Ch. 16, § 16.2 "Underinvestment as a Consequence of Debt Overhang" (Lily example), pp. 564–566; § 16.3 "How Chapter 11 Bankruptcy Mitigates Debt Holder–Equity Holder Incentive Problems", Result 16.11, p. 580. Lecture Topic 4B, slides 22–31.

2. Asset substitution: project A or B — Grinblatt & Titman 16.1, p. 591

Statement.

Source: Grinblatt, Ch. 16, Exercise 16.1, p. 591 (PDF 604) · Answer in book: no

A firm has \$100 million in cash on hand and a debt obligation of \$100 million due in the next period. With this cash, it can take on one of two projects—A or B—which cost \$100 million each. Assume that the firm cannot raise any additional outside funds. If the economy is favorable, project A will pay \$120 million and project B will pay \$101 million. If the economy is unfavorable, project A will pay \$60 million and project B will pay \$101 million. Assume that investors are risk neutral, there are no taxes or direct costs of bankruptcy, the riskless interest rate is zero, and the probability of each state is .5.

a. What is the *NPV* of each project?

b. Which project will equity holders want the managers to take? Why?

What is asked. (a) The net present value of each project for the firm as a whole. (b) Which project the shareholders, who control the firm, would pick, and the reason for their choice.

Approach. With risk-neutral investors and a zero riskless rate, a claim's value is its expected payoff. Debt and equity split each state's payoff: debt gets $\min\{X, F\}$, equity gets $\max\{X - F, 0\}$ (limited liability). Because the cash on hand exactly equals the face value, equity is an at-the-money call option on the project payoff, so it values variance. This is the lecture's *asset substitution* (risk-shifting) problem: comparing firm value (a) with equity value (b) shows the gap between what is efficient and what shareholders choose.

Notation.

- $j \in \{A, B\}$: project; $X_{j,s}$: its payoff in state $s \in \{\text{fav}, \text{unf}\}$, each with probability $\pi = 0.5$.
- $C = 100$: cash on hand, all spent on the project; $F = 100$: debt due next period; $r_f = 0$; risk-neutral investors.
- $V_j = \mathbb{E}[X_j]$: firm value under j; E_j, D_j: values of equity and debt under j.

Step 1 — (a) NPVs. We value each project's payoff and subtract the \$100 it costs. With $r_f = 0$ and risk neutrality the present value is the expected payoff:

$$\begin{aligned} V_A &= \pi X_{A,\text{fav}} + (1-\pi) X_{A,\text{unf}} \qquad (5)\\ &= (0.5)(120) + (0.5)(60) \\ &= 60 + 30 = 90, \\ NPV_A &= V_A - C \qquad (6)\\ &= (90) - (100) \\ &= -10, \\ V_B &= \pi X_{B,\text{fav}} + (1-\pi) X_{B,\text{unf}} \qquad (7)\\ &= (0.5)(101) + (0.5)(101) \\ &= 50.5 + 50.5 = 101, \\ NPV_B &= V_B - C \\ &= (101) - (100) \\ &= 1. \end{aligned}$$

Answer (a). $NPV_A = -\$10$ million, $NPV_B = +\$1$ million: only B creates value.

Step 2 — (b) Payoffs to each claim, state by state. The shareholders care about their own claim, not about firm value, so we split each project's payoff between the claims. Equity receives $\max\{X - F, 0\}$ and debt $\min\{X, F\}$:

	Equity		Debt	
	Favourable	Unfavourable	Favourable	Unfavourable
A	$120 - 100 = 20$	0	100	60
B	$101 - 100 = 1$	$101 - 100 = 1$	100	100

(In A's unfavourable state $\max\{60 - 100, 0\} = 0$ and $\min\{60, 100\} = 60$.) Taking expectations,

$$E_A = (0.5)(20) + (0.5)(0) = 10 + 0 = 10, \qquad D_A = (0.5)(100) + (0.5)(60) = 50 + 30 = 80, \qquad (8)$$

$$E_B = (0.5)(1) + (0.5)(1) = 0.5 + 0.5 = 1, \qquad D_B = (0.5)(100) + (0.5)(100) = 100. \qquad (9)$$

Check: $E_A + D_A = 10 + 80 = 90 = V_A$ and $E_B + D_B = 1 + 100 = 101 = V_B$.

Step 3 — (b) The choice. We compare the shareholders' payoffs and decompose the difference into a value effect and a transfer. Equity holders, who control the firm, want project A: $E_A = 10 > E_B = 1$, although A destroys value. Writing $E_j = V_j - D_j$ and using (8)–(9),

$$\begin{aligned} E_A - E_B &= (V_A - V_B) - (D_A - D_B) \qquad (10)\\ &= (90 - 101) - (80 - 100) \\ &= (-11) - (-20) \\ &= 9 : \end{aligned}$$

switching to A lowers firm value by 11 but moves 20 from the creditors to the shareholders.

Answer (b). The shareholders want project A ($E_A = \$10$ million $> E_B = \$1$ million), because limited liability lets them keep A's upside while the creditors bear its downside: the \$20 million transferred from debt exceeds the \$11 million of value destroyed.

Intuition. With \$100 due and \$100 in cash, the equity is an at-the-money call on the firm's assets. The safe project leaves it almost worthless (1); the risky one gives it the whole upside (20) while the creditors absorb the downside (60 instead of 100). This is the lecture's asset-substitution (risk-shifting) problem: limited liability makes levered equity prefer variance, even at negative NPV.

Common mistakes.

- Writing equity's payoff as $X - F$ in every state. In A's unfavourable state that gives -40; limited liability caps the loss at 0, and that cap is exactly what makes A attractive.
- Answering (b) with the NPVs of (a). Firm value and equity value rank the projects in opposite orders; (b) asks about the shareholders' claim.
- Forgetting that the \$100 of cash is spent on the project: the NPV subtracts it, but the payoff split in (b) uses only the project payoff and the \$100 face value.

Ref: Grinblatt & Titman Ch. 16, § 16.2 "The Asset Substitution Problem", pp. 569–572 (Results 16.7–16.8). Lecture Topic 4B, slides 2–8.

3. Debt overhang on vacant land — Berk 16.25, pp. 494–495

Statement.

Source: Berk, Ch. 16, Problem 16.25, pp. 494–495 (PDF 525) · Answer in book: no

Consider a firm whose only asset is a plot of vacant land, and whose only liability is debt of \$15 million due in one year. If left vacant, the land will be worth \$10 million in one year. Alternatively, the firm can develop the land at an upfront cost of \$20 million. The developed land will be worth \$35 million in one year. Suppose the risk-free interest rate is 10%, assume all cash flows are risk-free, and assume there are no taxes.

a. If the firm chooses not to develop the land, what is the value of the firm's equity today? What is the value of the debt today?

b. What is the NPV of developing the land?

c. Suppose the firm raises \$20 million from equity holders to develop the land. If the firm develops the land, what is the value of the firm's equity today? What is the value of the firm's debt today?

d. Given your answer to part (c), would equity holders be willing to provide the \$20 million needed to develop the land?

What is asked. (a) Today's values of equity and debt if the land stays vacant. (b) The NPV of developing the land, for the firm as a whole. (c) Today's values of equity and debt if the shareholders put in \$20 million and the land is developed. (d) Whether the shareholders gain from putting in that money, given (c).

Approach. All cash flows are riskless, so every claim is its payoff in one year discounted at $r_f = 10\%$; debt gets $\min\{\text{land value}, 15\}$ and equity the rest. This is the lecture's *debt overhang* (underinvestment) problem: the debt is under water, so part of the value a new project creates goes to the creditors, who pay none of its cost. Comparing the NPV (b) with the shareholders' gain (d) shows the wedge.

Notation.

- $F = 15$: face value of the debt, due in one year; $r_f = 0.10$; all cash flows riskless; no taxes.
- $L_0 = 10$: value of the vacant land in one year; $L_1 = 35$: value of the developed land in one year; $K = 20$: development cost paid today.
- E, D: values today of equity and debt.

Step 1 — (a) Land left vacant. We split the land's value in one year between the claims and discount. In one year the firm is worth $10 < 15$: the creditors take 10 and the shareholders get 0.

Discounting at r_f,

$$\begin{aligned} D &= \frac{\min\{L_0, F\}}{1+r_f} \\ &= \frac{\min\{10, 15\}}{1+0.10} \\ &= \frac{(10)}{(1.10)} \\ &= 9.0909, \\ E &= \frac{\max\{L_0 - F, 0\}}{1+r_f} = \frac{\max\{10-15, 0\}}{1.10} = \frac{(0)}{(1.10)} = 0. \end{aligned} \tag{11}$$

Answer (a). Equity is worth \$0; debt is worth \$9.09 million.

Step 2 — (b) NPV of developing. The NPV counts only the cash flows that change with the decision. Developing replaces land worth 10 by land worth 35, so the incremental cash flow in one year is $35 - 10 = 25$, against 20 paid today:

$$\begin{aligned} NPV &= -K + \frac{L_1 - L_0}{1+r_f} \\ &= -(20) + \frac{(35)-(10)}{(1.10)} \\ &= -20 + \frac{25}{1.10} \\ &= -20 + 22.7273 \\ &= 2.7273. \end{aligned} \tag{12}$$

(Ignoring the vacant-land value forgone would give $-20 + 35/1.10 = 11.8182$, the NPV measured against a firm with no land at all; the decision-relevant figure is (12).)

Answer (b). $NPV = \$2.73$ million > 0: developing creates value.

Step 3 — (c) Values if developed with \$20 of new equity. We repeat Step 1 with the developed land. The \$20 is spent on development, so it is not an asset in one year. In one year the firm is worth $35 \geq 15$: the debt is paid in full, $\min\{35, 15\} = 15$, and equity keeps $\max\{35 - 15, 0\} = 20$. Hence

$$D = \frac{(15)}{(1.10)} \tag{13}$$

$$\begin{aligned} &= 13.6364, \\ E &= \frac{(35)-(15)}{(1.10)} \\ &= \frac{20}{1.10} \\ &= 18.1818. \end{aligned} \tag{14}$$

Answer (c). Equity is worth \$18.18 million and debt \$13.64 million today.

Step 4 — (d) Will the shareholders put up \$20?. The shareholders compare what they hold after contributing with what they hold if they do nothing. Their gain is the equity value (14), minus

the equity value without development (Step 1), minus the contribution:

$$\begin{aligned}\Delta E &= E_{\text{dev}} - E_{\text{vacant}} - K \\ &= (18.1818) - (0) - (20) \\ &= -1.8182 < 0, \\ \Delta D &= D_{\text{dev}} - D_{\text{vacant}} \\ &= (13.6364) - (9.0909) \\ &= 4.5455.\end{aligned}$$

No: they would lose \$1.82 million. The sum $-1.8182 + 4.5455 = 2.7273$ is the NPV (12); the creditors capture more than all of it.

Answer (d). No. The shareholders would pay \$20 million for an equity stake worth \$18.18 million (a loss of \$1.82 million), while the debt gains \$4.55 million; the positive-NPV project is passed up.

Intuition. The debt is under water ($10 < 15$), so the first \$5 million of any value the project creates goes to the creditors, who pay none of its cost. The shareholders bear the \$20 million and receive only the value above the \$15 million face. This is Myers' debt overhang: leverage acts as a tax on new investment, and a positive-NPV project is passed up. Writing the debt down (renegotiation, as in the lecture) or senior new financing would fix it.

Common mistakes.

- Computing the NPV as $-20 + 35/1.10 = 11.82$: the vacant land worth \$10 million in one year is given up, so it is part of the project's cost.
- Answering (d) with $E_{\text{dev}} = 18.18 > 0$, "so yes". The shareholders must compare the stake with the \$20 million they put in.
- Valuing the debt in (a) at the face value $15/1.10$; the creditors can get at most what the land is worth, $\min\{10, 15\}$.

Ref: Berk Ch. 16, § 16.6 "Agency Costs and Information", "Equity-Debt Holder Conflicts" (underinvestment), pp. 483–484. Lecture Topic 4B, slides 19–30.

4. Short-term debt and incentive problems — Grinblatt & Titman 16.11, p. 593

Statement.

Source: Grinblatt, Ch. 16, Exercise 16.11, p. 593 (PDF 606) · Answer in book: no

Why are debt holder–equity holder incentive problems less severe for firms that borrow short term rather than long term?

What is asked. A single conceptual question: why the conflicts between shareholders and creditors (asset substitution, debt overhang, under-maintenance) do less damage when the firm finances itself with debt that matures soon and must be rolled over, rather than with long-term debt.

Approach. The conflicts exist because equity chooses the firm's strategy *after* the terms of the debt are fixed, so it can change the value of the creditors' claim (the lecture's asset substitution and debt overhang). We write equity as firm value minus debt value and ask what happens to the debt term when the debt is repriced after the choice is made: if new lenders price the debt knowing the strategy, the debt term no longer depends on the strategy and equity internalizes the full value consequences.

Notation.

- j: investment strategy (project choice, risk level, maintenance), chosen by the equity holders after the debt is in place; $X_j \geq 0$: resulting terminal firm value, $V_j = \mathbb{E}[X_j]$ (risk neutrality, $r_f = 0$).

- F: face value of the debt; L: amount the firm owes when its short-term debt is rolled over; D_j, E_j: values of debt and equity under j.

Step 1 — Where the conflict comes from. We locate the conflict in the debt term of equity's value. With long-term debt, F is fixed before j is chosen, and since $\max\{X-F,0\} = X - \min\{X,F\}$ in every state,

$$\begin{aligned} E_j &= \mathbb{E}[\max\{X_j - F, 0\}] \\ &= V_j - D_j, \qquad D_j = \mathbb{E}[\min\{X_j, F\}]. \end{aligned} \tag{15}$$

Equity can gain by picking a j that lowers D_j (more risk, less investment, payouts) even when V_j falls: comparing two strategies, $E_j - E_k = (V_j - V_k) - (D_j - D_k)$, and the second term is a transfer from creditors whenever $D_j < D_k$.

Step 2 — Debt that is repriced when the choice is made. We now let the debt be repriced after the choice and show the transfer term disappears. If the debt matures before or at the decision and must be rolled over, the new lenders see j and set the new face F_j so that their claim is worth what they lend, $\mathbb{E}[\min\{X_j, F_j\}] = L$. Substituting into (15),

$$E_j = V_j - (L), \tag{16}$$

and L does not depend on j: equity bears the full consequences of its choice and picks the value-maximizing strategy. Short-term debt is worth close to its face whatever the firm does, so there is little value to expropriate.

Step 3 — Illustration (Grinblatt & Titman 16.1). Take the firm of G&T Exercise 16.1: \$100 of debt, projects A (pays 120 or 60) and B (pays 101 in both states), each state with probability 0.5, $r_f = 0$. With long-term debt, equity is worth $0.5(20) + 0.5(0) = 10$ under A and $0.5(1) + 0.5(1) = 1$ under B, so it picks the value-destroying A. Suppose instead the \$100 had to be refinanced after the project is chosen. Firm values are $V_A = 0.5(120) + 0.5(60) = 90$ and $V_B = 0.5(101) + 0.5(101) = 101$; by (16) with $L = 100$,

$$E_A = (90) - (100) = -10, \qquad E_B = (101) - (100) = 1.$$

Under A the firm cannot raise \$100 (no claim on payoffs worth 90 is worth 100), so the equity holders choose B, the positive-NPV project, instead of A.

Step 4 — Debt overhang. Myers (1977): if the existing debt matures before the firm must raise money for a new project, the debt is repriced together with the project decision. The project's benefit to creditors (a safer firm) then shows up as a lower borrowing rate that accrues to equity, and positive-NPV projects are taken. In Unitron's case, the riskless 12% project would lower the firm's 13% borrowing rate and become attractive.

Step 5 — Asset substitution. The lending rate is renegotiated after the risk choice is made and seen, so a risky choice raises the firm's own cost of funds instead of transferring wealth from existing lenders. The threat of not being rolled over also disciplines the choice (lecture Topic 4B, slide 16).

Step 6 — The cost. Rolling over exposes the firm to refinancing and interest-rate risk: an unexpected rise in rates can push a highly levered firm into distress (hence the advice to pair short-term debt with interest-rate hedges).

Answer (16.11). Short-term debt is repriced (rolled over) after the shareholders' investment and risk choices are made and observed, so its terms reflect those choices: equity bears the full value consequences, $E_j = V_j - L$, and cannot shift wealth from creditors through risk-taking or under-investment. Short-term debt is also close to riskless, so there is little value to expropriate. The price is rollover and interest-rate risk.

Intuition. The conflicts exist because equity can change the value of a claim whose terms are already fixed. The shorter the maturity, the sooner those terms are reset at prices that reflect what

equity has done, and the less wealth equity can take from creditors. The incentive problem shrinks, and the firm takes on rollover risk in exchange.

Common mistakes.

- Saying short-term debt helps "because it is safer". The point is timing: the debt is priced after the decision, so the decision's effect on the creditors comes back to equity through the new rate.
- Confusing maturity with seniority. Seniority decides who is paid first in default; maturity decides when the terms are reset.
- Omitting the cost. Short maturities expose the firm to refinancing and interest-rate risk, so they are not a free cure.

Ref: Grinblatt & Titman Ch. 16, § 16.4 "The Use of Short-Term versus Long-Term Debt", pp. 584–585 (Result 16.12). Lecture Topic 4B, slide 16.

5. Random financing — Tirole 3.1, p. 144

Statement.

Source: Tirole, Ch. 3, Exercise 3.1, p. 144 (PDF 157) · Answer in book: yes, p. 571

Exercise 3.1 (random financing). Consider the fixed-investment model of Section 3.2. We know that if $A \geqslant \bar{A}$, where

$$I - \bar{A} = p_H \left(R - \frac{B}{\Delta p} \right),$$

it is both optimal and feasible for the borrower to sign a contract in which the project is undertaken for certain. We also noted that for $A < \bar{A}$, the borrower cannot convince investors to undertake the project with probability 1. With $A > 0$, the entrepreneur benefits from signing a "random financing contract," though.

(i) Consider a contract in which the borrower invests $\hat{A} \in [0, A]$ of her own money, the project is financed with probability x, and the borrower receives R_b in the case of success and 0 otherwise. Write the investors' breakeven condition.

(ii) Show that (provided the NPV, $p_H R - I$, is positive) it is optimal for the borrower to invest

$$\hat{A} = A.$$

How does the probability that the project is undertaken vary with A?

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \tag{3.6}$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. [...]

From Tirole §3.2.1, footnote 9, p. 116 (PDF 129):

"Random financing" contracts, in which the borrower brings equity in exchange for a probability between 0 and 1 of being financed may in some cases be optimal when the investment size is fixed (as it is here) or more generally in the presence of indivisibilities or increasing returns to scale (see Exercise 3.1). For simplicity, we focus on deterministic contracts.

What is asked. (i) Write the condition under which investors are willing to sign a contract in which the borrower contributes $\hat{A}$ of her cash, the project then goes ahead only with probability x, and she gets R_b on success. (ii) Show that a borrower with too little cash for certain financing ($A < \bar{A}$) should put *all* her cash in, and describe how the financing probability depends on A.

Approach. This is the Holmström–Tirole fixed-investment model of the lecture (Topic 4A). The two constraints are the borrower's incentive constraint (IC_b), which caps the *pledgeable income* at $\mathcal{P} = p_H(R - B/\Delta p)$, and the investors' breakeven constraint (IR_l). Below $\bar{A}$ the project cannot be financed for sure; a lottery over financing scales the investors' outlay and the pledgeable income by x but not the borrower's contribution, which convexifies the indivisible project. We write (IR_l) for the lottery, show (IC_b) must hold, and maximise the borrower's utility over $(x, \hat{A})$.

Notation.

- $I > 0$: fixed investment; $A \in (0, \bar{A})$: borrower's cash; $\hat{A} \in [0, A]$: the part she contributes; $x \in [0, 1]$: probability that the project is financed.
- $R > 0$: income on success (0 on failure); $R_b \geq 0$: borrower's reward on success; $R_l = R - R_b$: investors' claim on success.
- $p_H > p_L$: success probabilities when she behaves / misbehaves; $\Delta p = p_H - p_L$; $B > 0$: private benefit from misbehaving.
- $\mathcal{P} = p_H(R - B/\Delta p)$: pledgeable income; $\bar{A} = I - \mathcal{P}$: minimum cash for financing with probability 1 (the statement's definition); U_b: borrower's net utility. Rate of interest 0, everybody risk neutral.

Assumption. The borrower hands over $\hat{A}$ when the contract is signed, whether or not the lottery then selects the project (this is the reading in Tirole's answer). If $\hat{A}$ were paid only when the project goes ahead, the expected outlay would be $x(I - \hat{A})$ and the breakeven condition $x(I - \hat{A}) \leq x\mathcal{P}$ would require $\hat{A} \geq \bar{A} > A$: randomizing would achieve nothing.

Step 1 — (i) Investors' breakeven condition. Investors receive $\hat{A}$ for sure, pay I with probability x, and, if the borrower behaves, collect R_l with probability p_H when the project runs:

$$\underbrace{\hat{A}}_{\text{received}} + \underbrace{x\, p_H(R - R_b)}_{\text{expected repayment}} \geq \underbrace{xI}_{\text{expected outlay}}$$

$$x\, p_H(R - R_b) \geq xI - \hat{A}. \tag{17}$$

Answer (i). $x\, p_H(R - R_b) \geq xI - \hat{A}$ (with $R_b \geq B/\Delta p$ so that she behaves).

Step 2 — The contract must induce good behaviour. We check that a contract under which she shirks is never worth signing, so (IC_b) can be imposed. Suppose $(\Delta p)R_b < B$, so she shirks.

Investors break even only if $xp_L(R-R_b) \geq xI - \hat{A}$, i.e. $-\hat{A} \leq xp_L(R-R_b) - xI$. Her net utility is then

$$\begin{aligned} x(p_L R_b + B) - \hat{A} &\leq x(p_L R_b + B) + \big(xp_L(R-R_b) - xI\big) \\ &= xp_L R_b + xB + xp_L R - xp_L R_b - xI \\ &= x(p_L R + B - I) < 0 \end{aligned} \tag{18}$$

by (3.2), worse than not signing. So the contract satisfies (IC_b), $(\Delta p)R_b \geq B$, i.e. $R_b \geq B/\Delta p$.

Step 3 — Breakeven in terms of $\bar{A}$. We combine the two constraints into one feasibility condition on $(x, \hat{A})$. (IC_b) gives $R - R_b \leq R - B/\Delta p$, so the left side of (17) is at most $x\,p_H(R - B/\Delta p) = x\mathcal{P}$. A contract exists for $(x, \hat{A})$ only if

$$\begin{aligned} xI - \hat{A} &\leq x\,p_H\left(R - \frac{B}{\Delta p}\right) \\ xI - \hat{A} &\leq x(\mathcal{P}) \\ x(I - \mathcal{P}) &\leq \hat{A} \\ x(\bar{A}) &\leq \hat{A} \\ x &\leq \frac{\hat{A}}{\bar{A}}, \end{aligned} \tag{19}$$

substituting $I - \mathcal{P} = \bar{A}$ from the statement's definition (third to fourth line) and dividing by $\bar{A} > 0$ (last line). Conversely, if (19) holds, $R_b = B/\Delta p$ satisfies both constraints, so (19) is necessary and sufficient.

Step 4 — (ii) The borrower's objective. We write her utility and use the binding breakeven constraint to express it in terms of x only. She keeps $A - \hat{A}$ and, with probability x, earns R_b with probability p_H; net of the A she would consume anyway,

$$\begin{aligned} U_b &= x\,p_H R_b + (A - \hat{A}) - A \\ &= x\,p_H R_b - \hat{A}. \end{aligned} \tag{20}$$

Competitive investors make (17) bind; expanding and solving for $xp_H R_b$,

$$\begin{aligned} xp_H R - xp_H R_b &= xI - \hat{A} \\ xp_H R_b &= xp_H R - xI + \hat{A}. \end{aligned}$$

Substituting into (20),

$$\begin{aligned} U_b &= (xp_H R - xI + \hat{A}) - \hat{A} \\ &= x\,(p_H R - I). \end{aligned} \tag{21}$$

Step 5 — (ii) Optimal $\hat{A}$ and x. We maximise (21) over the feasible set (19). Differentiating,

$$\frac{\partial U_b}{\partial x} = p_H R - I > 0, \qquad \frac{\partial U_b}{\partial \hat{A}} = 0,$$

so U_b rises with x and $\hat{A}$ matters only through feasibility. By (19), the largest feasible x is $\hat{A}/\bar{A}$ (below 1 because $\hat{A} \leq A < \bar{A}$), which increases with $\hat{A}$: substituting it, $U_b = (\hat{A}/\bar{A})(p_H R - I)$, increasing in $\hat{A}$ on $[0, A]$. So she contributes everything:

$$\hat{A}^* = A,$$

$$x^* = \frac{A}{\bar{A}}, \tag{22}$$

$$U_b^* = \frac{A}{\bar{A}}\,(p_H R - I). \tag{23}$$

At x^* the binding (17), divided by $x^* = A/\bar{A}$, gives $p_H(R - R_b) = I - A/x^* = I - \bar{A} = \mathcal{P}$, so $R - R_b = R - B/\Delta p$ and $R_b = B/\Delta p$: (IC_b) binds.

Step 6 — (ii) How x^* varies with A. $x^* = A/\bar{A}$ rises linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$, and stays at 1 above $\bar{A}$. Utility (23) therefore rises smoothly to the full NPV instead of jumping at $\bar{A}$ as in (3.6).

Numerical illustration. $I = 100$, $R = 150$, $p_H = 0.8$, $p_L = 0.4$, $B = 30$, $A = 10$: (3.2) holds ($0.4(150) - 100 + 30 = -10 < 0$); $B/\Delta p = 75$; $\mathcal{P} = 0.8(150 - 75) = 60$; $\bar{A} = 100 - 60 = 40$; NPV $= 0.8(150) - 100 = 20$. Then $x^* = 10/40 = 0.25$ and $U_b^* = 0.25(20) = 5$; investors pay $0.25(100) - 10 = 15$ in expectation and receive $0.25(60) = 15$.

Answer (ii). $\hat{A}^* = A$; the financing probability is $x^* = A/\bar{A}$, rising linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$ (and $x^* = 1$ for $A \geq \bar{A}$); her utility is $x^*(p_H R - I)$.

Intuition. The obstacle is not the size of the NPV but the pledgeable income per unit of investment. A poor borrower can still buy a *fraction* of the deal: each unit of her cash covers the investors' shortfall on $1/\bar{A}$ units of probability of financing. Since she captures the whole NPV whenever the project runs, she wants as high a probability as her cash can buy and puts all of it in. Lotteries convexify the indivisible project.

Common mistakes.

- Writing the breakeven as $xp_H(R - R_b) \geq x(I - \hat{A})$, i.e. taking her contribution only when the project goes ahead. Then x cancels and randomisation achieves nothing.
- Using R instead of $R - B/\Delta p$ as the investors' maximal claim: without (IC_b) she shirks, and by (3.2) no one breaks even.
- Claiming she keeps some cash "for safety". Cash kept aside earns nothing extra, while each unit put in raises x by $1/\bar{A}$ and her utility by $(p_H R - I)/\bar{A} > 0$.

Book answer. Matches the book's answer (p. 571): breakeven $xI - A \leq xp_H(R - R_b)$, equivalently $x\bar{A} \leq A$; $U_b = x(p_H R - I)$, so $x^* = A/\bar{A}$, rising from 0 to 1 as A goes from 0 to $\bar{A}$.

Ref: Tirole Ch. 3, § 3.2.1–3.2.2 "The Fixed-Investment Model", "The Lenders' Credit Analysis", pp. 115–117 (footnote 9, p. 116). Lecture Topic 4A, slides 26–34.

6. Zymase: leverage and strategy choice — Berk 16.26, p. 495

Statement.

Source: Berk, Ch. 16, Problem *26, p. 495 (PDF 526) · Answer in book: no

*26. Zymase is a biotechnology start-up firm. Researchers at Zymase must choose one of three different research strategies. The payoffs (after taxes) and their likelihood for each strategy are shown below. The risk of each project is diversifiable.

Strategy	Probability	Payoff ($ million)
A	100%	75
B	50%	140
	50%	0
C	10%	300
	90%	40

a. Which project has the highest expected payoff?

b. Suppose Zymase has debt of $40 million due at the time of the project's payoff. Which strategy has the highest expected payoff for equity holders?

c. Suppose Zymase has debt of $110 million due at the time of the strategy's payoff. Which strategy has the highest expected payoff for equity holders?

d. If management chooses the strategy that maximizes the payoff to equity holders, what is the expected agency cost to the firm from having \$40 million in debt due? What is the expected agency cost to the firm from having \$110 million in debt due?

What is asked. (a) Which strategy gives the largest expected payoff to the firm as a whole. (b) and (c) Which strategy gives the shareholders the largest expected payoff when \$40 million, respectively \$110 million, of debt is due at the payoff date. (d) How much expected firm value is lost, in each case, when management follows the shareholders' preference instead of the value-maximising strategy (the expected agency cost of debt).

Approach. Split each strategy's payoff state by state into debt, $\min\{X, F\}$, and equity, $\max\{X - F, 0\}$, and take expectations. Equity is a call option on the firm, so as F rises it favours strategies whose payoff lies above F with some probability, even if their mean is lower: the lecture's *asset substitution* (excessive risk-taking). The agency cost is the drop in expected *firm* payoff caused by that choice, which is the loss the shareholders bear ex ante when creditors price the debt.

Notation.

- $j \in \{A, B, C\}$: research strategy; X_j: its after-tax payoff (\$ million), with the probabilities in the table.
- $F \in \{40, 110\}$: face value of the debt, due when the payoff arrives; equity gets $\max\{X_j - F, 0\}$, debt $\min\{X_j, F\}$.
- Agency cost: expected firm payoff of the value-maximizing strategy minus that of the strategy equity chooses.

Assumption. Risk is diversifiable, so every claim is discounted at the riskless rate, which is not given. All comparisons are in expected payoffs at the payoff date; present values are these divided by the same factor $1 + r_f$, so the rankings do not change.

Step 1 — (a) Expected firm payoffs. We weight each payoff by its probability:

$$\mathbb{E}[X_A] = (1.0)(75) = 75, \tag{24}$$
$$\mathbb{E}[X_B] = (0.5)(140) + (0.5)(0) = 70 + 0 = 70, \tag{25}$$
$$\mathbb{E}[X_C] = (0.1)(300) + (0.9)(40) = 30 + 36 = 66. \tag{26}$$

Strategy A has the highest expected payoff (and the lowest risk).

Answer (a). Strategy A, with an expected payoff of \$75 million (B: \$70 million; C: \$66 million).

Step 2 — (b) Debt of \$40 million. We split each state's payoff with $F = 40$: equity $\max\{X - 40, 0\}$ (e.g. $140 - 40 = 100$; $\max\{40 - 40, 0\} = 0$), debt $\min\{X, 40\}$:

Strategy, state (prob.)	Firm	Equity $\max\{X - 40, 0\}$	Debt $\min\{X, 40\}$
A (1.0)	75	35	40
B, high (0.5)	140	100	40
B, low (0.5)	0	0	0
C, high (0.1)	300	260	40
C, low (0.9)	40	0	40

$$\begin{aligned} \mathbb{E}[E_A] &= (1.0)(35) = 35, \\ \mathbb{E}[E_B] &= (0.5)(100) + (0.5)(0) = 50 + 0 = 50, \\ \mathbb{E}[E_C] &= (0.1)(260) + (0.9)(0) = 26 + 0 = 26. \end{aligned} \tag{27}$$

Equity prefers B. Debt then expects $(0.5)(40) + (0.5)(0) = 20$, against 40 under A or C.

Answer (b). Strategy B (expected equity payoff \$50 million, against \$35 million for A and \$26 million for C).

Step 3 — (c) Debt of \$110 million. Same split with $F = 110$: equity $\max\{X - 110, 0\}$ (e.g. $\max\{75 - 110, 0\} = 0$, $140 - 110 = 30$, $300 - 110 = 190$), debt $\min\{X, 110\}$:

Strategy, state (prob.)	Firm	Equity $\max\{X - 110, 0\}$	Debt $\min\{X, 110\}$
A (1.0)	75	0	75
B, high (0.5)	140	30	110
B, low (0.5)	0	0	0
C, high (0.1)	300	190	110
C, low (0.9)	40	0	40

$$\begin{aligned} \mathbb{E}[E_A] &= (1.0)(0) = 0, \\ \mathbb{E}[E_B] &= (0.5)(30) + (0.5)(0) = 15 + 0 = 15, \\ \mathbb{E}[E_C] &= (0.1)(190) + (0.9)(0) = 19 + 0 = 19. \end{aligned} \tag{28}$$

Equity prefers C. Debt then expects $(0.1)(110) + (0.9)(40) = 11 + 36 = 47$, against 75 under A and $(0.5)(110) + (0.5)(0) = 55$ under B.

Answer (c). Strategy C (expected equity payoff \$19 million, against \$15 million for B and \$0 for A).

Step 4 — (d) Expected agency costs. The agency cost is the expected firm payoff of the efficient strategy minus that of the strategy equity picks. The value-maximizing strategy is A (24). With $F = 40$ equity picks B (25); with $F = 110$ it picks C (26):

$$\text{cost}(40) = \mathbb{E}[X_A] - \mathbb{E}[X_B] = (75) - (70) = 5, \tag{29}$$

$$\text{cost}(110) = \mathbb{E}[X_A] - \mathbb{E}[X_C] = (75) - (66) = 9. \tag{30}$$

The expected agency cost is \$5 million with \$40 million of debt and \$9 million with \$110 million (in payoff-date dollars; divide by $1 + r_f$ for today's value).

Answer (d). \$5 million with \$40 million of debt; \$9 million with \$110 million of debt (expected payoff-date values).

Intuition. With \$40 million due, A leaves equity a safe \$35 million, but B's 50% chance of \$100 million is worth more to equity because the creditors bear the zero-payoff state. With \$110 million due, A leaves equity nothing and only the long shot C pays off above the face value often enough. More debt pushes equity towards riskier, lower-value strategies, and the agency cost grows with leverage. Creditors who anticipate the choice price it in, so the shareholders bear the cost ex ante.

Common mistakes.

- Measuring the agency cost as the creditors' loss (e.g. $40 - 20 = 20$ with $F = 40$). That is a transfer; the agency cost is the loss in *total* expected payoff, $75 - 70 = 5$.
- Giving equity $X - F$ in states where $X < F$ (C's low state with $F = 110$ gives -70, not 0), which makes the risky strategies look worse to equity than they are.
- Comparing equity's payoffs across debt levels instead of across strategies at a given debt level; each of (b) and (c) is a choice among A, B, C with F fixed.

Ref: Berk Ch. 16, § 16.6 "Agency Costs and Information", "Equity-Debt Holder Conflicts" (excessive risk-taking), pp. 483–484. Lecture Topic 4B, slides 2–13.

7. Ajax Manufacturing: debt overhang — Grinblatt & Titman 16.12–16.13, p. 593

Statement.

Source: Grinblatt–Titman, Ch. 16, Exercises 16.12 and 16.13, p. 593 (PDF 606) · Answer in book: no

16.12. Consider the case of Ajax Manufacturing which just completed an R&D project on widgets that required a \$70 million bond obligation. The R&D effort resulted in an investment opportunity that will cost \$75 million and generate cash flows of \$85 million in the event of a recession (*prob.* = 20%) and \$150 million if economic conditions are favorable (*prob.* = 80%). What is the *NPV* of the project assuming no taxes, no direct bankruptcy costs, risk neutrality, and a risk-free interest rate of zero? Can the firm fund the project if the original debt is a senior obligation that doesn't allow the firm to issue additional debt?

16.13. Assume now that if Ajax Manufacturing (see exercise 16.12) uses a more capital-intensive manufacturing process, it can produce a greater number of widgets at a lower variable cost. Given the greater fixed costs, the cash flows are only \$5 million in an unfavorable economy with the capital-intensive process but are \$170 million in a favorable economy. Hence, equity holders would receive \$100 million in the good state of the economy (\$170 million − \$70 million) and zero in a recession because \$5 million is less than the \$70 million debt obligation. Can the firm issue equity to fund the project?

What is asked. (16.12) The NPV of the widget project, and whether Ajax can raise the \$75 million when the existing \$70 million senior bond forbids new debt (so only new equity is possible). (16.13) Whether Ajax can raise the \$75 million in equity if it uses the riskier, capital-intensive process instead.

Approach. Risk neutrality and $r_f = 0$ make every value an expected payoff. New equity is junior to the bond, so the most new shareholders can be offered is the whole residual claim $\mathbb{E}[\max\{X - 70, 0\}]$; the project is fundable iff that is at least \$75 million. This is the lecture's *debt overhang*: the bond captures part of the project's value. In 16.13 a riskier payoff shrinks the bond's share (*asset substitution*), which can make the equity issue feasible.

Notation.

- $F = 70$: existing senior bond obligation, with a covenant forbidding additional debt; $I = 75$: investment cost.
- States: recession (probability $\pi = 0.2$) and favourable (probability 0.8); X_s: project cash flow in state s; risk-neutral investors, $r_f = 0$, no taxes or bankruptcy costs.
- V: project value; $E^{\max}$: value of the whole residual claim after the bond, the most new shareholders can be offered; D: bond value.

Assumption. The firm has no other assets: without the project, the bond receives nothing. In 16.13 the investment still costs $I = 75$ (the statement changes only the cash flows).

Step 1 — (16.12) NPV of the project. We value the project's cash flows and subtract their cost:

$$\begin{aligned} V &= \pi X_{\text{rec}} + (1-\pi) X_{\text{fav}} \\ &= (0.2)(85) + (0.8)(150) \\ &= 17 + 120 = 137, \end{aligned} \tag{31}$$

$$NPV = V - I = (137) - (75) = 62. \tag{32}$$

Step 2 — (16.12) Can it be funded?. We value the largest claim new shareholders can buy and compare it with the \$75 million they must pay. New debt is forbidden, so the money must come from equity, which is junior to the \$70 bond. Equity receives $\max\{85 - 70, 0\} = 15$ in a recession and $\max\{150 - 70, 0\} = 80$ in the favourable state:

$$\begin{aligned} E^{\max} &= (0.2)(15) + (0.8)(80) \\ &= 3 + 64 = 67 < 75 = I. \end{aligned} \tag{33}$$

Even 100% of the equity is worth less than the \$75 million it must raise: **no**. The bond, which would be paid in full ($85 \geq 70$), would be worth $D = 70$ instead of 0; the split $70 + (67 - 75) = 62$ is the NPV. The creditors capture more than the whole NPV, which is why the shareholders cannot pay for it.

Answer (16.12). $NPV = \$62$ million, but the firm **cannot** fund the project: all of the equity is worth \$67 million $<$ \$75 million.

Step 3 — (16.13) The capital-intensive variant. We repeat Steps 1–2 with the new cash flows, 5 (recession) and 170 (favourable):

$$\begin{aligned} V' &= (0.2)(5) + (0.8)(170) \\ &= 1 + 136 = 137, \\ NPV' &= (137) - (75) = 62, \end{aligned}$$

the same NPV but riskier. Equity receives $\max\{5 - 70, 0\} = 0$ and $170 - 70 = 100$:

$$\begin{aligned} E'^{\max} &= (0.2)(0) + (0.8)(100) \\ &= 80 \geq 75 = I. \end{aligned} \tag{34}$$

Yes: new shareholders pay \$75 million for a fraction $75/80 = 0.9375$ of the equity and the old shareholders keep $1 - 0.9375 = 6.25\%$, worth $0.0625(80) = 80 - 75 = 5$. The bond now receives $\min\{5, 70\} = 5$ and $\min\{170, 70\} = 70$, worth

$$\begin{aligned} D' &= (0.2)(5) + (0.8)(70) \\ &= 1 + 56 = 57, \end{aligned}$$

and $57 + 80 = 137 = V'$.

Answer (16.13). **Yes**: under the capital-intensive process the equity is worth \$80 million $\geq$ \$75 million, so new shareholders buy 93.75% of it for \$75 million.

Step 4 — Comparison. The two processes create the same value, 62. The risky one moves $70 - 57 = 13$ from the bondholders to the equity, which is exactly what lifts the equity's value from 67 to 80, above the \$75 million cost. The bondholders prefer the safe process, but under it nothing is built and they get 0, so they are better off with the risky one ($57 > 0$).

Intuition. Debt overhang: with a senior bond that cannot be diluted, new equity must pay for the whole project but receives only the value left after the bond, and here the bond absorbs too much of it. Raising risk shrinks the bond's share (asset substitution), and that makes the equity issue possible. Of the two distortions, the risk shift is the lesser here. A cleaner fix is to renegotiate the bond: under the safe process equity covers \$75 million whenever the face is cut to at most $137 - 75 = 62$, which the bondholders should accept since $62 > 57$ (the lecture's renegotiation argument).

Common mistakes.

- Answering "NPV > 0, so yes" in 16.12. Fundability depends on what the junior claim is worth to the new money, not on total value.
- Valuing equity as $V - F$. That gives $137 - 70 = 67$ for *both* processes and the wrong answer in 16.13; limited liability makes the recession payoff $\max\{5 - 70, 0\} = 0$, not -65.
- Forgetting that without the project the bond gets nothing here; that is why the bondholders prefer the risky process (57) to no project (0).

Ref: Grinblatt & Titman Ch. 16, § 16.2 "The Debt Overhang Problem" and "How the Seniority Structure of Debt Affects the Debt Overhang Problem", pp. 563–566 (Results 16.3–16.4); "The Asset Substitution Problem", pp. 569–572. Lecture Topic 4B, slides 19–31.

8. Project riskiness and credit rationing — Tirole 3.15, p. 150

Statement.

Source: Tirole, Ch. 3, Exercise 3.15, p. 150 (PDF 163) · Answer in book: yes, p. 578

Exercise 3.15 (project riskiness and credit rationing). Consider the basic, fixed-investment model (the investment is I, the entrepreneur borrows $I - A$; the probability of success is p_H (no private benefit) or $p_L = p_H - \Delta p$ (private benefit B), success (failure) yields verifiable profit R (respectively 0)). There are two variants, "A" and "B," of the projects, which differ only with respect to "riskiness":

$$p_H^A R^A = p_H^B R^B, \quad \text{but } p_H^A > p_H^B;$$

so project B is "riskier." The investment cost is the same for both variants and, furthermore,

$$p_H^A - p_L^A = p_H^B - p_L^B.$$

Which variant is less prone to credit rationing?

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\mathrm{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H (R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. [...]

What is asked. Two versions of the fixed-investment project have the same cost, the same expected income under good behaviour and the same effect of shirking on the success probability, but B succeeds less often with a larger payoff. Decide which version can be financed for a wider range of the entrepreneur's cash A, i.e. is less prone to credit rationing.

Approach. In the Holmström–Tirole fixed-investment model (lecture Topic 4A) a project is financed iff $A \geq \bar{A} = I - \mathcal{P}$, where $\mathcal{P} = p_H(R - B/\Delta p)$ is the *pledgeable income*. So the variant with the lower threshold $\bar{A}$, i.e. the higher pledgeable income, is less rationed. We write $\bar{A}$ as non-pledgeable rent minus NPV, use equal NPVs, and compare the rents.

Notation.

- $j \in \{A, B\}$: project variant; $p_H^j > p_L^j$: success probabilities with good / bad behaviour; $R^j > 0$: income on success (0 on failure).

- $\Delta p = p_H^A - p_L^A = p_H^B - p_L^B > 0$: common; $B > 0$: private benefit (common); I: common investment; A: borrower's cash.
- Maintained: $p_H^A R^A = p_H^B R^B$ (equal NPV) and $p_H^A > p_H^B$ (B riskier); (3.1), (3.2) hold for both variants.
- $\mathcal{P}^j = p_H^j(R^j - B/\Delta p)$: pledgeable income; $\bar{A}^j$: minimum cash for financing variant j (3.3).

Step 1 — The minimum stake is the same. We first derive the borrower's minimum reward in each variant. Variant j's incentive constraint is $p_H^j R_b^j \geq p_L^j R_b^j + B$, i.e. $(p_H^j - p_L^j)R_b^j \geq B$, i.e. $R_b^j \geq B/\Delta p$: the same for both variants, because B and Δp are common.

Step 2 — Financing thresholds. We compute each variant's threshold from (IR_l), $\mathcal{P}^j \geq I - A$. From (3.3),

$$\begin{aligned}\bar{A}^j &= I - \mathcal{P}^j \\ &= I - p_H^j\left(R^j - \frac{B}{\Delta p}\right) \\ &= I - p_H^j R^j + p_H^j \frac{B}{\Delta p} \\ &= p_H^j \frac{B}{\Delta p} - (p_H^j R^j - I). \end{aligned} \tag{35}$$

The first term is the non-pledgeable income: the expected rent the borrower must be left.

Step 3 — Compare. Subtracting (35) for B from that for A, and substituting $p_H^B R^B = p_H^A R^A$,

$$\begin{aligned}\bar{A}^A - \bar{A}^B &= \left[p_H^A \frac{B}{\Delta p} - (p_H^A R^A - I)\right] - \left[p_H^B \frac{B}{\Delta p} - ((p_H^A R^A) - I)\right] \\ &= p_H^A \frac{B}{\Delta p} - p_H^A R^A + I - p_H^B \frac{B}{\Delta p} + p_H^A R^A - I \\ &= (p_H^A - p_H^B)\frac{B}{\Delta p} > 0, \end{aligned} \tag{36}$$

positive because $p_H^A > p_H^B$, $B > 0$, $\Delta p > 0$. The riskier variant B is financed for every $A \geq \bar{A}^B$, including $A \in [\bar{A}^B, \bar{A}^A)$ where A is not: **variant B is less prone to credit rationing**.

Answer (3.15). Variant B, the riskier one: $\bar{A}^B = \bar{A}^A - (p_H^A - p_H^B)B/\Delta p < \bar{A}^A$, so every entrepreneur financed under A is financed under B, and those with $A \in [\bar{A}^B, \bar{A}^A)$ only under B (given that both variants need good behaviour, (3.2)).

Step 4 — Numerical illustration. $I = 100$, $\Delta p = 0.2$, $B = 9$; A: $p_H^A = 0.8$, $p_L^A = 0.6$, $R^A = 150$; B: $p_H^B = 0.6$, $p_L^B = 0.4$, $R^B = 200$. Both NPVs are $120 - 100 = 20$; (3.2) holds $(0.6(150) - 100 + 9 = -1 < 0$ and $0.4(200) - 100 + 9 = -11 < 0)$; $B/\Delta p = 45$. Then $\mathcal{P}^A = 0.8(105) = 84$, $\bar{A}^A = 16$; $\mathcal{P}^B = 0.6(155) = 93$, $\bar{A}^B = 7$; and $\bar{A}^A - \bar{A}^B = (0.2)(45) = 9$, as (36) says.

Step 5 — Beyond the maintained assumption (the book's extension). If (3.2) fails for the safe variant, a contract that leaves all income to investors ($R_b = 0$, the borrower misbehaves) may finance it: investors then receive $p_L^A R^A$, and the borrower, who gets the NPV with shirking, $p_L^A R^A + B - I$, when investors break even, wants it if that NPV is positive. The safe variant can then be financed and the risky one not when (in order) B cannot be financed with good behaviour, B cannot be financed with shirking, A can be financed with shirking, and A's NPV with shirking is positive:

$$I - A > p_H^B\left(R^B - \frac{B}{\Delta p}\right), \quad I - A > p_L^B R^B, \quad I - A \leq p_L^A R^A, \quad I < p_L^A R^A + B. \tag{37}$$

These conditions are compatible. With $A = 0$, $I = 120$, $\Delta p = 0.3$, $B = 45$, $p_H^A = 0.9$, $R^A = 200$, $p_H^B = 0.5$, $R^B = 360$ (both $p_H R = 180$): $p_H^B(R^B - B/\Delta p) = 0.5(210) = 105 < 120$; $p_L^B R^B =$

$0.2(360) = 72 < 120$; $p_L^A R^A = 0.6(200) = 120 \geq 120$; $p_L^A R^A + B = 165 > 120$. Here the risky variant is rationed and the safe one is funded. Good behaviour adds more pledgeable income when the success payoff is large, which favours the risky variant only when good behaviour is needed for funding.

Intuition. The borrower's rent $B/\Delta p$ is paid only on success. A safer project succeeds more often, so it pays out the rent more often and leaves less expected income to pledge: the non-pledgeable rent $p_H B/\Delta p$ is higher for the same NPV. Riskier projects with the same mean concentrate income in success, where it is pledged anyway.

Common mistakes.

- Importing the Stiglitz–Weiss intuition that riskier borrowers are rationed first. Here riskiness lowers the rent $p_H B/\Delta p$ and so *raises* pledgeable income.
- Computing the threshold as $I - p_H R$ (all income pledged), which is the same for both variants by $p_H^A R^A = p_H^B R^B$ and hides the difference.
- Forgetting that the comparison holds only while good behaviour is required for funding: if (3.2) fails for one variant, the book's extension (Step 5) can reverse the ranking.

Book answer. Matches the book's answer (p. 578): with the same minimum reward $B/\Delta p$, the safe project A is financed for a smaller range of A, i.e. it is more prone to credit rationing; the book then shows, with conditions (37), that the risky project may be the more rationed one if high-powered incentives are not necessarily called for. One slip in the book's prose: it says the third condition means that "the risky project" generates enough pledgeable income when investors get all the cash flow, but that condition, $I - A \leq p_L^A R^A$, is about the safe variant A.

Ref: Tirole Ch. 3, § 3.2.1–3.2.2 "The Fixed-Investment Model", "The Lenders' Credit Analysis", pp. 115–117. Lecture Topic 4A, slides 26–35 (determinants of credit rationing).

9. Entrepreneurial risk aversion — Tirole Exercise 3.2, p. 144

Statement.

Source: Tirole, Ch. 3, Exercise 3.2, p. 144 (PDF 157) · Answer in book: yes, p. 571

Exercise 3.2 (impact of entrepreneurial risk aversion). Consider the fixed-investment model developed in this chapter: an entrepreneur has cash amount A and wants to invest $I > A$ into a project. The project yields $R > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ ($\Delta p > 0$) if she shirks. The entrepreneur obtains private benefit B if she shirks and 0 otherwise. Assume that

$$I > p_H \left(R - \frac{B}{\Delta p} \right).$$

(Suppose that $p_L R + B < I$; so the project is not financed if the entrepreneur shirks.)

(i) In contrast with the risk-neutrality assumption of this chapter, assume that the entrepreneur has utility for consumption c:

$$u(c) = \begin{cases} c & \text{if } c \geqslant c_0, \\ -\infty & \text{otherwise.} \end{cases}$$

(Assume that $A \geqslant c_0$ to ensure that the entrepreneur is not in the "$-\infty$ range" in the absence of financing.)

Compute the minimum equity level $\bar{A}$ for which the project is financed by risk-neutral investors when the market rate of interest is 0. Discuss the difference between $p_H = 1$ and $p_H < 1$.

(ii) Generalize the analysis to risk aversion. Let $u(c)$ denote the entrepreneur's utility from consumption with $u' > 0$, $u'' < 0$. Conduct the analysis assuming either limited liability or the absence of limited liability.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. [...]

What is asked. (i) The entrepreneur is risk neutral above a subsistence level c_0 and infinitely averse to falling below it. Find the smallest cash $\bar{A}$ with which risk-neutral investors finance the project, and explain why the answer differs between a project that succeeds for sure under effort ($p_H = 1$) and one that can fail even under effort ($p_H < 1$). (ii) Redo the analysis for a general strictly concave utility, once with limited liability (her payments cannot be negative) and once without.

Approach. This is the fixed-investment model of the lecture (Topic 4A) with a risk-averse borrower. The contract pays her c^S on success and c^F on failure; she chooses it to maximise expected utility subject to the investors' breakeven constraint (IR_l) and her incentive constraint (IC_b), now written in utility. Pledgeable income is total expected income minus the cheapest expected payment that satisfies (IC_b) and the lower bound on c^F; a floor on consumption (subsistence in (i), limited liability in (ii)) raises that payment and so the minimum equity. In (ii) we write the Lagrangian and derive the first-order conditions, which give the Holmström–Shavell insurance-versus-incentives trade-off.

Notation.

- A: entrepreneur's cash ($A \geq c_0$ in (i)); $I > A$: fixed investment; $R > 0$: income in the case of success, 0 in the case of failure.
- p_H, $p_L = p_H - \Delta p$: success probabilities when she works / shirks ($0 < p_L < p_H \leq 1$); $B > 0$: private benefit from shirking.
- $NPV \equiv p_H R - I > 0$; $\bar{A}_{rn} \equiv I - p_H(R - B/\Delta p) > 0$: the risk-neutral threshold (3.3), positive by the assumption $I > p_H(R - B/\Delta p)$.
- c^S, c^F: the entrepreneur's consumption in the case of success / failure (her total payoff from the contract); $u(\cdot)$: utility of consumption; c_0: subsistence level in (i).
- $\mathcal{P}$: expected pledgeable income; $\bar{A}$: minimum cash for financing.
- $h(c) \equiv u^{-1}(u(c) + B/\Delta p)$: consumption in success that gives exactly the utility premium $B/\Delta p$ over consumption c in failure (part (ii)).

Assumption. The entrepreneur puts all of A into the project and the contract pays her c^S or c^F (keeping cash aside is the same as adding it to both payments). In (ii) the private benefit enters utility additively, $p\,u(c^S) + (1-p)\,u(c^F) + B$ when she shirks, as in the book's answer.

Step 1 — (i), $p_H < 1$: the programme. The entrepreneur offers the contract that maximises her expected utility subject to the investors' breakeven and her incentive constraint:

$$\max_{c^S, c^F} \; p_H\,u(c^S) + (1-p_H)\,u(c^F) \tag{38}$$

$$\text{s.t.} \quad p_H(R - c^S) + (1-p_H)(0 - c^F) \;\geq\; I - A, \tag{39}$$

$$p_H\,u(c^S) + (1-p_H)\,u(c^F) \;\geq\; p_L\,u(c^S) + (1-p_L)\,u(c^F) + B. \tag{40}$$

Failure has probability $1 - p_H > 0$ even under work, so $c^F < c_0$ would make (38) equal to $-\infty$: the optimum has $c^F \geq c_0$, and then $c^S \geq c_0$ too. On $[c_0, \infty)$ utility is linear, $u(c) = c$, so (40) reads

$$\begin{aligned} p_H c^S + (1-p_H)c^F &\geq p_L c^S + (1-p_L)c^F + B \\ (p_H - p_L)c^S - \big[(1-p_L) - (1-p_H)\big]c^F &\geq B \\ (p_H - p_L)c^S - (p_H - p_L)c^F &\geq B \\ (\Delta p)(c^S - c^F) &\geq B. \end{aligned} \tag{41}$$

Step 2 — (i), $p_H < 1$: which constraints bind, and $\bar{A}$. With linear utility the objective (38) is the expected payment $p_H c^S + (1-p_H)c^F$. If (39) were slack, raising c^S would raise the objective and relax (41); so (39) binds and the entrepreneur gets $A + NPV$. Financing is possible if and only if the cheapest contract that satisfies (41) and $c^F \geq c_0$ leaves the investors enough. Using $c^S \geq c^F + B/\Delta p$ from (41) and then $c^F \geq c_0$, any feasible contract costs

$$\begin{aligned} p_H c^S + (1-p_H)c^F &\geq p_H\Big(c^F + \frac{B}{\Delta p}\Big) + (1-p_H)c^F \\ &= c^F + p_H\frac{B}{\Delta p} \;\geq\; c_0 + p_H\frac{B}{\Delta p}, \end{aligned}$$

with equality iff both inequalities bind, so the cheapest contract is

$$c^F = c_0, \tag{42}$$

$$c^S = c_0 + \frac{B}{\Delta p}. \tag{43}$$

The pledgeable income is total expected income minus this minimum expected payment; substituting (42)–(43),

$$\begin{aligned} \mathcal{P} &= p_H R - \Big[p_H\Big(c_0 + \frac{B}{\Delta p}\Big) + (1-p_H)(c_0)\Big] \\ &= p_H\Big(R - \frac{B}{\Delta p}\Big) - c_0. \end{aligned} \tag{44}$$

(the middle step: $p_H c_0 + (1-p_H)c_0 = c_0$). Financing requires $\mathcal{P} \geq I - A$; using (44) and moving terms,

$$\begin{aligned} p_H\Big(R - \frac{B}{\Delta p}\Big) - c_0 &\geq I - A \\ A \;\geq\; \bar{A} &= I + c_0 - p_H\Big(R - \frac{B}{\Delta p}\Big) \\ &= \Big[I - p_H\Big(R - \frac{B}{\Delta p}\Big)\Big] + c_0 = \bar{A}_{rn} + c_0. \end{aligned} \tag{45}$$

Because $\bar{A}_{rn} > 0$, $\bar{A} > c_0$: entrepreneurs with $A \in [c_0, \bar{A})$ are rationed. The participation constraint holds, since she gets $A + NPV > A$. Illustration: with $p_H = 0.8$, $p_L = 0.4$, $B = 1$, $R = 10$, $I = 6.5$, $c_0 = 1$, (45) gives $\bar{A} = (6.5) + (1) - (0.8)(10 - 1/0.4) = 1.5$, against $\bar{A}_{rn} = 0.5$.

Step 3 — (i), $p_H = 1$. Now failure never happens on the equilibrium path; it happens only if she shirks, with probability $1 - p_L > 0$. Set $c^F < c_0$ (for example $c^F = 0$, feasible under limited liability when $c_0 > 0$). Shirking then yields $p_L\, u(c^S) + (1 - p_L)(-\infty) + B = -\infty$, so (40) holds for every $c^S \geq c_0$ and the incentive problem costs nothing. The cheapest contract is $c^S = c_0$, and

$$\mathcal{P} = R - c_0, \tag{46}$$

$$\bar{A} = I + c_0 - R \;<\; c_0, \tag{47}$$

where the inequality uses $R - I = NPV > 0$. Every entrepreneur with $A \geq c_0$ is financed: no rationing (with the numbers above, $\bar{A} = 6.5 + 1 - 10 = -2.5$). The threshold is discontinuous at $p_H = 1$: by (45), $\bar{A} \to I + c_0 - R + B/\Delta p$ as $p_H \to 1$, a jump of $B/\Delta p$. If $c_0 \leq 0$, limited liability rules out a punishment below c_0 and the risk-neutral analysis applies.

Answer (i). For $p_H < 1$: $\bar{A} = I + c_0 - p_H(R - B/\Delta p) = \bar{A}_{rn} + c_0$, so entrepreneurs with $A \in [c_0, \bar{A})$ are rationed. For $p_H = 1$: failure reveals shirking and can be punished with $c^F < c_0$ at no cost on the equilibrium path, so $\bar{A} = I + c_0 - R < c_0$ and every entrepreneur with $A \geq c_0$ is financed (no rationing).

Step 4 — (ii) The general programme. With $u' > 0$, $u'' < 0$, the contract solves

$$\max_{c^S, c^F} \; p_H\, u(c^S) + (1 - p_H)\, u(c^F) \tag{48}$$

$$\text{s.t.} \quad p_H(R - c^S) - (1 - p_H)c^F \;\geq\; I - A, \tag{IR$_l$}$$

$$(\Delta p)\big[u(c^S) - u(c^F)\big] \;\geq\; B, \tag{IC$_b$}$$

$$c^F \geq 0,\; c^S \geq 0 \quad \text{(only under limited liability)}, \tag{LL}$$

$$p_H\, u(c^S) + (1 - p_H)\, u(c^F) \;\geq\; u(A). \tag{PC}$$

(IC$_b$) is the risk-averse version of (IC$_b$): subtract $p_L u(c^S) + (1 - p_L)u(c^F) + B$ from (48). (PC) says she prefers the project to consuming A. In utility units $v^S = u(c^S)$, $v^F = u(c^F)$ the objective and (IC$_b$) are linear and the cost $p_H u^{-1}(v^S) + (1 - p_H)u^{-1}(v^F)$ is convex, so the problem is convex and the first-order conditions are sufficient.

Step 5 — (ii) Which constraints bind. We show both (IR$_l$) and (IC$_b$) bind, the second by deriving the first-order conditions. (a) (IR$_l$) binds: otherwise raise c^S, which raises (48) and relaxes (IC$_b$). (b) With multipliers $\lambda \geq 0$ on (IR$_l$) and $\mu \geq 0$ on (IC$_b$) (and (LL) slack), the Lagrangian is

$$\begin{aligned}\mathcal{L} = {} & p_H\, u(c^S) + (1 - p_H)\, u(c^F) + \lambda\big[p_H(R - c^S) - (1 - p_H)c^F - I + A\big] \\ & + \mu\big[(\Delta p)(u(c^S) - u(c^F)) - B\big].\end{aligned}$$

Differentiating with respect to c^S and c^F and setting the derivatives to zero,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c^S} &= p_H u'(c^S) - \lambda p_H + \mu(\Delta p)u'(c^S) = 0, \\ &\Longrightarrow \quad u'(c^S)(p_H + \mu\Delta p) = \lambda p_H, \\ \frac{\partial \mathcal{L}}{\partial c^F} &= (1 - p_H)u'(c^F) - \lambda(1 - p_H) - \mu(\Delta p)u'(c^F) = 0, \\ &\Longrightarrow \quad u'(c^F)(1 - p_H - \mu\Delta p) = \lambda(1 - p_H).\end{aligned}$$

Here $\lambda > 0$, because $\lambda = 0$ would make the first condition read $u'(c^S)(p_H + \mu\Delta p) = 0$, impossible with $u' > 0$. Dividing the first by $\lambda p_H\, u'(c^S)$ and the second by $\lambda(1 - p_H)\, u'(c^F)$,

$$\frac{1}{u'(c^S)} = \frac{1}{\lambda}\Big(1 + \mu\frac{\Delta p}{p_H}\Big), \tag{49}$$

$$\frac{1}{u'(c^F)} = \frac{1}{\lambda}\Big(1 - \mu\frac{\Delta p}{1 - p_H}\Big). \tag{50}$$

If $\mu = 0$, (49)–(50) give $u'(c^S) = u'(c^F)$, i.e. full insurance $c^S = c^F$, which violates (IC$_b$) because $B > 0$. So $\mu > 0$ and, by (49)–(50), $c^S > c^F$: the entrepreneur bears risk, but only as much as incentives require (Holmström–Shavell inverse-marginal-utility rule).

Step 6 — (ii) The solution. We reduce the two binding constraints to one equation in c^F. The binding (IC$_b$) is $u(c^S) = u(c^F) + B/\Delta p$, i.e. $c^S = h(c^F)$. The binding (IR$_l$) is $p_H R - p_H c^S - (1 - p_H)c^F = I - A$, i.e.

$$p_H c^S + (1 - p_H)c^F = A + (p_H R - I) = A + NPV.$$

Substituting $c^S = h(c^F)$,

$$p_H\, h(c^F) + (1 - p_H)\, c^F = A + NPV. \tag{51}$$

Differentiating the identity $u(h(c)) = u(c) + B/\Delta p$ with respect to c,

$$u'(h(c))\, h'(c) = u'(c) \quad \Longrightarrow \quad h'(c) = \frac{u'(c)}{u'(h(c))} \;\geq\; 1,$$

because $h(c) > c$ and u' is decreasing. Hence the left-hand side of (51) is increasing in c^F, so (51) defines a unique $c^F = c(A)$, increasing in A. Substituting $u(c^S) = u(c^F) + B/\Delta p$ into (48), her utility is

$$\begin{aligned} V(A) &= p_H\Big(u(c(A)) + \frac{B}{\Delta p}\Big) + (1 - p_H)\, u(c(A)) \\ &= u(c(A)) + \frac{p_H B}{\Delta p}. \end{aligned} \tag{52}$$

Step 7 — (ii) Without limited liability. Only (PC) can fail. Let CE be the certainty equivalent of the lottery (c^S, c^F), $u(CE) = V(A)$, and $\pi \equiv \mathbb{E}[c] - CE$ its risk premium; by (51), $\mathbb{E}[c] = A + NPV$. Then

$$\text{financing} \iff V(A) \geq u(A) \iff CE \geq A \iff \pi(A) \leq NPV. \tag{53}$$

A risk-neutral entrepreneur ($\pi = 0$) without limited liability is always financed (she can be punished with $c^F < 0$; there is no $\bar{A}$). A risk-averse one is financed only if the risk premium of the incentive lottery does not exceed the NPV: risk aversion alone can block positive-NPV projects. Whether this gives a threshold in A depends on how π varies with wealth (with B in utility units, a richer entrepreneur needs a larger money spread to produce the utility gap $B/\Delta p$).

Step 8 — (ii) With limited liability. (LL) requires $c(A) \geq 0$. Because the left-hand side of (51) is increasing, $c(A) \geq 0$ iff $A + NPV \geq p_H h(0)$, i.e.

$$A \;\geq\; A_{LL} \equiv I - p_H\big[R - h(0)\big]. \tag{54}$$

Below A_{LL} no contract satisfies (IR$_l$), (IC$_b$) and (LL): the cheapest incentive-compatible contract, $c^F = 0$, $c^S = h(0)$, already costs more than $A + NPV$. Above A_{LL} the solution is that of Step 6 with $c^F = c(A) \geq 0$, so the minimum equity is $\bar{A} = \min\{A \geq A_{LL} :\; V(A) \geq u(A)\}$. Two remarks. First, for $A > A_{LL}$ the entrepreneur receives $c^F > 0$ in failure: unlike the risk-neutral case, pure debt ($c^F = 0$) is no longer optimal, because she buys partial insurance. Second, risk neutrality gives back the book: with $u(c) = c$, $h(0) = B/\Delta p$ and (54) is (45) with $c_0 = 0$; part (i) is the case of a floor c_0 instead of 0. If u is normalised so that $u'(0) = 1$, concavity gives $h(0) \geq B/\Delta p$, so $A_{LL} \geq \bar{A}_{rn}$: risk aversion raises the equity requirement.

Answer (ii). Both (IR$_l$) and (IC$_b$) bind; the contract solves (51) with $u(c^S) = u(c^F) + B/\Delta p$, and the first-order conditions (49)–(50) give $c^S > c^F$ (partial insurance). Without limited liability, financing obtains iff the risk premium of the incentive lottery is at most the NPV, $\pi(A) \leq NPV$.

With limited liability, also $A \geq A_{LL} = I - p_H[R - h(0)]$, and $\bar{A} = \min\{A \geq A_{LL} : V(A) \geq u(A)\}$; above A_{LL} she gets $c^F > 0$, so pure debt is no longer optimal.

Intuition. Incentives need a spread between the success and failure payoffs. A risk-neutral entrepreneur bears that spread for free; a subsistence floor or limited liability bounds the punishment, so the spread must be paid as a rent in success, and a risk-averse entrepreneur must in addition be compensated for the risk. Both raise the stake she keeps and lower pledgeable income. The $p_H = 1$ case is the exception: failure then proves shirking, so an unbounded punishment off the equilibrium path removes the agency cost.

Common mistakes.

- Subtracting c_0 from the cash instead of adding it to the payments: the entrepreneur must be paid at least c_0 in *both* states, which costs investors c_0 in expectation; the two bookkeepings agree only if this is done consistently.
- Treating $p_H = 1$ like $p_H < 1$. With $p_H = 1$ failure never occurs under effort, so the $-\infty$ punishment is free; with $p_H < 1$ it would hit her on the equilibrium path.
- In (ii), carrying over the risk-neutral result that pure debt ($c^F = 0$) is optimal. A risk-averse borrower buys partial insurance, $c^F > 0$, whenever $A > A_{LL}$.

Book answer. Matches the book's answer (p. 571): $\bar{A} = I + c_0 - p_H(R - B/\Delta p)$ for $p_H < 1$, the programme of (ii) with binding incentive constraint. For $p_H = 1$ the book writes the pledgeable income as $p_H R$, i.e. with c_0 kept aside from A; this is (46)–(47) in other bookkeeping. The book only states the programme for (ii) and cites Holmström (1979) and Shavell (1979); Steps 6–8 are this solution's.

Ref: Tirole §3.2.1–3.2.2, pp. 115–117; answer p. 571.

10. Maximal incentives principle in the fixed-investment model — Tirole Exercise 3.18, p. 151

Statement.

Source: Tirole, Ch. 3, Exercise 3.18, p. 151 (PDF 164) · Answer in book: yes, p. 579

Exercise 3.18 (maximal incentives principle in the fixed-investment model). Pursue the analysis of Section 3.4.3, but for the fixed-investment model of Section 3.2: the investment cost I is given and the income is either R^S or R^F (instead of R or 0), where $R^S > R^F > 0$. We assume that

$$R^F < I - A,$$

so the project cannot be straightforwardly financed by bringing in net worth A and pledging the lower income R^F to lenders. Let

$$R \equiv R^S - R^F$$

denote the increase in income from the low to the high level. Show that the debt contract is optimal, but unlike in the variable-investment case it may not be uniquely optimal.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The

entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\mathrm{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\mathrm{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I-A}{p_H} \geqslant R - \frac{I-\bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. [...]

From Tirole §3.4.3 "Collateral Values: Outside Debt and the Maximal Incentives Principle", pp. 128–129 (PDF 141–142):

We now return to the indeterminacy of the financial structure (debt or equity) discussed earlier. It turns out that this indeterminacy was an artefact of the absence of profit in the case of failure.

Thus, assume that, for investment size I, the profit is $R^S I$ in the case of success and $R^F I$ in the case of failure, where R^F is now positive. $R^F I$ can be thought of as the salvage value of assets and

$$RI \equiv (R^S - R^F)I$$

as the increase in profit brought about by success. [...]

The model is otherwise the same as in the rest of Section 3.4: the private benefit (BI in the case of misbehavior, 0 otherwise) is also proportional to investment.

The generalization of the condition that the NPV per unit of investment is positive while the pledgeable income per unit of investment is negative ($p_H R > 1 > p_H(R - B/\Delta p)$) is

$$p_H R + R^F > 1 > p_H \left(R - \frac{B}{\Delta p} \right) + R^F.$$

A contract specifies an investment level I and a sharing rule, or equivalently a reward for the entrepreneur for each performance level: $\{R_b^S, R_b^F\}$, with $R_b^S, R_b^F \geqslant 0$ due to limited liability.

The optimal contract maximizes the entrepreneur's expected compensation,

$$U_b = \max_{\{R_b^S, R_b^F, I\}} \{p_H R_b^S + (1 - p_H) R_b^F - A\},$$

subject to two constraints (that will turn out to be binding at the optimum): the entrepreneur's incentive constraint,

$$(\Delta p)(R_b^S - R_b^F) \geqslant BI,$$

and the investors' breakeven constraint,

$$p_H(R^S I - R_b^S) + (1 - p_H)(R^F I - R_b^F) \geqslant I - A.$$

To show that the investors' breakeven constraint is binding, note that, if it were not, then the entrepreneur could increase R_b^S and R_b^F by an equal and small amount without affecting the incentive compatibility constraint. This uniform increase in compensation would raise the entrepreneur's payoff. As is now familiar, we conclude that the investors receive no surplus, and so (by substituting the breakeven constraint into the objective function) the entrepreneur's utility is equal to the NPV:

$$U_b = (p_H R + R^F - 1)I.$$

Because the NPV per unit of investment is positive, the entrepreneur therefore chooses the highest possible investment.

Next, note that the incentive constraint is binding (otherwise, the optimal investment would be infinite, which would violate the two constraints combined).

Lastly, suppose that $R_b^F > 0$ at the optimum. And consider a small increase $\delta R_b^S > 0$ in managerial compensation in the case of success together with a small decrease $\delta R_b^F < 0$ in the case of failure that keeps the investors' profitability constant:

$$p_H \delta R_b^S + (1 - p_H)\delta R_b^F = 0.$$

This small change (which is feasible only if $R_b^F > 0$) also keeps the objective function constant. But the incentive constraint is now slack, a contradiction. We thus conclude that at the optimum

$$R_b^F = 0.$$

Hence, an all-equity firm cannot be optimal: in the absence of debt, the entrepreneur would receive $R^F I$ times her share of stocks in the firm, and therefore would be rewarded even in the case of failure. By contrast, investors' holding debt $D \geqslant R^F I$ is an optimal financial structure. Using the fact that the two constraints are binding, the borrowing capacity is given by

$$R^F I + p_H \left(R - \frac{B}{\Delta p} \right) I = I - A$$

or

$$I = \frac{A}{1 - [p_H(R - B/\Delta p) + R^F]}. \qquad (3.16)$$

What is asked. In the fixed-investment model, failure now yields a positive income R^F (salvage value) instead of 0. Show that a debt contract (investors take everything in failure and a fixed amount in success) is an optimal financial structure, and show that, unlike the variable-investment model of Section 3.4.3, other contracts may be optimal too.

Approach. This is the lecture's "debt dominates equity" result (maximal incentives principle, Topic 4A). Set up the borrower's programme: maximise her expected reward subject to (IC_b), (IR_l) and limited liability. With a fixed investment her utility equals the NPV for *every* contract on which (IR_l) binds, so the contract matters only for feasibility, i.e. for *pledgeable income*. Paying her in failure buys no incentive, so the contract that maximises pledgeable income pays her nothing in failure: debt. Uniqueness fails because, when $A > \bar{A}$, the spare pledgeable income can be paid in either state.

Notation.

- I: fixed investment; A: entrepreneur's cash; $R^S > R^F > 0$: income in success / failure, with $R^F < I - A$; $R \equiv R^S - R^F$.
- p_H, $p_L = p_H - \Delta p$, B: as in Section 3.2.
- R_b^S, $R_b^F \geq 0$: entrepreneur's reward in success / failure (limited liability).
- $NPV \equiv p_H R^S + (1 - p_H)R^F - I = p_H R + R^F - I > 0$; U_b: entrepreneur's net utility.
- D: face value of debt; $\bar{A}$: minimum cash for financing.

Step 1 — Programme. The borrower designs the contract (R_b^S, R_b^F); her incentive constraint is $p_H R_b^S + (1 - p_H)R_b^F \geq p_L R_b^S + (1 - p_L)R_b^F + B$, which simplifies (collect R_b^S and R_b^F terms, $(1 - p_L) - (1 - p_H) = \Delta p$) to ($\text{IC}_b$):

$$\begin{aligned} U_b = \max_{R_b^S, R_b^F} \; & p_H R_b^S + (1 - p_H)R_b^F - A && (55) \\ \text{s.t.} \; & (\Delta p)(R_b^S - R_b^F) \;\geq\; B, && (\text{IC}_b) \\ & p_H(R^S - R_b^S) + (1 - p_H)(R^F - R_b^F) \;\geq\; I - A, && (\text{IR}_l) \\ & R_b^S \geq 0, \; R_b^F \geq 0. && (\text{LL}) \end{aligned}$$

Step 2 — The breakeven constraint binds; U_b is the NPV. If (IR_l) were slack, raising R_b^S would raise (55) and relax (IC_b). With (IR_l) binding, solve it for her expected reward,

$$\begin{aligned} p_H R^S - p_H R_b^S + (1-p_H)R^F - (1-p_H)R_b^F &= I - A \\ p_H R_b^S + (1-p_H)R_b^F &= p_H R^S + (1-p_H)R^F - (I-A), \end{aligned}$$

and substitute it into (55):

$$\begin{aligned} U_b &= [p_H R^S + (1-p_H)R^F - (I-A)] - A \\ &= p_H R^S + (1-p_H)R^F - I \\ &= NPV. \end{aligned} \tag{56}$$

Because I is fixed, (56) is the same for *every* feasible contract with binding (IR_l). The financial structure matters only for feasibility: it must deliver (IC_b) while leaving investors $I - A$.

Step 3 — Maximal pledgeable income and the financing condition. The constraint set is nonempty iff the largest expected income that can go to investors subject to (IC_b) and (LL) covers $I - A$, i.e. iff expected income minus the *smallest* expected reward compatible with (IC_b) and (LL) is at least $I - A$. Using $R_b^S \geq R_b^F + B/\Delta p$ from (IC_b),

$$\begin{aligned} p_H R_b^S + (1-p_H)R_b^F &\geq p_H\Big(R_b^F + \frac{B}{\Delta p}\Big) + (1-p_H)R_b^F \\ &= R_b^F + p_H\frac{B}{\Delta p} \;\geq\; p_H\frac{B}{\Delta p}, \end{aligned}$$

the last step by $R_b^F \geq 0$; both bounds are attained by $R_b^F = 0$, $R_b^S = B/\Delta p$. Each unit of failure reward costs one unit of expected income and buys no incentive. So

$$\begin{aligned} \mathcal{P} &= p_H R^S + (1-p_H)R^F - p_H\frac{B}{\Delta p} \\ &= p_H(R^S - R^F) + R^F - p_H\frac{B}{\Delta p} \\ &= R^F + p_H\Big(R - \frac{B}{\Delta p}\Big), \end{aligned} \tag{57}$$

and financing obtains iff $\mathcal{P} \geq I - A$, i.e.

$$A \;\geq\; \bar{A} \equiv I - R^F - p_H\Big(R - \frac{B}{\Delta p}\Big) = p_H\frac{B}{\Delta p} - NPV, \tag{58}$$

where the second form uses $NPV = p_H R + R^F - I$. The safe income R^F works exactly like cash on hand.

Step 4 — A debt contract is optimal. Let investors hold debt of face value D defined by

$$p_H D + (1-p_H)R^F = I - A. \tag{59}$$

In failure the firm defaults and investors take R^F; in success they get D. By $R^F < I - A$, (59) gives $D > R^F$, so the borrower gets $R_b^F = 0$. Substituting (59) into (58) (written as $p_H(R^S - B/\Delta p) + (1-p_H)R^F \geq I - A$),

$$\begin{aligned} p_H\Big(R^S - \frac{B}{\Delta p}\Big) + (1-p_H)R^F &\geq p_H D + (1-p_H)R^F \\ p_H\Big(R^S - \frac{B}{\Delta p}\Big) &\geq p_H D \\ R^S - \frac{B}{\Delta p} &\geq D \\ R_b^S = R^S - D &\geq \frac{B}{\Delta p}, \end{aligned} \tag{60}$$

(subtracting $(1-p_H)R^F$, then dividing by $p_H > 0$, then rearranging) so (IC_b) holds, (IR_l) binds by construction, and by (56) debt attains the maximum. It is optimal whenever financing is feasible.

Step 5 — Debt is not uniquely optimal. The optimal set is every $(R_b^S, R_b^F) \geq 0$ with binding (IR_l) and (IC_b). From the binding (IR_l), $R_b^S = [A + NPV - (1-p_H)R_b^F]/p_H$. Substituting this into (IC_b),

$$\begin{aligned}
(\Delta p)\Big(\frac{A + NPV - (1-p_H)R_b^F}{p_H} - R_b^F\Big) &\geq B \\
(\Delta p)\,\frac{A + NPV - (1-p_H)R_b^F - p_H R_b^F}{p_H} &\geq B \\
\frac{A + NPV - R_b^F}{p_H} &\geq \frac{B}{\Delta p} \\
A + NPV - R_b^F &\geq p_H \frac{B}{\Delta p} \\
R_b^F &\leq A + NPV - p_H\frac{B}{\Delta p} = A - \bar{A}, \qquad (61)
\end{aligned}$$

using (58) in the last line. So every $R_b^F \in [0, A-\bar{A}]$ is optimal (add $R_b^F \leq R^F$ if investors' claim must be nonnegative in each state). The contract is unique only at $A = \bar{A}$. For $A > \bar{A}$, a small failure reward (debt plus a little equity, or even all-equity when its implied R_b^F satisfies (61)) does as well. Illustration: $R^S = 12$, $R^F = 2$, $p_H = 0.8$, $p_L = 0.4$, $B = 1.6$, $I = 9$, $A = 3$. Then $B/\Delta p = 1.6/0.4 = 4$, $NPV = 0.8(12) + 0.2(2) - 9 = 9.6 + 0.4 - 9 = 1$, $\bar{A} = 0.8(4) - 1 = 2.2$, and by (59) $D = (9 - 3 - 0.2(2))/0.8 = 5.6/0.8 = 7$ (with $2 < 7 \leq 12 - 4 = 8$). The optimal set is $R_b^F \in [0,\, 3 - 2.2] = [0, 0.8]$; at $R_b^F = 0.8$, $R_b^S = (3 + 1 - 0.2(0.8))/0.8 = 3.84/0.8 = 4.8$ and (IC_b) binds: $0.4(4.8 - 0.8) = 1.6 = B$.

Step 6 — Contrast with variable investment. In Section 3.4.3 the entrepreneur's utility $(p_H R + R^F - 1)I$ rises with I, so she expands I until the financing constraint binds, the analogue of $A = \bar{A}$. Then the slack in (61) is zero, $R_b^F = 0$ is forced and debt is the unique optimum; any $R_b^F > 0$ would waste incentive power and reduce the borrowing capacity (3.16).

Answer (3.18). Debt with face value D given by $p_H D + (1-p_H)R^F = I - A$ (so $R^F < D \leq R^S - B/\Delta p$) is optimal whenever the project can be financed, $A \geq \bar{A} = I - R^F - p_H(R - B/\Delta p)$. It is not uniquely optimal: every contract with $R_b^F \in [0, A-\bar{A}]$ and binding (IR_l) gives the same utility NPV; the optimum is unique only at $A = \bar{A}$.

Intuition. Paying the entrepreneur in failure costs expected income but weakens incentives, so it is the worst way to reward her; debt, which gives her nothing in failure, maximises incentives per unit of pay. With a fixed project, once the project is funded there is no further use for pledgeable income, so any leftover slack can be paid out in either state.

Common mistakes.

- Concluding that all-equity is never optimal, as in Section 3.4.3. That needs the investment to expand until the financing constraint binds; with a fixed I and $A > \bar{A}$ it does not bind.
- Leaving R^F out of pledgeable income. The salvage value is not subject to moral hazard and works exactly like cash on hand: $\bar{A}$ falls one for one with R^F.
- Writing (IC_b) as $(\Delta p)R_b^S \geq B$, forgetting that a failure reward R_b^F weakens incentives: what matters is the spread $R_b^S - R_b^F$.

Book answer. Matches the book's answer (p. 579), including condition (1) there, $p_H(R - B/\Delta p) \geq I - (A + R^F)$, the debt implementation with $R^F < D < R^S$ and the non-uniqueness when (IR_l) holds with slack at $R_b^F = 0$; (61) makes the size of the optimal set explicit.

Ref: Tirole §3.2.2, pp. 116–117; §3.4.3, pp. 128–129; answer p. 579.

11. Retained-earnings benefit — Tirole Exercise 3.11, p. 148

Statement.

Source: Tirole, Ch. 3, Exercise 3.11, p. 148 (PDF 161) · Answer in book: yes, p. 575

Exercise 3.11 (retained-earnings benefit). An entrepreneur has at date 1 a project of fixed size with characteristics $\{I^1, R^1, p_H^1, p_L^1, B^1\}$ (see Section 3.2). This entrepreneur will at date 2 have a different fixed size project with characteristics $\{I^2, R^2, p_H^2, p_L^2, B^2\}$, which will then require new financing. So, we are considering only a short-term loan for the first project. Retained earnings from the first project can, however, be used to defray part of the investment cost of the second project. Assume that all the characteristics of the second project are known at date 1 except B^2, which is distributed on $[\underline{B}^2, \bar{B}^2]$ according to the cumulative distribution $F(B^2)$. Assume for simplicity that $\underline{B}^2 > \Delta p^2(p_H^2 R^2 - I^2)/p_H^2$. The characteristics of the second project become common knowledge at the beginning of date 2.

(i) Compute the shadow value of retained earnings. (Hint: what is the entrepreneur's gross utility in period 2?)

(ii) Show that it is possible that the first project is funded even though it would not be funded if the second project did not exist and even though the entrepreneur cannot pledge at date 1 income resulting from the second project.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]

Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]

Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.

[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

So, the borrower's utility jumps up at $A = \bar{A}$. [...]

From Tirole §3.4.2, p. 128 (PDF 141) — definition of the shadow value of equity:

Another important concept (which will be used, for example, in computing the value of retained earnings in a dynamic context) is the *shadow value v of equity (here cash).* The entrepreneur derives gross utility $v > 1$ from one more unit of equity. Letting $U_b^g \equiv A + U_b$ denote the borrower's gross utility, [...]

What is asked. (i) Find how much one extra unit of retained earnings (the entrepreneur's date-1 profit kept in the firm) is worth to her at date 2, given that it helps finance a second project whose private benefit B^2 is still random. (ii) Show, by argument or example, that the existence of the second project can get the first project financed even though no date-2 income can be pledged at date 1.

Approach. Two applications of the fixed-investment model (lecture Topic 4A, investment and internal funds). At date 2, retained earnings play the role of cash A in (IR_l): more of them lets the second project be financed for higher values of B^2. Their *shadow value* is the derivative of the entrepreneur's gross date-2 utility. At date 1, that continuation value is lost when project 1 fails, so it works as an extra reward for effort: the date-1 incentive constraint needs less cash reward, and date-1 *pledgeable income* rises.

Notation.

- Project $t \in \{1,2\}$: investment I^t, income R^t in success, probabilities p_H^t, $p_L^t = p_H^t - \Delta p^t$, private benefit B^t; $NPV^2 \equiv p_H^2 R^2 - I^2 > 0$.
- A^1: entrepreneur's cash at date 1; R_b^1: her date-1 reward in success (her retained earnings), 0 in failure.
- $B^2 \in [\underline{B}^2, \bar{B}^2]$: random, c.d.f. F, density f, revealed at the start of date 2.
- $\hat{B}^2(R)$: largest B^2 for which project 2 is financed with retained earnings R.
- $W(R)$: entrepreneur's expected gross date-2 utility with retained earnings R; $\mu(R) \equiv W'(R)$: shadow value of retained earnings (v in §3.4.2).

Assumption. No discounting; the entrepreneur's only date-2 wealth is R_b^1 (or 0), and date-2 lenders cannot be pledged date-1 income.

Step 1 — (i) Date-2 financing given retained earnings R. Project 2 is financed iff its pledgeable income covers the outside funding, (IR_l) for project 2:

$$p_H^2\Big(R^2 - \frac{B^2}{\Delta p^2}\Big) \geq I^2 - R. \tag{62}$$

Solving (62) for B^2 (expand, move $p_H^2 B^2/\Delta p^2$ to the right and $I^2 - R$ to the left, then multiply by $\Delta p^2/p_H^2 > 0$),

$$\begin{aligned} p_H^2 R^2 - \frac{p_H^2 B^2}{\Delta p^2} &\geq I^2 - R \\ \frac{p_H^2 B^2}{\Delta p^2} &\leq p_H^2 R^2 - I^2 + R = NPV^2 + R \\ B^2 \;\leq\; \hat{B}^2(R) &\equiv \frac{\Delta p^2}{p_H^2}(NPV^2 + R). \end{aligned} \tag{63}$$

By the assumption $\underline{B}^2 > \Delta p^2\, NPV^2/p_H^2$, (63) gives $\hat{B}^2(0) < \underline{B}^2$: with no retained earnings, project 2 is never financed.

Step 2 — (i) Gross date-2 utility. We follow the hint and compute her expected gross utility. If financed, project 2's lenders break even and the entrepreneur gets her cash plus NPV^2 (equation (3.6)); otherwise she keeps R. With probability $F(\hat{B}^2(R))$ it is financed, so

$$\begin{aligned} W(R) &= F\big(\hat{B}^2(R)\big)\big(R + NPV^2\big) + \Big[1 - F\big(\hat{B}^2(R)\big)\Big]R \\ &= R + F\big(\hat{B}^2(R)\big)\, NPV^2. \end{aligned} \tag{64}$$

Step 3 — (i) Shadow value. Differentiating (64) and using $d\hat{B}^2/dR = \Delta p^2/p_H^2$ from (63),

$$\begin{aligned} \mu(R) = W'(R) &= 1 + f\big(\hat{B}^2(R)\big)\,\frac{d\hat{B}^2}{dR}\, NPV^2 \\ &= 1 + f\big(\hat{B}^2(R)\big)\,\frac{\Delta p^2}{p_H^2}\, NPV^2 \;\geq 1. \end{aligned} \tag{65}$$

One unit of retained earnings is worth one unit of consumption plus the expected NPV it unlocks: it raises the probability of financing at the margin by $f\,\Delta p^2/p_H^2$.

Answer (i). $\mu(R) = 1 + f(\hat{B}^2(R))\,(\Delta p^2/p_H^2)\,(p_H^2 R^2 - I^2) \geq 1$, with $\hat{B}^2(R) = (\Delta p^2/p_H^2)(p_H^2 R^2 - I^2 + R)$ and gross utility $W(R) = R + F(\hat{B}^2(R))(p_H^2 R^2 - I^2)$.

Step 4 — (ii) Date-1 incentive constraint. By (64) and Step 1, $W(0) = 0 + F(\hat{B}^2(0))\,NPV^2 = 0$, so the entrepreneur works at date 1 iff

$$\begin{aligned} p_H^1 W(R_b^1) + (1 - p_H^1) W(0) &\geq p_L^1 W(R_b^1) + (1 - p_L^1) W(0) + B^1 \\ (p_H^1 - p_L^1) W(R_b^1) - (p_H^1 - p_L^1) W(0) &\geq B^1 \\ (\Delta p^1)\, W(R_b^1) &\geq B^1 \qquad (W(0) = 0) \\ (\Delta p^1)[R_b^1 + F(\hat{B}^2(R_b^1))\, NPV^2] &\geq B^1, \end{aligned} \tag{IC^1}$$

the last line substituting (64). Let $\tilde{R}_b^1$ be the smallest R_b^1 satisfying (IC^1). Whenever $F(\hat{B}^2(\tilde{R}_b^1)) > 0$,

$$\tilde{R}_b^1 = \frac{B^1}{\Delta p^1} - F(\hat{B}^2(\tilde{R}_b^1))\, NPV^2 < \frac{B^1}{\Delta p^1}. \tag{66}$$

Step 5 — (ii) Date-1 financing. The entrepreneur maximises $p_H^1 W(R_b^1) - A^1$ subject to (IC^1) and $p_H^1(R^1 - R_b^1) \geq I^1 - A^1$. Since W is increasing (65), the breakeven constraint binds, and financing is possible iff

$$p_H^1(R^1 - \tilde{R}_b^1) \;\geq\; I^1 - A^1. \tag{67}$$

By (66) the left-hand side exceeds the stand-alone pledgeable income $p_H^1(R^1 - B^1/\Delta p^1)$. So for A^1 with $p_H^1(R^1 - B^1/\Delta p^1) < I^1 - A^1 \leq p_H^1(R^1 - \tilde{R}_b^1)$, project 1 is funded only because project 2 exists, although none of project 2's income is pledged at date 1.

Step 6 — (ii) An example. Project 2: $I^2 = 20$, $R^2 = 40$, $p_H^2 = 0.75$, $p_L^2 = 0.25$, $B^2 \sim U[7, 9]$, so $NPV^2 = 10$ and $\Delta p^2 NPV^2/p_H^2 = 20/3 < 7$ (assumption holds; also $p_L^2 R^2 + B^2 < I^2$). By (63), $\hat{B}^2(R) = \frac{2}{3}(10 + R)$, which lies in $[7, 9]$ for $R \in [0.5, 3.5]$; there $F = (\hat{B}^2 - 7)/2$ and (64)–(65) give

$$\begin{aligned} W(R) &= R + \frac{\frac{2}{3}(10 + R) - 7}{2}\,(10) \\ &= R + \frac{(20 + 2R - 21)/3}{2}\,(10) = R + \frac{5(2R - 1)}{3} \\ &= \frac{3R + 10R - 5}{3} = \frac{13R - 5}{3}, \\ \mu &= 1 + \Big(\frac{1}{2}\Big)\Big(\frac{0.5}{0.75}\Big)(10) = 1 + \frac{10}{3} = \frac{13}{3}. \end{aligned}$$

Project 1: $I^1 = 4.5$, $R^1 = 10$, $p_H^1 = 0.5$, $p_L^1 = 0.25$, $B^1 = 0.5$, so $B^1/\Delta p^1 = 2$. Alone, the pledgeable income is $(0.5)(10 - 2) = 4$ and $\bar{A}^1 = 4.5 - 4 = 0.5$. With project 2, (IC^1) at equality reads $W(\tilde{R}_b^1) = B^1/\Delta p^1$, i.e. $(13\tilde{R}_b^1 - 5)/3 = 2$, so $13\tilde{R}_b^1 = 6 + 5 = 11$, $\tilde{R}_b^1 = 11/13 \approx 0.846154$ (inside $[0.5, 3.5]$, where the formula for W holds), and

$$p_H^1(R^1 - \tilde{R}_b^1) = (0.5)\Big(10 - \frac{11}{13}\Big) = \frac{119}{26} \approx 4.576923 > 4.5.$$

The threshold falls from 0.5 to $4.5 - 119/26 = -1/13 \approx -0.076923$: any $A^1 \in [0, 0.5)$ is now funded (at $A^1 = 0$, $R_b^1 = 10 - 4.5/0.5 = 1 \geq \tilde{R}_b^1$).

Answer (ii). Yes. Because project 2's rent is lost on failure, the date-1 reward needed for effort falls to $\tilde{R}_b^1 < B^1/\Delta p^1$ and date-1 pledgeable income rises to $p_H^1(R^1 - \tilde{R}_b^1) > p_H^1(R^1 - B^1/\Delta p^1)$. Any A^1 with $p_H^1(R^1 - B^1/\Delta p^1) < I^1 - A^1 \leq p_H^1(R^1 - \tilde{R}_b^1)$ is financed only thanks to project 2 (in the example, $A^1 \in [0, 0.5)$).

Intuition. Retained earnings are worth more than their face value because they are the net worth that gets the next project financed. That future rent is lost if project 1 fails, so it acts as a reward

for date-1 effort: part of the incentive is paid with continuation value instead of cash, and the cash saved becomes pledgeable to date-1 lenders.

Common mistakes.

- Valuing retained earnings at face ($\mu = 1$). Each unit also raises the chance that project 2 is financed, worth $f\left(\Delta p^2/p_H^2\right) NPV^2$ more.
- Writing the date-1 incentive constraint with R_b^1 alone, $(\Delta p^1)R_b^1 \geq B^1$: the reward for success is $W(R_b^1)$, which includes the continuation rent.
- Treating project 2's income as pledgeable at date 1. The effect works only through the entrepreneur's incentives; date-1 lenders are repaid out of R^1.

Book answer. Matches the book's answer (p. 575): $\mu = 1 + f(\hat{B}^2)(d\hat{B}^2/dR_b^1)NPV^2$ and the date-1 constraint $(\Delta p^1)[R_b^1 + F(\hat{B}^2(R_b^1))NPV^2] \geq B^1$. The book writes the constraint without $W(0)$; this is right because the assumption on $\underline{B}^2$ makes $W(0) = 0$ (Step 1). The numerical example is this solution's.

Ref: Tirole §3.2.2, pp. 116–117; §3.4.2, p. 128 (shadow value of equity); answer p. 575.

12. Renegotiation and debt forgiveness — Tirole Exercise 3.6, p. 146

Statement.

Source: Tirole, Ch. 3, Exercise 3.6, p. 146 (PDF 159) · Answer in book: yes, p. 573

Exercise 3.6 (renegotiation and debt forgiveness). When computing the multiplier k (given by equation (3.12)), we have assumed that it is optimal to specify a stake for the borrower large enough that the incentive constraint (IC_b) is satisfied. Because condition (3.8) implies that the project has negative NPV in the case of misbehavior, such a specification is clearly optimal when the contract cannot be renegotiated. The purpose of this exercise is to check in a rather mechanical way that the borrower cannot gain by offering a loan agreement in which (IC_b) is not satisfied, and which is potentially renegotiated before the borrower chooses her effort. While there is a more direct way to prove this result, some insights are gleaned from this pedestrian approach. Indeed, the exercise provides conditions under which the lender is willing to forgive debt in order to boost incentives (the analysis will bear some resemblance to that of liquidity shocks in Chapter 5, except that the lender's concession takes the form of debt forgiveness rather than cash infusion).[60]

(i) Consider a loan agreement specifying investment I and stake $R_b < BI/\Delta p$ for the borrower. Suppose that the loan agreement can be renegotiated after it is signed and the investment is sunk and before the borrower chooses her effort. Renegotiation takes place if and only if it is mutually advantageous. Show that the loan agreement is renegotiated if and only if

$$(\Delta p)RI - \frac{p_H BI}{\Delta p} + p_L R_b \geqslant 0.$$

(ii) Interpret the previous condition. In particular, show that it can be obtained directly from the general theory. Hint: consider a fictitious, "fixed-investment" project with income $(\Delta p)RI$, investment 0, and cash on hand $p_L R_b$.

(iii) Assume for instance that the entrepreneur makes a take-it-or-leave-it offer in the renegotiation (that is, the entrepreneur has the bargaining power). Compute the borrowing capacity when $R_b < BI/\Delta p$ and the loan agreement is renegotiated.

(iv) Use a direct, rational expectations argument to point out in a different way that there is no loss of generality in assuming $R_b \geqslant BI/\Delta p$ (and therefore no renegotiation).

[60] The phenomenon of debt renegotiation has been analyzed in a number of settings: see, for example, Bulow and Rogoff (1989a,b), Eaton and Gersovitz (1981), Fernandez and Rosenthal (1990), Gale and Hellwig (1989), Gromb (1994), Hart and Moore (1989, 1995), and Snyder (1994).

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", p. 115 (PDF 128) (assumptions kept in the variable-investment model):*

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which

is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130) (the "general theory" of the fixed-investment model used in part (ii); here I is the fixed investment, A the borrower's cash, R the income in the case of success and B the private benefit):

[...] Because she has stake R_b in the firm's income (she receives R_b in the case of success and 0 in the case of failure), the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

where "IR_l" stands for the lenders' individual rationality constraint (which we will also often call the "breakeven constraint" or the "participation constraint"). Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

From Tirole §3.4.1 "The Continuous-Investment Model", p. 127 (PDF 140):

The continuous-investment model of this section is the polar opposite of the fixed-investment one. The fixed-investment model depicts a situation in which returns are sharply decreasing beyond some investment level. In contrast, we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. [...]

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

As in Section 3.2, we assume that investment has positive NPV (net present value), here per unit of investment, if the borrower behaves,

$$p_H R > 1, \qquad (3.7)$$

but negative NPV otherwise,

$$1 > p_L R + B, \qquad (3.8)$$

so that unless one can control the agency problem the investment cannot be funded. We also make an assumption that guarantees that the equilibrium investment is finite:

$$p_H R < 1 + \frac{p_H B}{\Delta p}. \qquad (3.9)$$

Like inequality (3.5) in Section 3.2, inequality (3.9) has a simple interpretation: the expected net revenue per unit of investment, $p_H R - 1$, is lower than the per-unit agency cost, $p_H B / \Delta p$.

Finally, we keep assuming that the capital market is competitive. [...]

From Tirole §3.4.2 "The Lenders' Credit Analysis", pp. 127–128 (PDF 140–141):
Following the steps of Section 3.2, the incentive compatibility and the breakeven conditions are

$$(\Delta p)R_b \geqslant BI \qquad (\mathrm{IC}_b)$$

and

$$p_H(RI - R_b) \geqslant I - A. \qquad (\mathrm{IR}_l)$$

In equilibrium, competitive lenders make no profit on the contract that is most advantageous for the borrower; the borrower's net utility is therefore equal to the social surplus brought about by the investment:

$$U_b = (p_H R - 1)I. \qquad (3.10)$$

From (3.10) it is optimal for the borrower to invest as much as possible. The upper bound on investment and in turn her borrowing capacity ("outside financing capacity" or "debt capacity") are determined by constraints (IC_b) and (IR_l). Substituting (IC_b) into (IR_l), we obtain

$$I \leqslant kA, \qquad (3.11)$$

where

$$k = \frac{1}{1 - p_H(R - B/\Delta p)} > 1. \qquad (3.12)$$

The denominator of k is positive from (3.9). Furthermore, conditions (3.7) and (3.8) imply that $(\Delta p)R > B$, and therefore that the denominator of k is smaller than 1. This is important: the fact that $k > 1$ shows that the borrower can lever her wealth, k being the multiplier.

[...] Conditions (3.7) and (3.10) furthermore imply that it is optimal for the borrower to invest k times her cash A, that is, to borrow $d = (k - 1)$ times her level of cash, where

$$d = \frac{p_H(R - B/\Delta p)}{1 - p_H(R - B/\Delta p)}. \qquad (3.13)$$

The maximum loan, dA, is called "borrowing capacity." [...]

What is asked. In the variable-investment model, suppose the initial loan leaves the borrower too small a stake to work, $R_b < BI/\Delta p$, but can be renegotiated after the investment and before effort. (i) Show that a mutually beneficial renegotiation exists exactly when the stated inequality holds. (ii) Explain that inequality, and derive it as the financing condition of a fictitious fixed-investment project. (iii) If the borrower makes a take-it-or-leave-it renegotiation offer, find how much she can invest (her borrowing capacity) with such an initial contract. (iv) Argue directly that writing $R_b \geq BI/\Delta p$ from the start loses nothing.

Approach. Lecture Topic 4B, renegotiation. Without renegotiation the borrower shirks; renegotiation can only help if it restores (IC_b), and it happens iff both sides gain relative to the shirking status quo. The lender gives up part of his claim (*debt forgiveness*) and is paid back out of the extra *pledgeable income* that effort creates, so the condition has the form "pledgeable income ≥ investment − cash" of the fixed-investment model. For (iii) the lender's ex ante breakeven uses his renegotiated payoff; for (iv) rational expectations make the renegotiated contract one that could have been signed directly.

Notation.

- I: (sunk) investment; income RI in success, 0 in failure; A: borrower's cash.
- p_H, $p_L = p_H - \Delta p$; private benefit BI from shirking; (3.7) $p_H R > 1$, (3.8) $1 > p_L R + B$, (3.9) $p_H R < 1 + p_H B/\Delta p$; these imply $(\Delta p)R > B$.
- $R_b < BI/\Delta p$: borrower's stake in the initial contract; $\hat{R}_b$: renegotiated stake.
- $k = 1/[1 - p_H(R - B/\Delta p)]$: equity multiplier (3.12).

Step 1 — (i) Status quo. We compute each party's payoff if nothing is renegotiated; these are the outside options in the renegotiation. With $R_b < BI/\Delta p$, $(\Delta p)R_b < BI$: (IC_b) fails and the borrower shirks (success probability p_L, private benefit BI). Payoffs if the contract is not renegotiated:

$$U_b^0 = p_L R_b + BI, \tag{68}$$

$$U_l^0 = p_L(RI - R_b). \tag{69}$$

Step 2 — (i) Renegotiation must restore incentives. A new stake $\hat{R}_b < BI/\Delta p$ still induces shirking, so total surplus $p_L RI + BI$ is unchanged and the change is a pure transfer: one side loses. A mutually advantageous renegotiation therefore needs $\hat{R}_b \geq BI/\Delta p$, and both parties must gain relative to (68)–(69):

$$p_H \hat{R}_b \geq p_L R_b + BI, \tag{70}$$

$$p_H(RI - \hat{R}_b) \geq p_L(RI - R_b). \tag{71}$$

Step 3 — (i) The condition. For any $\hat{R}_b \geq BI/\Delta p$, (70) holds automatically, since $R_b < BI/\Delta p$:

$$p_H \hat{R}_b \geq p_H \frac{BI}{\Delta p} = BI + p_L \frac{BI}{\Delta p} > BI + p_L R_b.$$

The lender prefers the smallest admissible stake, $\hat{R}_b = BI/\Delta p$. So a mutually advantageous $\hat{R}_b$ exists iff (71) holds at $\hat{R}_b = BI/\Delta p$:

$$\begin{aligned} p_H\Big(RI - \frac{BI}{\Delta p}\Big) &\geq p_L(RI - R_b) \\ p_H RI - \frac{p_H BI}{\Delta p} &\geq p_L RI - p_L R_b \\ (p_H - p_L)RI - \frac{p_H BI}{\Delta p} + p_L R_b &\geq 0 \\ (\Delta p)RI - \frac{p_H BI}{\Delta p} + p_L R_b &\geq 0. \end{aligned} \tag{72}$$

Necessity: any renegotiated contract has $\hat{R}_b \geq BI/\Delta p$, and (71) is hardest to meet at larger $\hat{R}_b$. Sufficiency: if (72) holds, $\hat{R}_b = BI/\Delta p$ makes the lender weakly and the borrower strictly better off. The renegotiation is *debt forgiveness*: the lender's claim in success falls from $RI - R_b$ to $RI - \hat{R}_b$.

Answer (i). Renegotiation (to $\hat{R}_b \geq BI/\Delta p$) takes place iff $(\Delta p)RI - p_H BI/\Delta p + p_L R_b \geq 0$.

Step 4 — (ii) Interpretation via the general theory. Treat "inducing effort" as a fictitious fixed-investment project. Its income is the increase in expected output, $(\Delta p)RI$; it needs no new investment ($I_f = 0$); the borrower's minimum stake in it is $p_H\, BI/\Delta p$ (her expected rent under (IC_b)), and she brings as "cash" the expected income she gives up, $A_f \equiv p_L R_b$. Its pledgeable income is $(\Delta p)RI - p_H BI/\Delta p$, and the financing condition (IR_l), "pledgeable income $\geq$ investment $-$ cash", reads

$$(\Delta p)RI - \frac{p_H BI}{\Delta p} \;\geq\; 0 - p_L R_b, \tag{73}$$

which is (72). The total gain from restoring effort, $(\Delta p)RI - BI$, is always positive, yet renegotiation may fail: the lender can only be paid out of pledgeable income, and the borrower's initial stake $p_L R_b$ plays the role of net worth. A larger R_b makes forgiveness easier.

Answer (ii). The condition says the pledgeable part of the effort gain, $(\Delta p)RI - p_H BI/\Delta p$, must cover the lender's "outlay" $0 - p_L R_b$: it is (IR_l) for a fictitious project with income $(\Delta p)RI$, investment 0 and cash $p_L R_b$.

Step 5 — (iii) Borrowing capacity under renegotiation, borrower has all bargaining power. When (72) holds, the borrower offers the $\hat{R}_b$ that leaves the lender indifferent:

$$p_H(RI - \hat{R}_b) = p_L(RI - R_b), \quad \text{i.e.} \quad \hat{R}_b = RI - \frac{p_L}{p_H}(RI - R_b). \tag{74}$$

The lender therefore anticipates exactly (69), and his ex ante breakeven constraint is

$$p_L(RI - R_b) \;\geq\; I - A. \tag{75}$$

The borrower's utility is total surplus with effort minus the lender's payoff: $p_H RI - p_L(RI - R_b) - A$, which by the binding (75) equals $p_H RI - (I - A) - A = (p_H R - 1)I$, the NPV with effort. Since $p_H R > 1$ (3.7), she maximises I subject to (72), (75) and $0 \leq R_b < BI/\Delta p$. (72) bounds $p_L R_b$ from below and (75) (expanded: $p_L RI - p_L R_b \geq I - A$) bounds it from above:

$$\frac{p_H BI}{\Delta p} - (\Delta p)RI \;\leq\; p_L R_b \;\leq\; p_L RI - (I - A). \tag{76}$$

An R_b exists iff the lower bound is below the upper bound:

$$\begin{aligned}
\frac{p_H BI}{\Delta p} - (\Delta p)RI &\leq p_L RI - I + A \\
I - (\Delta p)RI - p_L RI + \frac{p_H BI}{\Delta p} &\leq A \\
I - p_H RI + \frac{p_H BI}{\Delta p} &\leq A \qquad (\Delta p + p_L = p_H) \\
I\Big[1 - p_H\Big(R - \frac{B}{\Delta p}\Big)\Big] &\leq A \\
I &\leq kA.
\end{aligned} \tag{77}$$

If $p_H B/\Delta p \geq (\Delta p)R$, the lower bound in (76) is nonnegative and the capacity is exactly kA: at $I = kA$ both (72) and (75) bind, $R_b < BI/\Delta p$ (by $(\Delta p)R > B$) and (74) gives $\hat{R}_b = BI/\Delta p$. If $p_H B/\Delta p < (\Delta p)R$, the constraint $R_b \geq 0$ binds first, and (75) with $R_b = 0$ gives $I \leq A/(1 - p_L R) < kA$. Either way the capacity is at most kA, the capacity without renegotiation.

Answer (iii). The borrowing capacity is $I = kA$, $k = 1/[1 - p_H(R - B/\Delta p)]$, when $p_H B/\Delta p \geq (\Delta p)R$ (then $p_L R_b = kA[p_H B/\Delta p - (\Delta p)R]$ and $\hat{R}_b = BI/\Delta p$), and $I = A/(1 - p_L R) < kA$ otherwise; it never exceeds the capacity kA of an incentive-compatible initial contract.

Step 6 — (iv) Rational expectations. Take any initial contract (I, R_b) with $R_b < BI/\Delta p$. If it is not renegotiated, the borrower shirks; the NPV is $(p_L R + B - 1)I < 0$ by (3.8), and since the lender breaks even the borrower's utility is that negative NPV: worse than not investing. If it is renegotiated to $\hat{R}_b \geq BI/\Delta p$, the final contract $(I, \hat{R}_b)$ satisfies (IC_b), and by rational expectations the lender's ex ante breakeven uses his final payoff, $p_H(RI - \hat{R}_b) \geq I - A$, which is ($\text{IR}_l$). So $(I, \hat{R}_b)$ could have been signed directly, with the same effort and the same payoffs. Nothing is lost by restricting attention to $R_b \geq BI/\Delta p$, which is renegotiation-proof.

Answer (iv). Any initial contract with $R_b < BI/\Delta p$ either is not renegotiated (negative NPV, worse than not investing) or ends as an incentive-compatible contract that satisfies (IR_l) and could have been signed at the outset; so restricting to $R_b \geq BI/\Delta p$ loses nothing.

Intuition. A lender facing a borrower who is about to shirk is willing to forgive debt, because the higher stake raises the success probability and the lender keeps part of that gain. He forgives only if his remaining claim after forgiveness, the pledgeable part of the effort gain, beats what he gets from a shirking borrower. Anticipating this, an initial contract that relies on forgiveness can at best reproduce the incentive-compatible contract: the multiplier k is the right one.

Common mistakes.

- Arguing that renegotiation always happens because restoring effort raises total surplus ($(\Delta p)RI > BI$). The lender gains only the *pledgeable* part of that surplus.
- Writing the lender's ex ante breakeven in (iii) as $p_H(RI - R_b) \geq I - A$, i.e. his initial claim with effort. He anticipates the renegotiation and values his final payoff, which with a take-it-or-leave-it offer by the borrower is $p_L(RI - R_b)$.
- Mixing the fixed- and variable-investment notation: here the private benefit is BI and the minimum stake is $BI/\Delta p$, not $B/\Delta p$.

Book answer. Matches the book's answer to (i)–(ii) (pp. 573–574), including the fictitious-project reading with cash $\hat{A} = p_L R_b$. The book gives no answer to (iii)–(iv); Steps 5–6 are this solution's.

Ref: Tirole §3.4.1–3.4.2, pp. 127–128; §3.2.2, pp. 116–117; answer pp. 573–574.

13. Loan size and collateral requirements — Tirole Exercise 4.17, p. 195

Statement.

Source: Tirole, Ch. 4, Exercise 4.17, p. 195 (PDF 208) · Answer in book: yes, p. 591

Exercise 4.17 (loan size and collateral requirements). An entrepreneur with limited wealth A finances a variable-investment project. A project of size $I \in \mathbb{R}$ if successful yields $R(I)$, where $R(0) = 0$, $R' > 0$, $R'' < 0$, $R'(0) = \infty$, $R'(\infty) = 0$. The probability of success is p_H if the entrepreneur behaves (she then receives no private benefit) and $p_L = p_H - \Delta p$ if she misbehaves (she then receives private benefit BI).

The entrepreneur can pledge an arbitrary amount of collateral with cost $C \geqslant 0$ to the entrepreneur and value $\phi(C)$ for the investors with $\phi(0) = 0$, $\phi' > 0$, $\phi'' < 0$, $\phi'(0) = 1$, $\phi'(\infty) = 0$.

The entrepreneur is risk neutral and protected by limited liability and the investors are competitive, risk neutral, and demand a rate of return equal to 0.

Assume that the first-best policy does not yield enough pledgeable income. (This first-best policy is $C^* = 0$ and I^* given by $p_H R'(I^*) = 1$. Thus, the assumption is $p_H[R(I^*) - BI^*/\Delta p] < I^* - A$.)

Assume that the entrepreneur pledges collateral only in the case of failure (on this, see Section 4.3.5), and that the investors' breakeven constraint is binding. Show that as A decreases or the agency cost (as measured by B or, keeping p_H constant, $p_H/\Delta p$) increases, the optimal investment size decreases and the optimal collateral increases.

Context needed to solve it *From Tirole §3.4.1 "The Continuous-Investment Model", p. 127 (PDF 140):*

The continuous-investment model of this section is the polar opposite of the fixed-investment one. [...] we now assume that there are constant returns to scale in the investment technology. An investment $I \in [0, \infty)$ yields income RI, proportional to I, in the case of success, and 0 in the case of failure. The borrower's private benefit from misbehaving is also taken to be proportional to investment. As before, the borrower has a choice between behaving, in which case she derives no private benefit and the probability of success is p_H, and misbehaving, that is, enjoying private benefit BI, and reducing the probability of success to $p_L = p_H - \Delta p < p_H$. [...]

The borrower initially has cash A, and must therefore borrow $I - A$ to finance a project of size I. A loan agreement specifies that the lenders (who as before are assumed to make no profit) and the borrower receive 0 each in the case of failure, and R_l and R_b, respectively, in the case of success, where $R_l + R_b = RI$.

As in Section 3.2, we assume that investment has positive NPV (net present value), here per unit of investment, if the borrower behaves,

$$p_H R > 1, \qquad (3.7)$$

but negative NPV otherwise,

$$1 > p_L R + B, \qquad (3.8)$$

so that unless one can control the agency problem the investment cannot be funded. We also make an assumption that guarantees that the equilibrium investment is finite:

$$p_H R < 1 + \frac{p_H B}{\Delta p}. \qquad (3.9)$$

[...] Finally, we keep assuming that the capital market is competitive. [...]
From Tirole §3.4.2 "The Lenders' Credit Analysis", pp. 127–128 (PDF 140–141):
Following the steps of Section 3.2, the incentive compatibility and the breakeven conditions are

$$(\Delta p)R_b \geqslant BI \qquad (\text{IC}_b)$$

and

$$p_H(RI - R_b) \geqslant I - A. \qquad (\text{IR}_l)$$

In equilibrium, competitive lenders make no profit on the contract that is most advantageous for the borrower; the borrower's net utility is therefore equal to the social surplus brought about by the investment:

$$U_b = (p_H R - 1)I. \qquad (3.10)$$

From (3.10) it is optimal for the borrower to invest as much as possible. The upper bound on investment and in turn her borrowing capacity ("outside financing capacity" or "debt capacity") are determined by constraints (IC_b) and (IR_l). Substituting (IC_b) into (IR_l), we obtain

$$I \leqslant kA, \qquad (3.11)$$

where

$$k = \frac{1}{1 - p_H(R - B/\Delta p)} > 1. \qquad (3.12)$$

[...] the fact that $k > 1$ shows that the borrower can lever her wealth, k being the multiplier. [...]
From Tirole §4.3.5 "Pledging Existing Wealth", p. 170 (PDF 183):
[...] Suppose that the entrepreneur can pledge an arbitrary amount C,

$$0 \leqslant C \leqslant C^{\max},$$

conditional on failing (we will later check that conditional collateral dominates unconditional collateral). Investors value collateral C at βC, where $\beta < 1$, when they seize it.

The borrower's net utility, as usual, is equal to the NPV. The NPV is equal to its value in the absence of collateral, $p_H R - I$, minus the deadweight loss associated with collateral pledging. This deadweight loss is equal to $(1 - \beta)C$ times the probability, $1 - p_H$, that the firm fails. And so

$$U_b = p_H R - I - (1 - p_H)(1 - \beta)C.$$

The NPV is maximized when $C = 0$; the borrower will therefore not pledge collateral unless she needs to. Thus, if A denotes the borrower's cash on hand and $A \geqslant \bar{A}$, where

$$p_H\left(R - \frac{B}{\Delta p}\right) = I - \bar{A},$$

then $C = 0$.

By contrast, firms with weaker balance sheets, i.e., $A < \bar{A}$, need to pledge collateral in order to raise funds.[22] Under collateral pledging, the incentive constraint becomes[23]

$$(\Delta p)(R_b + C) \geqslant B,$$

since the borrower loses both her reward R_b and the collateral when she fails (her stake is just larger). The investors' breakeven condition becomes

$$p_H(R - R_b) + (1 - p_H)\beta C \geqslant I - A,$$

or, using the incentive compatibility constraint,

$$p_H\left(R - \frac{B}{\Delta p}\right) + p_H C + (1 - p_H)\beta C \geqslant I - A.$$

Note that the pledging of collateral raises pledgeable income both directly (term $(1-p_H)\beta C$) and indirectly through the reduction in entrepreneurial reward (term $p_H C$).[24] To minimize the deadweight loss, the borrower pledges the minimum amount of collateral that allows investors to break even:

$$C(A) = \frac{(I - A) - p_H(R - B/\Delta p)}{p_H + (1 - p_H)\beta}.$$

Note that $C(A)$ is a decreasing function of A: among firms that pledge collateral, those with the weakest balance sheet pledge more collateral. [...]

22. We assume that $C^{\max}$ is small enough that the NPV remains positive even if the borrower pledges all assets: $p_H R - I - (1 - p_H)(1 - \beta)C^{\max} \geqslant 0$.
23. A different way of writing this constraint is $p_H R_b + (1 - p_H)(-C) \geqslant p_L R_b + (1 - p_L)(-C) + B$.
24. This latter term (and the validity of the analysis) rests on the condition that $R_b \geqslant 0$, which we will assume (this is guaranteed by imposing $B/\Delta p \geqslant C^{\max}$). For large amounts of collateral, it is no longer possible to substitute collateral for reward, since the latter would become negative and violate limited liability.

What is asked. The project has decreasing returns, $R(I)$ concave, and the entrepreneur can pledge costly collateral C that investors seize on failure. Assuming the first best cannot be financed, show that a poorer entrepreneur (lower A) or a worse agency problem (higher B, or higher $p_H/\Delta p$ with p_H fixed) leads to a smaller optimal investment and to more collateral.

Approach. This combines the variable-investment model (§3.4) with collateral pledging (§4.3.5); lecture Topic 4A, collateral reading. Collateral raises *pledgeable income* twice: investors get $\phi(C)$ on failure, and the borrower's stake grows, so her cash reward can shrink by C ($p_H C$). It costs a deadweight loss $C - \phi(C)$ on failure. We eliminate R_b with the binding (IC_b), write the Lagrangian with multiplier μ on (IR_l), derive the first-order conditions, and do comparative statics on the resulting system with the implicit-function theorem.

Notation.

- $I \geq 0$: investment; $R(I)$: income in success ($R(0) = 0$, $R' > 0$, $R'' < 0$, Inada).
- p_H, $p_L = p_H - \Delta p$; private benefit BI; $b \equiv p_H B/\Delta p$: per-unit agency cost (rises with B and, with p_H fixed, with $p_H/\Delta p$).
- $C \geq 0$: collateral pledged in failure, cost C to the entrepreneur, value $\phi(C)$ to investors ($\phi(0) = 0$, $\phi' > 0$, $\phi'' < 0$, $\phi'(0) = 1$, $\phi'(\infty) = 0$).
- $R_b \geq 0$: reward in success; A: cash; I^*: first-best size, $p_H R'(I^*) = 1$.
- $\mu \geq 0$: multiplier of the breakeven constraint; $\theta \equiv \mu/(1+\mu) \in [0, 1)$; $\varepsilon(I) \equiv -IR''(I)/R'(I) > 0$: curvature of R.

Assumption. The limited-liability constraint $R_b \geq 0$, i.e. $C \leq BI/\Delta p$, does not bind (as in footnote 24 of §4.3.5).

Step 1 — Programme. We set up the problem and eliminate R_b. The entrepreneur gets R_b on success and $-C$ on failure, so (IC_b) is $p_H R_b - (1 - p_H)C \geq p_L R_b - (1 - p_L)C + BI$, i.e. $(\Delta p)R_b + (\Delta p)C \geq BI$, i.e. $(\Delta p)(R_b + C) \geq BI$. Investors get $R(I) - R_b$ in success and $\phi(C)$ in failure. Collateral destroys $C - \phi(C) \geq 0$ in failure, so with binding breakeven her utility is the NPV:

$$\max_{I, C, R_b} U_b = p_H R(I) - I - (1 - p_H)\big[C - \phi(C)\big] \tag{78}$$

$$\text{s.t.} \quad (\Delta p)(R_b + C) \;\geq\; BI, \tag{IC_b}$$

$$p_H\big[R(I) - R_b\big] + (1 - p_H)\phi(C) \;\geq\; I - A. \tag{IR_l}$$

R_b does not enter (78) and lowers (IR_l), so (IC_b) binds: $R_b = BI/\Delta p - C$. Substituting into (IR_l),

$$p_H\Big[R(I) - \frac{BI}{\Delta p} + C\Big] + (1 - p_H)\phi(C) \geq I - A$$

$$p_H R(I) - \frac{p_H B}{\Delta p} I + p_H C + (1 - p_H)\phi(C) - I + A \geq 0$$

$$G(I, C) \equiv p_H R(I) - bI + p_H C + (1 - p_H)\phi(C) - I + A \;\geq\; 0. \tag{79}$$

Collateral raises pledgeable income directly (ϕ) and by replacing reward ($p_H C$). The objective is concave and G is concave in (I, C), so the Kuhn–Tucker conditions are sufficient.

Step 2 — First-order conditions. We derive the conditions that characterise the optimum. With multiplier $\mu \geq 0$ on (79), the Lagrangian is

$$\begin{aligned}\mathcal{L}(I, C, \mu) = {} & p_H R(I) - I - (1 - p_H)\big[C - \phi(C)\big] \\ & + \mu\big[p_H R(I) - bI + p_H C + (1 - p_H)\phi(C) - I + A\big].\end{aligned}$$

Differentiating with respect to I and C and setting the derivatives to zero,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial I} &= p_H R'(I) - 1 + \mu\big[p_H R'(I) - b - 1\big] = 0, \\ \frac{\partial \mathcal{L}}{\partial C} &= -(1 - p_H)\big[1 - \phi'(C)\big] + \mu\big[p_H + (1 - p_H)\phi'(C)\big] = 0.\end{aligned}$$

The I-condition. Collect the terms in $p_H R'(I) - 1$:

$$\begin{aligned}(1 + \mu)\big[p_H R'(I) - 1\big] - \mu b &= 0 \\ p_H R'(I) - 1 &= \frac{\mu}{1 + \mu}\, b.\end{aligned}$$

The C-condition. Expand and collect the terms in $\phi'(C)$:

$$\begin{aligned}-(1 - p_H) + (1 - p_H)\phi'(C) + \mu p_H + \mu(1 - p_H)\phi'(C) &= 0 \\ (1 + \mu)(1 - p_H)\phi'(C) &= (1 - p_H) - \mu p_H \\ \phi'(C) &= \frac{1}{1 + \mu} - \frac{\mu}{1 + \mu}\,\frac{p_H}{1 - p_H}.\end{aligned}$$

Writing $\theta = \mu/(1 + \mu)$, so that $1/(1 + \mu) = 1 - \theta$,

$$p_H R'(I) - 1 = \theta\, b, \tag{80}$$

$$\phi'(C) = 1 - \theta - \theta\frac{p_H}{1 - p_H} = 1 - \theta\,\frac{(1 - p_H) + p_H}{1 - p_H} = 1 - \frac{\theta}{1 - p_H}. \tag{81}$$

If $\mu = 0$, (80)–(81) give $I = I^*$, $C = 0$, which violates (79) by the assumption $p_H[R(I^*) - BI^*/\Delta p] < I^* - A$. So $\theta > 0$: (80) gives $I < I^*$ and (81) gives $\phi'(C) < 1 = \phi'(0)$, i.e. $C > 0$; finiteness of C needs $\theta < 1 - p_H$. These are the book's conditions, since $\theta = \mu/(1 + \mu)$.

Step 3 — Monotonicity in θ. The FOCs define I and C as functions of θ (and b); we find how they move and then pin θ down with the binding breakeven. Totally differentiating (80), $p_H R''(I)\, dI = b\, d\theta + \theta\, db$; and (81), $\phi''(C)\, dC = -d\theta/(1 - p_H)$. With $R'' < 0$, $\phi'' < 0$ and $Q \equiv 1/(p_H|R''(I)|) > 0$, dividing by $p_H R'' = -1/Q$ and by ϕ'':

$$\frac{\partial I}{\partial \theta} = -bQ < 0, \qquad \frac{\partial I}{\partial b} = -\theta Q < 0, \tag{82}$$

$$C'(\theta) = -\frac{1}{(1 - p_H)\phi''(C)} > 0. \tag{83}$$

Let $g(\theta; A, b) \equiv G(I(\theta, b), C(\theta))$, the slack of (79) along the FOCs; the optimum is the root $g = 0$. Using (80), $p_H R' = 1 + \theta b$, so $G_I = p_H R' - b - 1 = (1 + \theta b) - b - 1 = -(1 - \theta)b < 0$; also $G_C = p_H + (1 - p_H)\phi' > 0$. Then, by the chain rule and (82)–(83),

$$\begin{aligned}g_\theta &= G_I \frac{\partial I}{\partial \theta} + G_C C'(\theta) \\ &= \big(-(1 - \theta)b\big)\big(-bQ\big) + G_C C'(\theta) \\ &= (1 - \theta)b^2 Q + G_C C'(\theta) \;> 0.\end{aligned} \tag{84}$$

g is < 0 at $\theta = 0$ (first best not pledgeable) and $\to +\infty$ as $\theta \to 1 - p_H$ ($C \to \infty$), so the root $\theta^* \in (0, 1 - p_H)$ is unique.

Step 4 — Comparative statics in A. $g_A = 1$, so by the implicit-function theorem and (84)

$$\frac{d\theta^*}{dA} = -\frac{g_A}{g_\theta} = -\frac{1}{g_\theta} < 0. \tag{85}$$

Combining with (82)–(83) by the chain rule:

$$\frac{dI}{dA} = \frac{\partial I}{\partial \theta}\,\frac{d\theta^*}{dA} = (-bQ)\Big(-\frac{1}{g_\theta}\Big) = \frac{bQ}{g_\theta} > 0,$$
$$\frac{dC}{dA} = C'(\theta)\,\frac{d\theta^*}{dA} = -\frac{C'(\theta)}{g_\theta} < 0.$$

As A decreases, θ^* (hence $\mu = \theta/(1-\theta)$) rises, the investment size falls and the collateral rises, as claimed.

Answer (balance sheet). As A decreases, the shadow cost μ rises, investment I falls and collateral C rises: $dI/dA > 0$, $dC/dA < 0$.

Step 5 — Comparative statics in the agency cost b. Both B and $p_H/\Delta p$ (with p_H fixed) enter only through $b = p_H B/\Delta p$, so it suffices to vary b. Now b also enters $I(\theta, b)$; by (79), $G_b = -I$, so

$$\begin{aligned} g_b &= G_b + G_I \frac{\partial I}{\partial b} \\ &= -I + \big(-(1-\theta)b\big)\big(-\theta Q\big) = -I + \theta(1-\theta)bQ, \end{aligned} \tag{86}$$

and by the implicit-function theorem $d\theta^*/db = -g_b/g_\theta$. *Investment.* Using (82), (84) and (86) (with $D \equiv G_C C' > 0$),

$$\begin{aligned} \frac{dI}{db} &= \frac{\partial I}{\partial \theta}\frac{d\theta^*}{db} + \frac{\partial I}{\partial b} = (-bQ)\Big(-\frac{g_b}{g_\theta}\Big) + (-\theta Q) = -Q\Big(\theta - b\,\frac{g_b}{g_\theta}\Big) \\ &= -Q\,\frac{\theta g_\theta - b g_b}{g_\theta} = -Q\,\frac{\theta D + bI}{g_\theta} \;< 0, \end{aligned} \tag{87}$$

where the numerator is $\theta g_\theta - b g_b = \theta(1-\theta)b^2 Q + \theta D + bI - \theta(1-\theta)b^2 Q = \theta D + bI > 0$. So I falls with B and with $p_H/\Delta p$, as claimed. *Collateral.* $dC/db = C'(\theta)\,d\theta^*/db = -C'(\theta) g_b/g_\theta$ has the sign of $-g_b$, and $-g_b > 0$ iff $I > \theta(1-\theta)bQ$, i.e. (multiplying by $1/Q = p_H|R''|$) iff $p_H|R''|I > \theta(1-\theta)b$. By the definition of ε and (80), $p_H|R''|I = \varepsilon(I)\,p_H R'(I) = \varepsilon(I)(1+\theta b)$, so

$$\frac{dC}{db} > 0 \iff \varepsilon(I)\,(1+\theta b) \;>\; \theta(1-\theta)\,b. \tag{88}$$

This holds when R is concave enough (for instance $\varepsilon \geq 1$, since $\theta(1-\theta)b < 1 + \theta b$), but it can fail when R is close to linear. Counterexample: $R(I) = 3I^{0.95}$ ($\varepsilon = 0.05$), $\phi(C) = \ln(1+C)$, $p_H = 0.5$, $\Delta p = 0.25$, $A = 1$. For $B = 0.5$ ($b = 1$) the solution is $\theta^* = 0.3276$, $I = 4.1203$, $C = 1.9003$; for $B = 2$ ($b = 4$) it is $\theta^* = 0.1238$, $I = 0.3815$, $C = 0.3292$. Collateral *falls* as the agency cost rises; at $b = 1$, (88) reads $0.05(1 + 0.3276) = 0.0664 > 0.3276(1 - 0.3276)(1) = 0.2203$, false. (The first best is not pledgeable in both cases and $C \leq BI/\Delta p$.)

Answer (agency cost). Investment always falls with the agency cost, $dI/db < 0$. Collateral rises with it iff $\varepsilon(I)(1+\theta b) > \theta(1-\theta)b$ (e.g. whenever $\varepsilon \geq 1$); the statement's claim for C is not general, as the counterexample shows.

Intuition. A weaker balance sheet tightens the financing constraint; the entrepreneur pays for funding on both margins she controls, investing less (which saves outside funds) and pledging more collateral (which adds pledgeable income at a deadweight cost). A higher agency cost also lowers

investment, but it has a second effect: each unit of investment now absorbs more pledgeable income, so the downsizing itself releases funds. When output is nearly linear in I, downsizing is cheap and strong enough that less collateral is needed. Only the balance-sheet comparative static is unambiguous.

Common mistakes.

- Dropping the indirect effect of collateral, the term $p_H C$: collateral also lets the cash reward R_b fall by C without breaking (IC_b).
- Treating b as a pure shift of the constraint. It also enters the I-condition (80), which is why the sign of dC/db is ambiguous.
- Ignoring limited liability, $R_b = BI/\Delta p - C \geq 0$: the substitution of collateral for reward is valid only while it holds.

Book answer. Agrees with the book's answer (p. 591): the FOCs (80)–(81) and "as the balance sheet deteriorates, μ increases, I decreases, and C increases". The book's answer does not claim that collateral rises with the agency cost (it says only that borrowing rises when the agency cost falls). The statement's claim for C holds only under (88); the claim for I holds in general (87).

Ref: Tirole §4.3.5, p. 170; §3.4.2, pp. 127–128; answer p. 591.

14. Dealing with asset substitution — Tirole Exercise 7.3, pp. 323–324

Statement.

Source: Tirole, Ch. 7, Exercise 7.3, pp. 323–324 (PDF 336) · Answer in book: yes, p. 602

Exercise 7.3 (dealing with asset substitution). Consider the fixed-investment model with a probability that the investment must be resold (redeployed) at an intermediate date because, say, it is learned that there is no demand for the product. The timing is summarized in Figure 7.16.

Figure 7.16 (timeline, left to right; p. 324, PDF 337)

Step 1	Step 2	Step 3 (demand revealed)	Step 4	Step 5
Financing. Firm borrows $I - A$.	Asset substitution?	With probability $1-x$: verifiable liquidation value L. With probability x: continue $\rightarrow$	Moral hazard (choice of probability of success).	Outcome (R or 0).

An entrepreneur has cash A and wants to invest a fixed amount $I > A$ into a project. The shortfall must be raised in a competitive capital market. The project yields R with probability p and 0 with probability $1-p$, provided that there is a demand for the product (which has probability x and is revealed at the intermediate stage; the final profit is always 0 if there is no demand, and so it is then optimal to liquidate at the intermediate stage). Investors and entrepreneur are risk neutral, the latter is protected by limited liability, and the market rate of interest is 0.

(i) In a first step, ignore the possibility of asset substitution. The liquidation value is $L = L_0$, and the probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ if she shirks (in which case she obtains a private benefit B). Assume that the NPV of the project is positive if the entrepreneur works, and negative if she shirks.

Assume that $A \geqslant \bar{A}$, where

$$(1-x)L_0 + xp_H\left(R - \frac{B}{\Delta p}\right) = I - \bar{A} \qquad (1)$$

(and that $L_0 \leqslant p_H(R - B/\Delta p)$).

- Interpret (1).

- Compute the entrepreneur's expected utility.
- What is the class of optimal contracts (or, at least, characterize the optimal contract for $A = \bar{A}$)?

(ii) Suppose now that, before the state of demand is realized, but after the investment is sunk, the entrepreneur can engage in asset substitution. She can reallocate funds between asset maintenance (value of L) and future profit (as characterized by the probability of success, say).

More precisely, suppose that the entrepreneur chooses L and

- the probability of success is $p_H + \tau(L)$ if the entrepreneur behaves and $p_L + \tau(L)$ if she misbehaves;
- the function τ is decreasing and strictly concave;
- $$\tau(L_0) = 0 \quad \text{and} \quad \tau'(L_0)R = -\frac{1-x}{x}; \qquad (2)$$
- the entrepreneur secretly chooses L (multitasking).

Consider contracts in which

- liquidation occurs if and only if there is no demand (hence, with probability x);
- the entrepreneur receives $r_b(L)$ if the assets are liquidated, and R_b if they are not and the project is successful (and 0 if the project fails).

Interpret (2). Compute the minimum level of A such that the threat of (excessive) asset substitution is innocuous. Interpret the associated optimal contract. (Hint: what is the optimal asset maintenance (liquidation value)? Note that, in order to induce the entrepreneur to choose this value, in the case of liquidation you may pay $r_b(L) = r_b$ if L is at the optimal level and 0 otherwise.)

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):
[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\mathrm{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\mathrm{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

To make things interesting, we will assume that

$$\bar{A} > 0 \iff p_H R - I < p_H \frac{B}{\Delta p}, \qquad (3.4)$$

otherwise even a borrower with no wealth of her own would be able to obtain credit. [...]
Thus, *the borrower must have enough assets in order to be granted a loan.* Note that, if $A < \bar{A}$, the project has positive NPV and yet is not funded. [...] There is *credit rationing.* [...]
Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

Her stake,

$$R_b = R - R_l = R - \frac{I - A}{p_H} \geqslant R - \frac{I - \bar{A}}{p_H} = \frac{B}{\Delta p}, \qquad (3.5)$$

then induces her to behave.
[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H (R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*
So, the borrower's utility jumps up at $A = \bar{A}$. [...]

What is asked. (i) Without asset substitution: explain the threshold equation (1), give the entrepreneur's expected utility, and describe the optimal contracts (at least at $A = \bar{A}$). (ii) Now she can secretly run down maintenance (lower liquidation value L) to raise the success probability. Explain condition (2), find the smallest A for which this threat does no harm, and describe the contract that achieves it.

Approach. The fixed-investment model with a verifiable liquidation value (lecture Topic 4B, asset substitution and remedies). Liquidation value is not subject to moral hazard, so all of it is pledgeable; continuation income is pledgeable only up to $R - B/\Delta p$. A debt-like claim pays the entrepreneur only in continuation, so she prefers to shift value from liquidation to continuation: *asset substitution*. To prevent it she must be paid something in liquidation, conditional on good maintenance; that payment is a rent, which lowers *pledgeable income* and raises the minimum equity. We write (IR_l), (IC_b) and a no-substitution constraint, and minimise her rent.

Notation.

- $I > A$: fixed investment; A: cash; $x \in (0,1)$: probability of demand; with probability $1-x$ there is no demand and the assets are liquidated for L.
- R: income in success; p_H, $p_L = p_H - \Delta p$, B: as in Section 3.2.
- L_0: liquidation value without asset substitution; $\tau(L)$: change in success probability when maintenance yields liquidation value L ($\tau' < 0$, $\tau'' < 0$, $\tau(L_0) = 0$).
- r_b: entrepreneur's payment on liquidation; R_b: payment on continuation and success.
- U_b^*: entrepreneur's utility; $\bar{A}$: threshold (1); A^*: threshold with asset substitution.

Assumption. The statement's "(hence, with probability x)" for liquidation is read as probability $1-x$, the probability of no demand. $L \geq 0$ is chosen from $[0, \infty)$.

Step 1 — (i) Interpreting (1). The liquidation value L_0 is verifiable and not subject to moral hazard, so all of it can go to investors. On continuation only $R - R_b$ with $R_b \geq B/\Delta p$ can. The left-hand side of (1) is the maximal expected pledgeable income,

$$\mathcal{P}_0 = (1-x)L_0 + xp_H\Big(R - \frac{B}{\Delta p}\Big), \tag{89}$$

and (1) sets it equal to the investors' outlay at the threshold: financing is feasible iff $A \geq \bar{A} = I - \mathcal{P}_0$. The side condition $L_0 \leq p_H(R - B/\Delta p)$ says that, when there is demand, investors also prefer continuation (their maximal continuation claim beats L_0): there is no conflict over the continuation decision.

Answer (i), interpret (1). The left side of (1) is the maximal expected pledgeable income (all of L_0 on liquidation, $R - B/\Delta p$ on success after continuation); (1) defines $\bar{A}$ as the cash at which it just covers the investors' outlay $I - A$.

Step 2 — (i) Utility and optimal contracts. Competitive investors break even, so the entrepreneur receives the expected value created minus the investment, net of her own cash: $U_b^* = [(1-x)L_0 + xp_H R] - (I - A) - A$, i.e.

$$U_b^* = (1-x)L_0 + xp_H R - I. \tag{90}$$

A contract (r_b, R_b), liquidating iff there is no demand, is optimal iff

$$(1-x)(L_0 - r_b) + xp_H(R - R_b) = I - A, \tag{91}$$

$$R_b \geq \frac{B}{\Delta p}, \quad r_b \geq 0. \tag{92}$$

For $A > \bar{A}$ there is a continuum (the slack $A - \bar{A}$ can be paid as r_b or as extra R_b). At $A = \bar{A}$, subtracting (1), $I - \bar{A} = (1-x)L_0 + xp_H(R - B/\Delta p)$, from (91),

$$-(1-x)\, r_b - xp_H\Big(R_b - \frac{B}{\Delta p}\Big) = 0,$$

and both terms are ≤ 0 by (92), so each is zero: $r_b = 0$ and $R_b = B/\Delta p$. This is a debt contract with face value $D = R - B/\Delta p \geq L_0$ (since $L_0 \leq p_H(R - B/\Delta p) \leq D$). On liquidation the firm defaults and investors take L_0; on success they are repaid D.

Answer (i), utility and contracts. $U_b^* = (1-x)L_0 + xp_H R - I$. Optimal contracts: any (r_b, R_b) with $r_b \geq 0$, $R_b \geq B/\Delta p$ and binding (91); at $A = \bar{A}$ the unique one is debt with face $D = R - B/\Delta p$ ($r_b = 0$, $R_b = B/\Delta p$).

Step 3 — (ii) Interpreting (2). With maintenance level L (and effort), the NPV is

$$NPV(L) = (1-x)L + x\big[p_H + \tau(L)\big]R - I. \tag{93}$$

Differentiating, $NPV'(L) = (1-x) + x\tau'(L)R$ and $NPV''(L) = x\tau''(L)R < 0$, so it is strictly concave, and its first-order condition is $(1-x) + x\tau'(L)R = 0$, i.e. (dividing by x) $\tau'(L)R = -(1-x)/x$, which (2) says holds at L_0. So L_0 is the unique efficient maintenance level: one more unit of liquidation value, worth $1-x$ in expectation, costs exactly $x|\tau'|R$ of expected continuation income. $\tau(L_0) = 0$ normalises the success probabilities so that $NPV(L_0) = U_b^*$.

Answer (ii), interpret (2). (2) says L_0 is the efficient maintenance level: it maximises the NPV, and at L_0 the success probabilities are those of part (i).

Step 4 — (ii) The entrepreneur's maintenance choice. Under the step contract ($r_b(L) = r_b$ if $L \geq L_0$, 0 otherwise) and anticipating effort at stage 4 (her (IC$_b$), $(\Delta p)R_b \geq B$, does not depend on L), she chooses L to maximise

$$U(L) = (1-x)\, r_b(L) + x\big[p_H + \tau(L)\big]R_b. \tag{94}$$

For $L < L_0$ she gets 0 on liquidation and τ is decreasing, so the best deviation is $L = 0$; any $L > L_0$ lowers τ for no gain. So L_0 is chosen iff $U(L_0) \geq U(0)$:

$$\begin{aligned} (1-x)r_b + x\big[p_H + \tau(L_0)\big]R_b &\geq (1-x)(0) + x\big[p_H + \tau(0)\big]R_b \\ (1-x)r_b + xp_H R_b &\geq xp_H R_b + x\tau(0)R_b \qquad (\tau(L_0) = 0) \\ (1-x)\, r_b &\geq x\,\tau(0)\, R_b. \end{aligned} \tag{NS}$$

A debt contract ($r_b = 0$) violates (NS): the entrepreneur would run the assets down to gamble on continuation.

Step 5 — (ii) Minimum A. Financing at the efficient L_0 requires the investors' breakeven (91) with (IC$_b$) and (NS). The entrepreneur's expected rent $(1-x)r_b + xp_H R_b$ is minimised by (NS) at equality (any larger r_b is pure rent); substituting $(1-x)r_b = x\tau(0)R_b$, the rent is $x\tau(0)R_b + xp_H R_b = x\big[p_H + \tau(0)\big]R_b$, increasing in R_b, so $R_b = B/\Delta p$, the lowest value (IC$_b$) allows. Then

$$r_b = \frac{x\,\tau(0)}{1-x}\,\frac{B}{\Delta p}. \tag{95}$$

Substituting $R_b = B/\Delta p$ and (95) into (91) at equality,

$$\begin{aligned} I - A^* &= (1-x)(L_0 - r_b) + xp_H\Big(R - \frac{B}{\Delta p}\Big) \\ A^* &= \Big[I - (1-x)L_0 - xp_H\Big(R - \frac{B}{\Delta p}\Big)\Big] + (1-x)\, r_b \\ &= \bar{A} + (1-x)\, r_b = \bar{A} + x\,\tau(0)\,\frac{B}{\Delta p}, \end{aligned} \tag{96}$$

using (1) for the bracket and (95) for $(1-x)r_b$. For $A \geq A^*$, asset substitution is innocuous: the firm is financed, maintains L_0 and the entrepreneur gets U_b^*. For $A \in [\bar{A}, A^*)$ the efficient contract is not fundable. Illustration: $x = 0.6$, $p_H = 0.8$, $p_L = 0.4$, $B = 1$, $R = 10$, $L_0 = 3$, $I = 5.5$, $\tau(L) = (9 - L^2)/90$ (which satisfies (2): $\tau'(3)R = -(6/90)(10) = -2/3 = -(0.4/0.6)$): $U_b^* = 0.4(3) + 0.6(0.8)(10) - 5.5 = 1.2 + 4.8 - 5.5 = 0.5$, $B/\Delta p = 2.5$, $\bar{A} = 5.5 - 1.2 - 0.6(0.8)(10 - 2.5) = 5.5 - 1.2 - 3.6 = 0.7$, $\tau(0) = 9/90 = 0.1$, $r_b = (0.6)(0.1)(2.5)/(0.4) = 0.375$, $A^* = 0.7 + (0.4)(0.375) = 0.85$.

Step 6 — (ii) The optimal contract. It is debt plus a reward for good maintenance: the entrepreneur keeps $B/\Delta p$ on success and receives r_b on liquidation, but only if the assets are in good condition (a severance payment conditional on $L \geq L_0$). Giving her a stake in the liquidation value counters her bias toward the continuation state, where her debt-like claim pays.

Answer (ii), minimum A and contract. $A^* = \bar{A} + x\,\tau(0)\,B/\Delta p$. The contract is debt ($R_b = B/\Delta p$ on success) plus a payment $r_b = x\tau(0)B/[(1-x)\Delta p]$ on liquidation, made only if the assets were maintained at L_0.

Intuition. A claim that pays only in success makes the entrepreneur care only about continuation profit, so she cuts maintenance to raise the success probability: risk shifting at the expense of investors, who own the liquidation value. To stop it, she must share in the liquidation value. That share is an extra rent, so it lowers pledgeable income and raises the equity needed, by the expected gain from the most aggressive substitution, $x\,\tau(0)B/\Delta p$.

Common mistakes.

- Reading "liquidation … (hence, with probability x)" literally; liquidation happens when there is no demand, with probability $1-x$ (Figure 7.16).
- Keeping the debt contract of part (i) in part (ii). With $r_b = 0$ the no-substitution constraint fails and she runs the assets down to $L = 0$.
- Paying r_b on liquidation unconditionally. A payment that does not depend on L leaves her maintenance incentive unchanged; it must be conditional on $L \geq L_0$.

Book answer. Matches the book's answer (p. 602): the same interpretation of (1), U_b^*, the debt contract at $A = \bar{A}$, the step-function contract with $R_b = B/\Delta p$, the no-substitution inequality $(1-x)r_b + xp_H B/\Delta p \geq x[p_H + \tau(0)]B/\Delta p$ and the threshold $I - A^* = (1-x)(L_0 - r_b) + xp_H(R - B/\Delta p)$. The statement's "with probability x" for liquidation looks like a typo for $1-x$.

Ref: Tirole §3.2.2, pp. 116–117; §7.2.3 "Effort and Risk Taking", p. 314; answer p. 602.

15. A borrower-friendly bankruptcy court — Tirole Exercise 4.9, pp. 190–191

Statement.

Source: Tirole, Ch. 4, Exercise 4.9, pp. 190–191 (PDF 203) · Answer in book: yes, p. 586

Exercise 4.9 (borrower-friendly bankruptcy court). Consider the timing described in Figure 4.7.

Figure 4.7 (timeline, left to right; p. 191, PDF 204)

Date 0	Date 1		Date 2
Investment, contract. Entrepreneur has cash A, must borrow $I - A$.	Short-term profit realized. Liquidate → Liquidation value L (known); or Continue →	Moral hazard: Effort decision (high or low).	Final profit: Success (profit R) or failure (profit 0)?

The project, if financed, yields random and verifiable short-term profit $r \in [0, \bar{r}]$ (with a continuous density and *ex ante* mean $E[r]$). After r is realized and cashed in, the firm either liquidates (sells its assets), yielding some known liquidation value $L > 0$, or continues. Note that (the random) r and (the deterministic) L are not subject to moral hazard. If the firm continues, its prospects improve with r (so r is "good news" about the future). Namely, the probability of success is $p_H(r)$ if the entrepreneur works between dates 1 and 2 and $p_L(r)$ if the entrepreneur shirks. Assume that $p_H' > 0$, $p_L' > 0$, and

$$p_H(r) - p_L(r) \equiv \Delta p$$

is independent of r (so shirking reduces the probability of success by a fixed amount independent of prospects). As usual, one will want to induce the entrepreneur to work if continuation obtains. It is convenient to use the notation

$$\rho_1(r) \equiv p_H(r)R \quad \text{and} \quad \rho_0(r) \equiv p_H(r)\left[R - \frac{B}{\Delta p}\right].$$

Investors are competitive and demand an expected rate of return equal to 0. Assume

$$\rho_1(r) > L \quad \text{for all } r \qquad (1)$$

and

$$E[r] + L > I - A > E[r + \rho_0(r)]. \qquad (2)$$

(i) Argue informally that, in the optimal contract for the borrower, the short-term profit and the liquidation value (if the firm is liquidated) ought to be given to investors.

Argue that, in the case of continuation, $R_b = B/\Delta p$. (If you are unable to show why, take this fact for granted in the rest of the question.)

Interpret conditions (1) and (2).

(ii) Write the borrower's optimization program.

Assume (without loss of generality) that the firm continues if and only if $r \geqslant r^*$ for some $r^* \in (0, r)$. Exhibit the equation defining r^*.

(iii) Argue that this optimal contract can be implemented using, inter alia, a short-term debt contract at level $d = r^*$. Interpret "liquidation" as a "bankruptcy."

How does short-term debt vary with the borrower's initial equity? Explain.

(iv) Suppose that, when the decision to liquidate is taken, the firm must go to a bankruptcy court. The judge mechanically splits the bankruptcy proceeds L equally between investors and the borrower.

Define $\hat{r}$ by

$$\rho_0(\hat{r}) \equiv \tfrac{1}{2}L.$$

Assume first that

$$r^* > \hat{r}$$

(where r^* is the value found in question (ii)).

Show that the borrower-friendly court actually prevents the borrower from having access to financing. (Note: a diagram may help.)

(v) Continuing on question (iv), show that when

$$r^* < \hat{r},$$

the borrower-friendly court either prevents financing or increases the probability of bankruptcy, and in all cases hurts the borrower and not the lenders.

Context needed to solve it The statement defines its own model; "as usual" refers to the fixed-investment model's moral-hazard setup (private benefit B, limited liability, incentive constraint), copied here.

From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the

entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\mathrm{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

What is asked. (i) Argue that investors should get the short-term profit and the liquidation value, that the entrepreneur's continuation reward should be exactly $B/\Delta p$, and explain assumptions (1) and (2). (ii) Write the borrower's optimisation problem and the equation for the continuation cutoff r^*. (iii) Show that short-term debt $d = r^*$ implements the contract (default = liquidation) and how d depends on her cash A. (iv) A court splits the liquidation proceeds equally with the borrower; show that if $r^* > \hat{r}$ she loses access to financing. (v) If $r^* < \hat{r}$, show that the court either blocks financing or makes bankruptcy more likely, hurting the borrower but not lenders.

Approach. Lecture Topic 4B, bankruptcy design. The fixed-investment model with an intermediate verifiable profit r and a liquidation option. Everything not subject to moral hazard (r, L) is *pledgeable income* at no incentive cost; continuation income is pledgeable only up to $\rho_0(r)$. When the project cannot be financed with continuation in every state, the borrower buys funding by accepting inefficient liquidation where it raises pledgeable income most cheaply, which gives a cutoff rule. We write the Lagrangian, maximise it state by state, and use the investors' breakeven condition to find r^*; the court changes investors' payoff from liquidation from L to $L/2$.

Notation.

- $r \in [0, \bar{r}]$: verifiable short-term profit, density $f > 0$, c.d.f. F; $L > 0$: liquidation value.
- $p_H(r)$, $p_L(r) = p_H(r) - \Delta p$: success probabilities ($p_H' > 0$); R: final profit in success; B: private benefit.
- $\rho_1(r) = p_H(r)R$: continuation value; $\rho_0(r) = p_H(r)(R - B/\Delta p)$: pledgeable continuation value; both increasing in r.
- $\ell(r) \in \{0, 1\}$: liquidation policy; $R_b(r)$: reward on continuation and success; r^*: optimal cutoff; d: short-term debt.
- $\hat{r}$: $\rho_0(\hat{r}) = L/2$; r^{**}: cutoff under the borrower-friendly court.
- $P(c)$, $P_c(c)$: investors' expected income with cutoff c under an enforcing / borrower-friendly court.

Assumption. The court's equal split of L cannot be undone by contract.

Step 1 — (i) What goes to investors; $R_b = B/\Delta p$; conditions (1)–(2). r and L are not subject to moral hazard, so giving them to investors raises pledgeable income without hurting incentives. On continuation, (IC$_b$) is $[p_H(r) - p_L(r)]R_b \geq B$, i.e. $R_b \geq B/\Delta p$; any excess is wasted pledgeable income, which would force more liquidation (proved in Step 2). Condition (1), $\rho_1(r) > L$, says continuation is always efficient. Condition (2): always liquidating raises enough, $E[r]+L > I-A$, but always continuing does not, $E[r+\rho_0(r)] < I-A$. Financing therefore needs inefficient liquidation in some states.

Answer (i). r and L go to investors because pledging them costs no incentives; $R_b = B/\Delta p$ because any larger reward wastes pledgeable income. (1): continuation is always efficient. (2): liquidating always would raise enough funds, continuing always would not, so some inefficient liquidation is the price of financing.

Step 2 — (ii) Programme and the cutoff. With binding breakeven the borrower's utility is the NPV:

$$\max_{\ell(\cdot),R_b(\cdot)} \; E[r] + \int_0^{\bar r} \big[\ell(r)L + (1-\ell(r))\rho_1(r)\big] f(r)\, dr - I \tag{97}$$

$$\text{s.t.} \;\; E[r] + \int_0^{\bar r} \big[\ell(r)L + (1-\ell(r))p_H(r)(R - R_b(r))\big] f(r)\, dr \;\geq\; I - A, \tag{98}$$

$$R_b(r) \geq B/\Delta p. \tag{99}$$

Let μ be the multiplier of (98). $\mu > 0$: the unconstrained optimum, $\ell \equiv 0$ by (1), violates (98) by (2). R_b does not enter (97) and lowers (98), so $R_b(r) = B/\Delta p$ and the investors' continuation income is $p_H(r)(R - B/\Delta p) = \rho_0(r)$. The Lagrangian is then

$$\mathcal{L} = (1+\mu)E[r] - I - \mu(I-A) + \int_0^{\bar r} \Big\{\ell(r)\,(1+\mu)L + \big(1-\ell(r)\big)\,[\rho_1(r) + \mu\rho_0(r)]\Big\} f(r)\, dr,$$

which is linear in $\ell(r)$ state by state. Maximising the integrand pointwise: liquidate iff $(1+\mu)L > \rho_1(r) + \mu\rho_0(r)$, i.e.

$$\mu[L - \rho_0(r)] \;>\; \rho_1(r) - L. \tag{100}$$

The left-hand side decreases and the right-hand side increases in r, so the policy is a cutoff: liquidate iff $r < r^*$. At the cutoff (100) holds with equality and $\rho_1(r^*) > L$, so $\mu[L - \rho_0(r^*)] = \rho_1(r^*) - L > 0$ and, with $\mu > 0$, $\rho_0(r^*) < L$.

Step 3 — (ii) The equation for r^*. With cutoff c, investors receive

$$P(c) = E[r] + L\,F(c) + \int_c^{\bar r} \rho_0(r) f(r)\, dr, \tag{101}$$

and the NPV, $E[r] + LF(c) + \int_c^{\bar r} \rho_1 f\, dr - I$, falls with c by (1). So r^* is the smallest root of

$$E[r] + L\,F(r^*) + \int_{r^*}^{\bar r} \rho_0(r) f(r)\, dr = I - A. \tag{102}$$

By (2), $P(0) = E[r+\rho_0(r)] < I - A < E[r] + L = P(\bar r)$, so by continuity $r^* \in (0, \bar r)$ (the statement's "$(0, r)$" should read $(0, \bar r)$).

Answer (ii). $\max_{\ell(\cdot),R_b(\cdot)}$ (97) s.t. (98)–(99); continue iff $r \geq r^*$, with r^* the smallest root of $E[r] + LF(r^*) + \int_{r^*}^{\bar r} \rho_0(r) f(r)\, dr = I - A$.

Step 4 — (iii) Short-term debt and the effect of A. Implementation: short-term debt $d = r^*$ due at date 1, any excess $r - d$ also paid to investors (for example as a payout to outside shareholders), and a long-term claim of $R - B/\Delta p$ on success. If $r \geq d$ the debt is paid and the

firm continues; if $r < d$ it defaults, creditors take control and liquidate ("bankruptcy"). They do not renegotiate: for $r < r^*$, $\rho_0(r) < \rho_0(r^*) < L$, and the borrower has no cash to bribe them. By Leibniz's rule applied to (101),

$$\frac{dP}{dc} = L\,f(c) - \rho_0(c)f(c) = \big[L - \rho_0(c)\big]f(c).$$

Differentiating (102), $P(r^*) = I - A$, with respect to A:

$$\begin{aligned}\big[L - \rho_0(r^*)\big]f(r^*)\,dr^* &= -dA \\ \frac{dr^*}{dA} &= -\frac{1}{[L - \rho_0(r^*)]f(r^*)} \;< 0. \end{aligned} \tag{103}$$

Short-term debt falls with initial equity: a weaker balance sheet needs more pledgeable income, so the firm must accept bankruptcy in more states.

Answer (iii). Short-term debt $d = r^*$ (default and liquidation iff $r < d$), plus a long-term claim $R - B/\Delta p$ on success, implements the optimum; $dd/dA = dr^*/dA = -1/\{[L - \rho_0(r^*)]f(r^*)\} < 0$.

Step 5 — (iv) Borrower-friendly court with $r^* > \hat{r}$. Liquidation now yields investors only $L/2$. For any policy, investors' income in state r (besides r) is at most $\max\{L/2, \rho_0(r)\}$. Compare with the enforcing-court contract state by state. For $r < r^*$, $\max\{L/2, \rho_0(r)\} < L$, since $L/2 < L$ and $\rho_0(r) < \rho_0(r^*) < L$. For $r \geq r^* > \hat{r}$, $\rho_0(r) > \rho_0(\hat{r}) = L/2$, so the maximum is $\rho_0(r)$, the same as before. Hence

$$E[r] + \int_0^{\bar{r}} \max\Big\{\frac{L}{2}, \rho_0(r)\Big\} f(r)\,dr \;<\; E[r] + LF(r^*) + \int_{r^*}^{\bar{r}} \rho_0 f\,dr = I - A, \tag{104}$$

using (102). No policy lets investors break even: the borrower loses access to financing.

Answer (iv). With $r^* > \hat{r}$, the investors' maximal expected income under the court is below $I - A$: no contract is financed.

Step 6 — (v) Court with $r^* < \hat{r}$. Cutoff policies are again optimal (same argument as (100) with $L/2$ for investors). Investors' income with cutoff c is

$$P_c(c) = E[r] + \frac{L}{2}F(c) + \int_c^{\bar{r}} \rho_0(r) f(r)\,dr, \qquad P_c'(c) = \Big[\frac{L}{2} - \rho_0(c)\Big] f(c), \tag{105}$$

which rises up to $\hat{r}$ and falls after it, so its maximum is $P_c(\hat{r})$. If $P_c(\hat{r}) < I - A$, there is no financing. Otherwise the borrower picks the smallest root r^{**} of $P_c(c) = I - A$. For $0 < c \leq r^*$, $P_c(c) < P(c) \leq I - A$ by (101) and (102), so $r^{**} \in (r^*, \hat{r}]$: bankruptcy is more likely. Lenders break even in either regime, so they are not hurt. The borrower gets the NPV, which falls with the cutoff (Step 3), so $NPV(r^{**}) < NPV(r^*)$: she is hurt. More directly, with an enforcing court she could have written the $L/2$ split into her contract; the friendly court only removes options.

Answer (v). If $P_c(\hat{r}) < I - A$ there is no financing; otherwise the cutoff rises to $r^{**} \in (r^*, \hat{r}]$ (more bankruptcy). Lenders still break even; the borrower's utility, the NPV, falls.

Step 7 — Numerical illustration. $r \sim U[0,1]$, $p_H(r) = 0.5 + 0.4r$, $\Delta p = 0.2$, $R = 10$, $B = 1.2$ ($R - B/\Delta p = 4$), $L = 4.5$. Then $\rho_1(r) = 5 + 4r > L$, $\rho_0(r) = 2 + 1.6r$, $E[r] = 0.5$, $P(0) = E[r + \rho_0] = 3.3$, $P(1) = E[r] + L = 5$, so (2) holds for $I - A \in (3.3, 5)$; $\hat{r} = (2.25 - 2)/1.6 = 0.15625$, and the court's maximal income is $P_c(\hat{r}) = 3.319531$. With $f = 1$ on $[0,1]$, (101) and (105) integrate to

$$\begin{aligned} P(c) &= 0.5 + 4.5c + 2(1 - c) + 0.8(1 - c^2) = 3.3 + 2.5c - 0.8c^2, \\ P_c(c) &= 0.5 + 2.25c + 2(1 - c) + 0.8(1 - c^2) = 3.3 + 0.25c - 0.8c^2, \end{aligned}$$

so $P_c(\hat{r}) = 3.3 + 0.25(0.15625) - 0.8(0.15625)^2 = 3.3 + 0.0390625 - 0.0195313 = 3.319531$. (iv) $I - A = 4$: $P(r^*) = 4$ gives $0.8r^{*2} - 2.5r^* + 0.7 = 0$, smallest root $r^* = (2.5 - \sqrt{6.25 - 2.24})/1.6 = 0.3109 > \hat{r}$;

$\rho_0(r^*) = 2 + 1.6(0.3109) = 2.4975 < L$; and $3.319531 < 4$: no financing. (v) $I - A = 3.31$: $2.5r^* - 0.8r^{*2} = 0.01$ gives $r^* = 0.0040 < \hat{r}$; under the court $0.25r^{**} - 0.8r^{**2} = 0.01$ gives $r^{**} = (0.25 - \sqrt{0.0625 - 0.032})/1.6 = 0.0471$, and the NPV loss is $\int_{r^*}^{r^{**}} (\rho_1 - L)\, dr = \int_{r^*}^{r^{**}} (0.5 + 4r)\, dr = 0.0260$.

Intuition. Liquidation is the investors' collateral: it is what they get when the firm cannot pay. A court that hands half of it to the borrower lowers the value of default to creditors. To keep them whole the contract must liquidate more often, which destroys value, or financing breaks down. Lenders still break even, so the whole cost of the borrower-friendly rule falls on the borrower it was meant to protect.

Common mistakes.

- Liquidating whenever $\rho_0(r) < L$. Liquidation is always inefficient here (1); it is used only as far as needed to break even, so the cutoff comes from the binding breakeven, not from $\rho_0 = L$.
- Concluding that a court that hands the borrower half of L helps her. Lenders must still break even, so the borrower pays for the transfer through more liquidation or no financing.
- Signing dr^*/dA wrong: more cash means less pledgeable income is needed, so fewer liquidations and *less* short-term debt.

Book answer. Matches the book's answer (pp. 586–587): the reasons for pledging r and L, $R_b = B/\Delta p$, the programme and r^* as the lowest root of breakeven with $L \geq \rho_0(r^*)$, the debt implementation $d = r^*$, $dr^*/dA < 0$, and (iv)–(v) as read from the book's Figures 2–3 (bankruptcy on $[0, r^{**}]$ with $r^* < r^{**} \leq \hat{r}$, lenders not hurt, borrower hurt). The statement's "$(0, r)$" is a typo for $(0, \bar{r})$.

Ref: Tirole §3.2.2, pp. 116–117; Exercise 4.9, pp. 190–191; answer pp. 586–587.

16. Default as insurance: the default cutoff — Original exercise

Statement.

Origin: original exercise on default as insurance (Indarte 2023), built on Prof. Ricca's Topic 4B slides 41–44 (Aula/Topic4_B.pdf) · Answer: in the solutions file

A household enters next period owing a debt payment $D = 12$. It holds liquid assets $A = 5$ (so $A < D$) that lenders cannot seize, and it receives an income Y that is uniformly distributed on $[\underline{Y}, \bar{Y}] = [5, 25]$, so that $F(a) = \Pr[Y \leq a] = (a - 5)/20$ for $a \in [5, 25]$. The household is risk-averse with utility of consumption $u(C) = \sqrt{C}$ (so $u(0) = 0$, $u' > 0$, $u'' < 0$). If it defaults, a court lets it keep the minimum income level $E = 4$ (note $E < \underline{Y}$); creditors receive the garnished income $Y - E$, up to the amount due D; and the household bears a non-pecuniary cost of default $\sigma = 1$, measured in utility units and subtracted from u. If it repays, it consumes $A + Y - D$; if it defaults, it consumes $A + E$. There is no later period and no other decision.

(a) For which incomes is the household unable to repay (cash-flow default)? What is the probability of that event?

(b) For incomes at which it can repay, write the condition under which it nevertheless chooses to default (strategic default). Show that $u(A + E) - \sigma > 0$ and find the cutoff Y^* defined by $u(A + Y^* - D) = u(A + E) - \sigma$. Check that $D - A < Y^* < D + E$, and explain why the second inequality matters for what the creditors receive in default.

(c) Compute the default probability $p = F(Y^*)$ and split it into its cash-flow and strategic parts.

(d) For $Y = 6$, $Y = 9$ and $Y = 20$: does the household repay or default? Give its consumption, its utility (net of σ if it defaults) and the creditors' receipts.

(e) Compute the creditors' expected receipts conditional on default, $E[\,Y - E \mid Y < Y^*\,]$, and the household's consumption floor. In two sentences, explain in what sense default works as insurance here.

Context needed to solve it The model is the one on Prof. Ricca's Topic 4B slides 41–44 (Aula/Topic4_B.pdf), based on Indarte (2023), "Moral hazard versus liquidity in household bankruptcy", *Journal of Finance* 78(5).

From the slides (41–44): In the next period, an individual has a debt payment D; liquid assets $A < D$, which lenders cannot seize; and income $Y \in [\underline{Y}, \bar{Y}]$ distributed according to a cdf $F(\cdot)$, $\Pr[Y \leq a] = F(a)$. The individual is risk-averse with utility $u(C)$, where $u(0) = 0$, $u' > 0$ and $u'' < 0$. In case of default, courts determine that the individual retains a minimum income level E, where $E < \underline{Y}$; creditors receive $Y - E$ (wage garnishment) up to the amount due D. Default entails a non-pecuniary cost $\sigma > 0$. If $A + Y < D$, individuals cannot repay (cash-flow default). If $A + Y \geq D$, individuals can choose not to repay (strategic default): if they repay, they consume $A + Y - D$; if they default, they consume $A + E$. When $A + Y \geq D$ (or $Y \geq D - A$), the individual defaults whenever

$$u(A + Y - D) < u(A + E) - \sigma.$$

As long as $u(A + E) - \sigma > 0$, there is a $Y^* > D - A$ such that $u(A + Y^* - D) = u(A + E) - \sigma$, and the individual defaults whenever $Y < Y^*$. Default acts as insurance whenever income drops below a given level: it guarantees a consumption of $A + E$ when income is too low. Since the individual is risk-averse, default can increase welfare by making the individual bear less income risk. The default probability is $p = \Pr[Y \leq Y^*] = F(Y^*)$.

Uniform distribution on $[a, b]$: density $1/(b - a)$, $E[Y \mid Y < y] = (a + y)/2$ for $y \in [a, b]$.

Data (full statement: list, exercise 16; built on Topic 4B slides 41–44, Indarte 2023). Debt payment $D = 12$; unseizable liquid assets $A = 5$; income $Y \sim U[5, 25]$, so $F(a) = (a - 5)/20$; utility $u(C) = \sqrt{C}$; in default the household keeps $E = 4$, creditors take $Y - E$ up to D, and the household bears a utility cost $\sigma = 1$.

What is asked. (a) Find the incomes at which the household simply does not have the cash to pay, and their probability. (b) Among the incomes at which it could pay, find where it prefers not to: derive the cutoff Y^* and check that it sits between $D - A$ and $D + E$. (c) Turn the cutoff into a default probability and split it into "cannot pay" and "will not pay". (d) Apply the rule to three concrete incomes. (e) Find what creditors expect to collect when default happens, and say why default is a form of insurance.

Approach. This is the lecture's *default as insurance* model (slides 41–44). In every income state the household compares two numbers: the utility of repaying, $u(A + Y - D)$, and the utility of defaulting, $u(A + E) - \sigma$. The first rises with Y and the second does not depend on Y, so there is a single income Y^* below which default wins. The default probability is then just $F(Y^*)$.

Notation.

- $D = 12$: debt payment due; $A = 5$: liquid assets creditors cannot seize; $Y \in [5, 25]$: income, cdf F.
- $E = 4$: income the court lets the household keep in default; $Y - E$: garnished income (capped at D).
- $\sigma = 1$: non-pecuniary cost of default, in utility units; $u(C) = \sqrt{C}$: utility of consumption $C \geq 0$.
- $c_r \equiv A + Y - D$: consumption if the household repays; $c_d \equiv A + E = 9$: consumption if it defaults.
- Y^*: default cutoff; $p = F(Y^*)$: default probability.

Step 1 — (a) Cash-flow default. The household can pay only if its liquid assets plus income cover the debt, so it is forced into default when

$$\begin{aligned} A + Y &< D \\ Y &< D - A \\ Y &< 12 - 5 = 7. \end{aligned} \tag{106}$$

With the uniform cdf,

$$\Pr[Y < 7] = F(7) = \frac{7 - 5}{20} = \frac{2}{20} = 0.1. \tag{107}$$

Answer (a). Cash-flow default for $Y < 7$, probability 0.1.

Step 2 — (b) The strategic-default condition and the cutoff. For $Y \geq 7$ the household can pay and picks the option with higher utility. Repaying gives

$$u(A + Y - D) = \sqrt{5 + Y - 12} = \sqrt{Y - 7}, \tag{108}$$

and defaulting gives

$$u(A + E) - \sigma = \sqrt{5 + 4} - 1 = \sqrt{9} - 1 = 3 - 1 = 2 > 0. \tag{109}$$

So it defaults strategically when $\sqrt{Y - 7} < 2$. The cutoff equates the two:

$$\begin{aligned} \sqrt{Y^* - 7} &= 2 \\ Y^* - 7 &= 2^2 = 4 \\ Y^* &= 11. \end{aligned} \tag{110}$$

Bounds: $D - A = 7 < 11$, and $D + E = 12 + 4 = 16 > 11$. The first says that some households who *can* pay do not. The second matters for creditors: in every default state $Y < 11$, so the garnishment is $Y - E < 11 - 4 = 7 < 12 = D$; the cap at D never binds, creditors take all of $Y - E$, and the household keeps exactly $A + E$, as the slides assume. (If $Y - E$ exceeded D, "default" would leave the household with $A + Y - D$, the same as repaying, minus σ, and it would never choose it.)

Answer (b). Strategic default iff $7 \leq Y$ and $\sqrt{Y - 7} < 2$; $u(A + E) - \sigma = 2 > 0$; $Y^* = 11$, with $7 < 11 < 16$.

Step 3 — (c) Default probability. The household defaults for every $Y < Y^*$, whatever the reason:

$$p = F(Y^*) = F(11) = \frac{11 - 5}{20} = \frac{6}{20} = 0.3. \tag{111}$$

Of this, $F(7) = 0.1$ is cash-flow default (107) and the rest is strategic:

$$F(11) - F(7) = 0.3 - 0.1 = 0.2. \tag{112}$$

Answer (c). $p = 0.3$: 0.1 cash-flow default and 0.2 strategic default.

Step 4 — (d) Three incomes. We apply (106) and (110).

- $Y = 6$: $A + Y = 5 + 6 = 11 < 12$, so it cannot pay (cash-flow default). It consumes $A + E = 9$, utility $\sqrt{9} - 1 = 2$; creditors receive $Y - E = 6 - 4 = 2$.
- $Y = 9$: $A + Y = 14 \geq 12$, so it can pay. Repaying gives $C = 5 + 9 - 12 = 2$ and $u = \sqrt{2} = 1.414214 < 2$, so it defaults strategically: $C = 9$, utility 2; creditors receive $9 - 4 = 5$.
- $Y = 20$: repaying gives $C = 5 + 20 - 12 = 13$ and $u = \sqrt{13} = 3.605551 > 2$, so it repays; creditors receive $D = 12$.

Answer (d). $Y = 6$: cash-flow default ($C = 9$, $u = 2$, creditors 2). $Y = 9$: strategic default ($C = 9$, $u = 2$, creditors 5). $Y = 20$: repays ($C = 13$, $u = 3.605551$, creditors 12).

Step 5 — (e) Recovery in default and the consumption floor. Conditional on default, Y is uniform on $[5, 11)$, so

$$\begin{aligned} E[Y \mid Y < 11] &= \frac{5 + 11}{2} = 8, \\ E[Y - E \mid Y < 11] &= 8 - 4 = 4. \end{aligned} \tag{113}$$

The household's consumption never falls below $A + E = 9$: in all states $Y < 11$ it consumes 9.

Answer (e). Creditors expect to collect 4 per defaulted loan (out of 12 due). The consumption floor is $A + E = 9$. Default insures: it swaps the low and risky consumption $A + Y - D$ of bad income states for a fixed $A + E$, paid for by the creditors' losses (and by σ); a risk-averse household values that.

Intuition. The household holds a put option on its debt. Exercising it costs σ and yields the consumption floor $A + E$, so it is exercised exactly when repayment would leave consumption below the level at which the floor, net of σ, is better. Two thirds of the defaults (0.2 of 0.3) are strategic: these households could pay but the insurance is worth more to them than the stigma.

Common mistakes.

- Calling all default "inability to pay". Here only $Y < 7$ is cash-flow default; on $[7, 11)$ default is a choice.
- Comparing consumption instead of utility: σ is a utility cost, so the comparison is $\sqrt{Y-7}$ versus $3 - 1$, not $Y - 7$ versus 9.
- Giving creditors D or 0 in default. They receive the garnishment $Y - E$, which is below D in every default state here.

Ref: Prof. Ricca, Topic 4B slides 41–44 (Aula/Topic4_B.pdf); Indarte (2023), *Journal of Finance* 78(5). No book exercise exists.

17. Default as insurance: comparative statics, the price of credit and ex ante welfare — Original exercise

Statement.

Origin: original exercise on default as insurance (Indarte 2023) and the ex ante cost of soft debt, built on Prof. Ricca's Topic 4B slides 37–44 (Aula/Topic4_B.pdf) · Answer: in the solutions file

Baseline. A household owes a debt payment D next period. It holds liquid assets $A = 5$ that lenders cannot seize, and its income Y is uniform on $[5, 25]$, with cdf $F(a) = (a - 5)/20$ and density $F'(a) = 1/20$. Its utility of consumption is $u(C) = \sqrt{C}$. In default it keeps the minimum income $E = 4$ and consumes $A + E$; creditors receive $Y - E$, up to D; and the household bears a non-pecuniary cost σ (in utility units, subtracted from u). If it repays it consumes $A + Y - D$. It defaults if and only if $Y < Y^*$, where Y^* solves

$$u(A + Y^* - D) = u(A + E) - \sigma, \qquad Y^* \in (D - A,\ D + E),$$

and the default probability is $p = F(Y^*)$. In the baseline $D = 12$ and $\sigma = 1$, which gives $Y^* = 11$ and $p = 0.3$. Lenders are competitive and risk-neutral, the riskless rate is 0, and a lender values a claim at its expected repayment.

(a) Differentiate the cutoff equation implicitly and derive dY^*/dD, dY^*/dA, $dY^*/d\sigma$ and dY^*/dE for a general increasing and strictly concave u, with their signs. (Hint: first show that $\sigma > 0$ implies $A + Y^* - D < A + E$.) Evaluate each at the baseline, give the effect on the default probability, $dp/dx = F'(Y^*)\, dY^*/dx$, and check that your dp/dE matches the expression on slide 44. Explain why more unseizable liquid assets A *lower* the default probability.

(b) At the baseline ($D = 12$, $\sigma = 1$), compute the value Q of the lender's claim (its expected repayment) and the promised interest rate $D/Q - 1$ that a borrower pays if the amount lent is Q.

(c) The household must raise $L = 9.6$ at date 0 from these lenders; what it does with the money is the same in every regime and is ignored. Compare two legal regimes.

- Regime N (no default): courts enforce repayment in full in every state. This is feasible because $A + Y \geq 10 > 9.6$, so the debt is riskless and $D_N = 9.6$.
- Regime B (bankruptcy as above, $\sigma = 1$): lenders set the face value D_B so that $Q(D_B) = 9.6$. Show that $D_B = 12$.

For each regime compute the household's expected utility (net of σ in default states) and its certainty equivalent $u^{-1}(EU)$, and the expected consumption $E[C]$. Which regime does the household prefer ex ante?

(d) Now let $\sigma = 0.1$, everything else unchanged. Show that, with $c_r \equiv \left(\sqrt{A+E} - \sigma\right)^2$, the cutoff is $Y^* = D - A + c_r$ and the lender's value satisfies $40\,Q(D) = -D^2 + 42D + (c_r - 9)^2 - 1$ (as long as $5 \le Y^* \le 25$). Find D_B from $Q(D_B) = 9.6$, then Y^*, p, the promised rate $D_B/9.6 - 1$, the expected utility and its certainty equivalent. Compare with Regime N.

(e) Explain why bankruptcy raises ex ante welfare in (d) but lowers it in (c), although in both cases lenders price the anticipated default and the household's expected consumption is the same as in Regime N. *(Harder, numerical.)* Find, to two decimals, the cost $\bar{\sigma}$ at which the household is indifferent between Regime B and Regime N.

Context needed to solve it The model is the one on Prof. Ricca's Topic 4B slides 41–44 (Aula/Topic4_B.pdf), based on Indarte (2023); slides 37–40 give the ex ante side.

From the slides (41–44): In the next period, an individual has a debt payment D; liquid assets $A < D$, which lenders cannot seize; and income $Y \in [\underline{Y}, \bar{Y}]$ with cdf $F(\cdot)$. The individual is risk-averse with utility $u(C)$, $u(0) = 0$, $u' > 0$, $u'' < 0$. In case of default, courts determine that the individual retains a minimum income level $E < \underline{Y}$; creditors receive $Y - E$ (wage garnishment) up to the amount due D. Default entails a non-pecuniary cost $\sigma > 0$. If $A + Y < D$, individuals cannot repay (cash-flow default); if $A + Y \ge D$, they can choose not to repay (strategic default). If they repay, they consume $A + Y - D$; if they default, they consume $A + E$. The individual defaults whenever $Y < Y^*$, where $u(A + Y^* - D) = u(A + E) - \sigma$. Default acts as insurance: it guarantees a consumption of $A + E$ when income is too low, and since the individual is risk-averse, default can increase welfare by making the individual bear less income risk. The default probability is $p = F(Y^*)$. An increase in the generosity of debt relief E provides more insurance, but it also increases default rates (and thus interest rates in equilibrium):

$$\frac{\partial p}{\partial E} = F'(Y^*)\,\frac{u'(A+E)}{u'(A+Y^*-D)} > 0.$$

From the slides (39–40): By undermining incentives, the prospect of renegotiation can increase the default risk of debt contracts, leading to credit rationing and higher costs. The prospect of renegotiation can be ex ante inefficient; public policy should take into account costs and benefits.

Tools: implicit-function theorem, $dY^*/dx = -(\partial G/\partial x)/(\partial G/\partial Y^*)$ for $G(Y^*, x) = 0$; $\int y\,dy = y^2/2$, $\int (a+y)^{1/2} dy = \frac{2}{3}(a+y)^{3/2}$. A random consumption C_1 that has the same mean as C_2 and crosses it once from above (higher in low states, lower in high states) is less risky: every risk-averse household prefers it.

Data (full statement: list, exercise 17; built on Topic 4B slides 37–44). $A = 5$, $E = 4$, $Y \sim U[5, 25]$ ($F(a) = (a-5)/20$, $F' = 1/20$), $u(C) = \sqrt{C}$, default cost σ in utility; baseline $D = 12$, $\sigma = 1$, $Y^* = 11$, $p = 0.3$. Competitive risk-neutral lenders, riskless rate 0. The household must raise $L = 9.6$ in parts (c)–(e).

What is asked. (a) Find how the cutoff and the default probability move with the debt, the protected assets, the stigma cost and the exemption, first in general and then at the baseline. (b) Price the baseline claim: what is it worth to a lender, and what promised rate does that imply? (c) Compare, from the household's point of view before income is known, a world where debt must always be repaid with the baseline bankruptcy system, when in both the lender is paid fairly for lending 9.6. (d) Redo the bankruptcy side with a much smaller stigma cost. (e) Explain why the ranking flips, and find the stigma cost at which it flips.

Approach. Part (a) is the implicit-function theorem applied to the cutoff equation of slide 43; the sign of each derivative comes from concavity of u. Parts (b)–(e) add the lenders' side: a competitive lender prices *anticipated* default, so the household pays for its insurance through a higher face value. Because lenders break even and σ is a utility (not a resource) cost, the household's expected consumption is the same with and without bankruptcy. Bankruptcy then only reshapes consumption risk (a gain, by risk aversion) at the price of the deadweight σ in default states. Welfare depends on which is larger. This is the ex ante side of soft debt on slides 37–40.

Notation.

- $c_r \equiv A + Y^* - D$: consumption at the cutoff if the household repays; $c_d \equiv A + E$: consumption in default.
- $G(Y^*, x) \equiv u(A + Y^* - D) - u(A + E) + \sigma = 0$: the cutoff equation; $x \in \{D, A, \sigma, E\}$.

- $Q(D)$: lender's value of a claim with face D (expected repayment); $D/Q - 1$: promised rate.
- Regime N: no default, debt D_N; Regime B: bankruptcy, debt D_B with $Q(D_B) = L = 9.6$.
- EU: expected utility net of σ in default states; $CE = u^{-1}(EU) = EU^2$: certainty equivalent.

Step 1 — A sign lemma. All signs in (a) rest on the household consuming less at the cutoff when it repays than when it defaults. From $G = 0$ and $\sigma > 0$,

$$u(c_r) = u(c_d) - \sigma < u(c_d) \implies c_r < c_d \implies u'(c_r) > u'(c_d), \tag{114}$$

the first implication because u is increasing, the second because u is strictly concave.

Step 2 — (a) Implicit differentiation. The partial derivatives of G are

$$G_{Y^*} = u'(c_r), \quad G_D = -u'(c_r), \quad G_A = u'(c_r) - u'(c_d), \quad G_\sigma = 1, \quad G_E = -u'(c_d). \tag{115}$$

By the implicit-function theorem, $dY^*/dx = -G_x/G_{Y^*}$, so

$$\frac{dY^*}{dD} = -\frac{-u'(c_r)}{u'(c_r)} = 1 > 0, \tag{116}$$

$$\frac{dY^*}{dA} = -\frac{u'(c_r) - u'(c_d)}{u'(c_r)} = \frac{u'(c_d)}{u'(c_r)} - 1 < 0, \tag{117}$$

$$\frac{dY^*}{d\sigma} = -\frac{1}{u'(c_r)} < 0, \tag{118}$$

$$\frac{dY^*}{dE} = -\frac{-u'(c_d)}{u'(c_r)} = \frac{u'(c_d)}{u'(c_r)} > 0, \tag{119}$$

where the sign of (117) uses $u'(c_d)/u'(c_r) < 1$ from (114).

Step 3 — Baseline values. At the baseline $c_r = A + Y^* - D = 5 + 11 - 12 = 4$ and $c_d = 5 + 4 = 9$. With $u'(C) = 1/(2\sqrt{C})$:

$$u'(4) = \frac{1}{2 \cdot 2} = 0.25, \qquad u'(9) = \frac{1}{2 \cdot 3} = \frac{1}{6}, \qquad \frac{u'(9)}{u'(4)} = \frac{1/6}{1/4} = \frac{2}{3}. \tag{120}$$

Substituting in (116)–(119):

$$\frac{dY^*}{dD} = 1, \quad \frac{dY^*}{dA} = \frac{2}{3} - 1 = -\frac{1}{3}, \quad \frac{dY^*}{d\sigma} = -\frac{1}{0.25} = -4, \quad \frac{dY^*}{dE} = \frac{2}{3}.$$

Multiplying by $F'(Y^*) = 1/20$ gives the effects on p:

$$\frac{dp}{dD} = \frac{1}{20} = 0.05, \quad \frac{dp}{dA} = -\frac{1}{3 \cdot 20} = -\frac{1}{60}, \quad \frac{dp}{d\sigma} = -\frac{4}{20} = -0.2, \quad \frac{dp}{dE} = \frac{2}{3 \cdot 20} = \frac{1}{30}. \tag{121}$$

Slide 44's formula is $\partial p/\partial E = F'(Y^*)\, u'(A + E)/u'(A + Y^* - D)$, which is $F'(Y^*)$ times (119): $\frac{1}{20} \cdot \frac{2}{3} = \frac{1}{30}$, the same.

Why more A lowers default: an extra unit of protected assets raises consumption by one unit in *both* options. The utility gain is $u'(c_r)$ if the household repays and $u'(c_d)$ if it defaults, and $u'(c_r) > u'(c_d)$ because repaying leaves it poorer at the margin. Repaying gains more, so the household at the old cutoff now strictly prefers to repay, and Y^* falls. A richer household needs the insurance less.

Answer (a). $dY^*/dD = 1$, $dY^*/dA = u'(c_d)/u'(c_r) - 1 < 0$, $dY^*/d\sigma = -1/u'(c_r) < 0$, $dY^*/dE = u'(c_d)/u'(c_r) > 0$. Baseline: 1, $-1/3$, -4, 2/3; $dp/dD = 0.05$, $dp/dA = -1/60$, $dp/d\sigma = -0.2$, $dp/dE = 1/30$, matching slide 44.

Step 4 — (b) The value of the baseline claim. The lender receives D when $Y \geq Y^*$ and $Y - E$ when $Y < Y^*$, so

$$\begin{aligned} Q(D) &= [1 - F(Y^*)]D + \frac{1}{20}\int_5^{Y^*} (y - E)\, dy \\ &= (1 - 0.3)(12) + \frac{1}{20}\left[\frac{(y-4)^2}{2}\right]_5^{11} \\ &= 8.4 + \frac{1}{20}\cdot\frac{7^2 - 1^2}{2} = 8.4 + \frac{48}{40} = 8.4 + 1.2 = 9.6. \end{aligned} \tag{122}$$

A borrower who receives 9.6 and promises 12 pays

$$\frac{D}{Q} - 1 = \frac{12}{9.6} - 1 = 1.25 - 1 = 0.25. \tag{123}$$

Answer (b). $Q = 9.6$; promised rate 25%, all of it a default premium (the riskless rate is 0).

Step 5 — (c) Regime N. With full enforcement the debt is riskless, so $D_N = L = 9.6$ and the household consumes $C_N = A + Y - 9.6 = Y - 4.6 \in [0.4, 20.4]$. Its expected utility is

$$\begin{aligned} EU_N &= \frac{1}{20}\int_5^{25} \sqrt{y - 4.6}\, dy = \frac{1}{20}\cdot\frac{2}{3}\Big[(y-4.6)^{3/2}\Big]_5^{25} \\ &= \frac{1}{30}\left[(20.4)^{3/2} - (0.4)^{3/2}\right] = \frac{92.139373 - 0.252982}{30} = \frac{91.886391}{30} = 3.062880, \end{aligned} \tag{124}$$

so $CE_N = 3.062880^2 = 9.381232$. Expected consumption: $E[C_N] = A + E[Y] - 9.6 = 5 + 15 - 9.6 = 10.4$.

Step 6 — (c) Regime B with $\sigma = 1$. By (b), $Q(12) = 9.6$, and Q increases in D below $D = 21$ (Step 7), so $D_B = 12$ and $Y^* = 11$, $p = 0.3$. In default ($Y < 11$) utility is $u(9) - 1 = 2$; when repaying, $C = Y - 7$:

$$\begin{aligned} EU_B &= p\,[u(A + E) - \sigma] + \frac{1}{20}\int_{11}^{25} \sqrt{y - 7}\, dy \\ &= 0.3(2) + \frac{1}{30}\left[18^{3/2} - 4^{3/2}\right] = 0.6 + \frac{76.367532 - 8}{30} = 0.6 + 2.278918 = 2.878918, \end{aligned} \tag{125}$$

so $CE_B = 2.878918^2 = 8.288168$. Expected consumption:

$$E[C_B] = 0.3(9) + \frac{1}{20}\int_{11}^{25} (y - 7)\, dy = 2.7 + \frac{1}{20}\cdot\frac{18^2 - 4^2}{2} = 2.7 + \frac{308}{40} = 2.7 + 7.7 = 10.4.$$

Answer (c). $D_N = 9.6$ (rate 0), $EU_N = 3.062880$, $CE_N = 9.381232$; $D_B = 12$ (rate 25%), $EU_B = 2.878918$, $CE_B = 8.288168$; $E[C] = 10.4$ in both. The household prefers Regime N.

Step 7 — (d) The general pricing formula. From $\sqrt{A + Y^* - D} = \sqrt{A + E} - \sigma$, squaring,

$$A + Y^* - D = \left(\sqrt{A + E} - \sigma\right)^2 \equiv c_r \quad \Longrightarrow \quad Y^* = D - A + c_r. \tag{126}$$

Write $Y^* = D + k$ with $k \equiv c_r - A = c_r - 5$. Multiplying (122) by 40 and evaluating the integral as in Step 4,

$$\begin{aligned} 40\,Q &= 2(25 - Y^*)D + (Y^* - 4)^2 - 1 \\ &= 2(25 - D - k)D + (D + k - 4)^2 - 1 \\ &= 50D - 2D^2 - 2kD + D^2 + 2(k - 4)D + (k - 4)^2 - 1 \\ &= -D^2 + 42D + (k - 4)^2 - 1, \end{aligned} \tag{127}$$

and $k-4 = c_r - 9$. (Check at $\sigma = 1$: $c_r = 4$, $-144+504+25-1 = 384 = 40(9.6)$.) The slope $42-2D$ is positive for $D < 21$, so the lender accepts the smallest root. With $\sigma = 0.1$: $c_r = (3-0.1)^2 = 8.41$ and $(c_r - 9)^2 = 0.3481$. Setting $40Q = 384$,

$$\begin{aligned} -D^2 + 42D + 0.3481 - 1 &= 384 \\ D^2 - 42D + 384.6519 &= 0 \\ D_B = 21 - \sqrt{441 - 384.6519} &= 21 - \sqrt{56.3481} = 21 - 7.506537 = 13.493463. \end{aligned} \tag{128}$$

Then, from (126),

$$\begin{aligned} Y^* &= 13.493463 - 5 + 8.41 = 16.903463, \\ p &= \frac{16.903463 - 5}{20} = 0.595173, \\ \frac{D_B}{9.6} - 1 &= 1.405569 - 1 = 0.405569. \end{aligned}$$

The formula applies: $5 \le Y^* \le 25$, and the garnishment cap does not bind since $Y^* - E = 12.903463 < D_B$ (in general $D_B - (Y^* - E) = 9 - c_r > 0$).

Step 8 — (d) Expected utility with $\sigma = 0.1$. In default utility is $u(9) - 0.1 = 2.9$; when repaying $C = 5 + y - D_B$, which equals $c_r = 8.41$ at $y = Y^*$ and $30 - D_B = 16.506537$ at $y = 25$:

$$\begin{aligned} EU_B &= p(2.9) + \frac{1}{30}\Big[(30 - D_B)^{3/2} - c_r^{3/2}\Big] \\ &= 0.595173(2.9) + \frac{16.506537^{3/2} - 8.41^{3/2}}{30} \\ &= 1.726002 + \frac{67.063152 - 24.389}{30} = 1.726002 + 1.422472 = 3.148474, \end{aligned} \tag{129}$$

so $CE_B = 3.148474^2 = 9.912888$, above $CE_N = 9.381232$. Expected consumption is again 10.4 (Step 9).

Answer (d). $D_B = 13.493463$, $Y^* = 16.903463$, $p = 0.595173$, promised rate 40.56%; $EU_B = 3.148474$, $CE_B = 9.912888 > 9.381232$: now the household prefers bankruptcy (Regime B).

Step 9 — (e) Why the ranking flips. First, expected consumption is the same in both regimes because the lender breaks even and σ costs no resources:

$$E[C_B] = A + E[Y] - E[\text{payment to lender}] = 5 + 15 - 9.6 = 10.4 = E[C_N].$$

Second, C_B is less risky than C_N. Their difference is

$$C_B - C_N = \begin{cases} (A+E) - (A+Y-9.6) = 13.6 - Y, & Y < Y^*, \\ (A+Y-D_B) - (A+Y-9.6) = 9.6 - D_B < 0, & Y \ge Y^*, \end{cases}$$

which falls with Y and changes sign once, from positive to negative. Two consumption plans with the same mean and a single crossing are ranked by every risk-averse household: the one higher in bad states wins, so $E[u(C_B)] > E[u(C_N)]$. Write the gap as an insurance gain minus a deadweight cost:

$$EU_B - EU_N = \underbrace{E[u(C_B)] - EU_N}_{\text{insurance gain}>0} - \underbrace{\sigma p}_{\text{stigma cost}}. \tag{130}$$

With $\sigma = 1$: $E[u(C_B)] = 2.878918 + 0.3 = 3.178918$, gain $= 3.178918 - 3.062880 = 0.116038 < \sigma p = 0.3$. With $\sigma = 0.1$: $E[u(C_B)] = 3.148474 + 0.059517 = 3.207991$, gain $= 0.145111 > \sigma p = 0.059517$. Bisection on $EU_B(\sigma) = EU_N$, recomputing $D_B(\sigma) = 21 - \sqrt{56 + (c_r - 9)^2}$ each time, gives $\bar{\sigma} = 0.264820$.

Answer (e). Lenders price anticipated default, so bankruptcy is actuarially fair insurance in consumption terms: same mean (10.4), less risk. It raises welfare when the insurance gain exceeds the expected stigma cost σp: yes for $\sigma = 0.1$ ($0.145 > 0.060$), no for $\sigma = 1$ ($0.116 < 0.3$). Indifference at $\bar{\sigma} \approx 0.26$.

Intuition. Default is a state-contingent repayment: households pay a lot when income is high and little when it is low. Lenders anticipate this and charge for it (25% or 40.6% instead of 0%), so the household pays the full cost of the insurance. What it gets is a smoother consumption profile, which a risk-averse household values. What society loses is only the stigma σ each time default happens. A lower σ makes default more frequent (p doubles) and credit dearer, yet welfare rises: more default is not bad in itself. This model has no effort choice; with moral hazard at the borrowing stage (slides 37–40), softer default would also weaken incentives and add a further ex ante cost.

Common mistakes.

- Comparing the regimes at the same face value D. At a fixed D the default option can only help the household; the real comparison re-prices the debt so the lender breaks even.
- Reading a higher default rate or a higher interest rate as lower welfare. Both rise in (d) while welfare rises.
- Signing dY^*/dA positive ("more assets, more protection, more default"). The assets are kept in both options; with concave u they make repaying relatively more attractive.

Ref: Prof. Ricca, Topic 4B slides 37–44 (Aula/Topic4_B.pdf); Indarte (2023). No book exercise exists.

18. Karlan–Zinman: which comparison identifies selection and which identifies moral hazard — Original exercise

Statement.

Origin: original exercise on separating adverse selection from moral hazard (Karlan–Zinman 2009), built on Prof. Ricca's Topic 4B slides 45–58 (Aula/Topic4_B.pdf) · Answer: in the solutions file

A lender makes one-period loans of size $I = 1$ to penniless, risk-neutral individuals whose outside option is worth 0. Half the population is of a safe type $\theta = S$ and half of a risky type $\theta = R$; the lender cannot tell them apart. A borrower's project succeeds with probability $\pi(\theta, e)$ and then yields $Y(\theta)$; it fails with probability $1 - \pi(\theta, e)$ and yields 0. Effort e is either low (L, cost 0) or high (H, cost $c = 0.17$), is chosen by the borrower and is not observed by the lender. A borrower who is indifferent chooses H.

Type	$Y(\theta)$	$\pi(\theta, H)$	$\pi(\theta, L)$
S (safe)	1.35	0.9	0.8
R (risky)	1.80	0.7	0.5

(The two types' projects need not have the same mean.) On success the borrower repays $1 + r_c$, where r_c is the contract rate; the cost of defaulting when the project succeeds is large enough that she always repays when she can (no strategic default). On failure she defaults and bears a cost $C_b(r_f)$, the forgone value of future loans, which depends on the rate r_f at which future loans will be offered: $C_b = 0.25$ if future loans carry the normal rate r, and $C_b = 0.45$ if the lender has promised future loans at a low rate to borrowers who repay (the dynamic repayment incentive). So a borrower of type θ who has taken a loan at contract rate r_c and faces default cost C_b and effort e gets

$$\pi(\theta, e)\left[Y(\theta) - (1 + r_c)\right] - \left[1 - \pi(\theta, e)\right] C_b - \text{cost}(e).$$

There are two rates, $r_L = 10\%$ and $r_H = 30\%$. Timing: - Stage 1: each individual receives a randomly assigned offer rate $r_o \in \{r_L, r_H\}$ and decides whether to take the loan, believing that the contract rate will be $r_c = r_o$ and that future loans carry the normal rate ($C_b = 0.25$). - Stage 2: among those who took up at $r_o = r_H$, a random half is surprised with a lower contract rate $r_c = r_L$; the rest keep $r_c = r_H$. Independently,

a random half of all borrowers is told that future loans will be at the low rate if they repay ($C_b = 0.45$). Each borrower then chooses effort. - Stage 3: the project succeeds or fails, and the borrower repays or defaults.

This creates three groups, each split by the dynamic incentive: group A ($r_o = r_c = r_L$), group B ($r_o = r_H$, $r_c = r_L$) and group C ($r_o = r_c = r_H$).

(a) Show that a borrower chooses H if and only if $[\pi(\theta, H) - \pi(\theta, L)]\,[Y(\theta) - (1 + r_c) + C_b] \geq c$. Find the effort of each type for each of the four combinations of $r_c \in \{r_L, r_H\}$ and $C_b \in \{0.25, 0.45\}$.

(b) Which types take up the loan at $r_o = r_L$ and at $r_o = r_H$? Give the take-up rate at each offer rate.

(c) Build the table of default rates (share of borrowers who default) for the six cells: groups A, B, C, each without and with the dynamic incentive.

(d) Which pair of cells differs only in who selected in, and so identifies hidden information (adverse selection)? Which pair differs only in the contract rate, and which only in the dynamic incentive, so that they identify hidden action (moral hazard)? Compute each effect in percentage points and explain why the dynamic incentive matters in group C but not in groups A and B.

(e) An analyst without an experiment compares borrowers who were offered and paid r_H with those who were offered and paid r_L (cells C and A, no dynamic incentive). Compute the difference in default rates and decompose it into a selection part and a moral-hazard part. Why can this comparison alone not tell the two apart?

(f) For the risky type, find the highest contract rate r_c at which it still chooses H, without and with the dynamic incentive. Relate this bound to the Holmström–Tirole incentive constraint $(\Delta p)R_b \geq B$ and to pledgeable income.

Context needed to solve it The design and model are those of Prof. Ricca's Topic 4B slides 45–58 (Aula/Topic4_B.pdf), based on Karlan and Zinman (2009), "Observing unobservables: Identifying information asymmetries with a consumer credit field experiment", *Econometrica* 77(6). The exercise replaces the slides' continuous type and effort with two types and two effort levels.

From the slides (45–47): A positive correlation between interest rates and default can arise due to adverse selection (those with a relatively high probability of default are more likely to accept a high rate) or moral hazard (a high interest rate can undermine incentives and increase the default probability). Karlan and Zinman randomize interest rates across three dimensions: the interest rate offered in a direct-mail solicitation (the offer rate), the actual interest rate on the loan contract (the contract rate), and the interest rate on future loans conditional on not defaulting (the dynamic repayment incentive). The design compares borrowers who select in at different rates and then face identical repayment incentives (adverse selection) and borrowers who select at identical rates and then face different repayment incentives (moral hazard). It separates hidden information, defined as ex ante selection effects, which include both selection on unobserved risk and selection on unobserved anticipated effort, from hidden action, ex post incentive effects. It cannot disentangle the two sources of hidden information.

From the slides (48–52): Stage 1: the individual decides whether to take up a loan at an exogenous rate r_o, believing that future loans will be offered at a normal rate r. Stage 2: the lender randomly lowers the rate for some borrowers to $r_c < r_o$; with an independent randomization, it lowers the rate of future loans to $r_f < r$ for some borrowers (conditional on repaying). Given r_c and r_f the borrower chooses effort. Stage 3: cash flows are realized and the borrower decides whether to repay. A penniless risk-neutral individual has a project that requires $I = 1$; with success probability $\pi(\theta_i, e)$ it yields $Y(\theta_i)$, otherwise 0. The borrower bears a state-specific cost of default $C_i(r_f)$, $i = g, b$, decreasing in r_f (the forgone discounted profits from future loans). With no strategic default, the borrower solves $\max_e \pi(\theta_i, e)[Y(\theta_i) - (1 + r_c)] + (1 - \pi(\theta_i, e))[-C_b(r_f)] - e$. Hidden action effect 1: effort is decreasing in r_c. Hidden action effect 2: effort increases with the cost of default, so repayment increases when r_f decreases.

From the slides (55–56): Hidden information effect: a higher offer rate leads to a riskier pool of clients and thus a higher default rate. Experiment: the offer rate is low or high, $r_o \in \{r_L, r_H\}$. Of those who take up $r_o = r_H$, some are randomly surprised with a lower contract rate $r_c = r_L$, while the rest keep $r_o = r_c = r_H$. Hidden information: compare those who were offered and accepted $r_o = r_c = r_L$ with those who were offered and accepted $r_o = r_H$ but ended up with $r_c = r_L$. Hidden action: compare those who took up the high rate and kept it, $r_o = r_c = r_H$, with those who took up $r_o = r_H$ but ended up with $r_c = r_L$.

From Tirole §3.2.2, pp. 116–117 (Holmström–Tirole fixed-investment model, Topic 4A): the borrower, who receives R_b on success and 0 on failure, behaves (success probability p_H) rather than misbehaves (success probability p_L, private benefit B) if $(\Delta p)R_b \geq B$, $\Delta p = p_H - p_L$. The highest income in the case of success that can be pledged to lenders is $R - B/\Delta p$, and the pledgeable income is $p_H(R - B/\Delta p)$.

Data (full statement: list, exercise 18; built on Topic 4B slides 45–58). Loan $I = 1$; types S, R (half each); $Y(S) = 1.35$, $Y(R) = 1.80$; $\pi(S,H) = 0.9$, $\pi(S,L) = 0.8$, $\pi(R,H) = 0.7$, $\pi(R,L) = 0.5$; effort cost $c = 0.17$ for H, 0 for L (ties go to H); default cost $C_b = 0.25$ (normal future rate) or 0.45 (dynamic incentive); $r_L = 10\%$, $r_H = 30\%$; outside option 0; no strategic default. Groups: A ($r_o = r_c = r_L$), B ($r_o = r_H$, $r_c = r_L$), C ($r_o = r_c = r_H$), each split by the dynamic incentive.

What is asked. (a) Derive when a borrower works hard and apply it to both types for each contract rate and default cost. (b) Find who borrows at each offer rate, using what borrowers expect when they sign. (c) Fill in the default rate of each of the six experimental cells. (d) Pick the comparisons that change only one thing, and measure the selection and the two incentive effects. (e) Show what goes wrong when high-rate and low-rate borrowers are compared without the experiment. (f) Find the highest rate that keeps the risky type working and connect it to Holmström–Tirole.

Approach. The experiment of slides 46 and 56 turns a correlation into separate effects by randomising three things at different times. Take-up is decided at stage 1, given the *offer* rate, so the offer rate shapes the pool (hidden information). Effort is chosen at stage 2, given the *contract* rate and the dynamic incentive, so these act on behaviour (hidden action). A comparison identifies one effect if the two cells differ only in the variable that drives that effect. We solve the model backwards: effort first, then take-up, then the default table.

Notation.

- $\theta \in \{S,R\}$: type; $e \in \{H,L\}$: effort; $\pi(\theta,e)$: success (repayment) probability; $\Delta\pi_\theta \equiv \pi(\theta,H) - \pi(\theta,L)$, so $\Delta\pi_S = 0.1$, $\Delta\pi_R = 0.2$.
- r_o: offer rate (stage 1); r_c: contract rate (stage 2); C_b: borrower's cost of default, 0.25 or 0.45.
- $g_\theta(r_c, C_b) \equiv \Delta\pi_\theta[Y(\theta) - (1+r_c) + C_b]$: gain from H before its cost c.
- d: default rate of a cell, the share of its borrowers who default, $1-\pi$ averaged over the pool.

Step 1 — (a) The effort condition. We compare the borrower's payoff under H and under L at given r_c and C_b:

$$\begin{aligned} U(H) - U(L) &= [\pi(\theta,H) - \pi(\theta,L)]\,[Y(\theta) - (1+r_c)] - [1-\pi(\theta,H)]C_b + [1-\pi(\theta,L)]C_b - c \\ &= \Delta\pi_\theta\,[Y(\theta) - (1+r_c)] + \Delta\pi_\theta\, C_b - c \\ &= \Delta\pi_\theta\,[Y(\theta) - (1+r_c) + C_b] - c, \end{aligned} \tag{131}$$

so H is chosen iff $g_\theta(r_c, C_b) \geq c = 0.17$. Effort rises with the stake in success, which falls with r_c (hidden action effect 1) and rises with C_b (hidden action effect 2). The gains:

$(r_c,\ C_b)$	$(10\%,\ 0.25)$	$(10\%,\ 0.45)$	$(30\%,\ 0.25)$	$(30\%,\ 0.45)$
g_S	$0.1(0.50) = 0.05$	$0.1(0.70) = 0.07$	$0.1(0.30) = 0.03$	$0.1(0.50) = 0.05$
effort of S	L	L	L	L
g_R	$0.2(0.95) = 0.19$	$0.2(1.15) = 0.23$	$0.2(0.75) = 0.15$	$0.2(0.95) = 0.19$
effort of R	H	H	L	H

Answer (a). H iff $\Delta\pi_\theta[Y(\theta) - (1+r_c) + C_b] \geq c$. S always chooses L. R chooses H except at $r_c = 30\%$ with $C_b = 0.25$, where it chooses L.

Step 2 — (b) Take-up. At stage 1 an individual expects $r_c = r_o$ and $C_b = 0.25$ and anticipates its own effort from Step 1. The payoff $\pi[Y - (1+r_o)] - (1-\pi)(0.25) - \text{cost}(e)$ is:

$$\begin{aligned} S,\ r_o = 10\%\ (L): &\quad 0.8(1.35 - 1.10) - 0.2(0.25) = 0.20 - 0.05 = 0.15 \ \geq 0, \\ S,\ r_o = 30\%\ (L): &\quad 0.8(1.35 - 1.30) - 0.2(0.25) = 0.04 - 0.05 = -0.01 \ < 0, \\ R,\ r_o = 10\%\ (H): &\quad 0.7(1.80 - 1.10) - 0.3(0.25) - 0.17 = 0.49 - 0.075 - 0.17 = 0.245 \ \geq 0, \\ R,\ r_o = 30\%\ (L): &\quad 0.5(1.80 - 1.30) - 0.5(0.25) = 0.25 - 0.125 = 0.125 \ \geq 0. \end{aligned}$$

(The other effort gives less: R at 10% with L gets $0.5(0.70) - 0.125 = 0.225$; at 30% with H, $0.7(0.50) - 0.075 - 0.17 = 0.105$.)

Answer (b). At $r_o = 10\%$ both types borrow (take-up 100%); at $r_o = 30\%$ only R borrows (take-up 50%). The safe type leaves first: its success payoff barely covers the higher repayment, while the risky type's larger $Y(R)$ keeps the loan worth taking.

Step 3 — (c) The default table. A cell's default rate averages $1 - \pi(\theta, e)$ over the types in its pool, at the effort of Step 1 for that cell's r_c and C_b. Group A has both types; B and C only R.

Cell	Pool	No dyn. incentive ($C_b = 0.25$)	Dyn. incentive ($C_b = 0.45$)
A: $r_o = r_c = 10\%$	S, R	$\frac{1}{2}(0.2) + \frac{1}{2}(0.3) = 0.25$	$\frac{1}{2}(0.2) + \frac{1}{2}(0.3) = 0.25$
B: $r_o = 30\%$, $r_c = 10\%$	R	$1 - 0.7 = 0.30$	$1 - 0.7 = 0.30$
C: $r_o = r_c = 30\%$	R	$1 - 0.5 = 0.50$	$1 - 0.7 = 0.30$

Answer (c). Default rates: A 25% / 25%; B 30% / 30%; C 50% / 30% (without / with the dynamic incentive).

Step 4 — (d) Identification. Each comparison holds everything but one variable fixed.

- *Selection (hidden information): B vs A.* Same contract rate (10%) and same dynamic incentive; only the offer rate, hence the pool, differs:

$$d_B - d_A = 0.30 - 0.25 = +0.05 \text{ (5 pp), in both halves.} \tag{132}$$

- *Repayment burden (hidden action 1): C vs B.* Same offer rate, so the same pool (R only); only r_c differs:

$$d_C - d_B = 0.50 - 0.30 = +0.20 \text{ (20 pp) without, } 0.30 - 0.30 = 0 \text{ with the dynamic incentive.} \tag{133}$$

- *Dynamic incentive (hidden action 2): "yes" vs "no" within a group.* Same pool and same r_c; only C_b differs:

$$\text{A: } 0.25 - 0.25 = 0, \qquad \text{B: } 0.30 - 0.30 = 0, \qquad \text{C: } 0.30 - 0.50 = -0.20 \ (-20 \text{ pp}). \tag{134}$$

The dynamic incentive changes behaviour only where an incentive constraint is close to binding. In C without it, R's gain 0.15 is below $c = 0.17$, and $C_b = 0.45$ lifts it to 0.19. In A and B, R already works ($0.19 \geq 0.17$) and S is far from working (0.05 or 0.07), so a higher C_b changes nobody's choice.

Answer (d). Selection: B vs A, +5 pp. Repayment burden: C vs B, +20 pp without the dynamic incentive and 0 with it. Dynamic incentive: within C, −20 pp; 0 in A and B.

Step 5 — (e) The naive comparison. Without the dynamic incentive,

$$\begin{aligned} d_C - d_A &= 0.50 - 0.25 = 0.25 \\ &= \underbrace{(d_B - d_A)}_{\text{selection}=0.05} + \underbrace{(d_C - d_B)}_{\text{moral hazard}=0.20}. \end{aligned} \tag{135}$$

Outside the experiment the offer rate always equals the contract rate, so a high-rate borrower is both from a worse pool *and* facing a heavier repayment. There is no cell like B with $r_o \neq r_c$, so the 25 pp cannot be split. The surprise contract rate is what creates cell B.

Answer (e). 25 pp = 5 pp selection + 20 pp moral hazard; inseparable without a cell where $r_o \neq r_c$.

Step 6 — (f) The highest rate that keeps R working. From (131) with $\theta = R$:

$$\begin{aligned} 0.2\big[1.80 - (1+r_c) + C_b\big] &\geq 0.17 \\ 1.80 - (1+r_c) + C_b &\geq \frac{0.17}{0.2} = 0.85 \\ 1 + r_c &\leq 1.80 + C_b - 0.85. \end{aligned} \tag{136}$$

With $C_b = 0.25$: $1 + r_c \leq 1.20$, so $r_c \leq 20\%$. With $C_b = 0.45$: $1 + r_c \leq 1.40$, so $r_c \leq 40\%$. This is the Holmström–Tirole constraint $(\Delta p)R_b \geq B$ with $R_b = Y(R) - (1 + r_c)$ the borrower's stake on success, c in place of B, plus the default cost: $\Delta\pi_R(R_b + C_b) \geq c$. The maximal repayment on success, $Y - c/\Delta\pi + C_b$, is the analogue of $R - B/\Delta p$, and the expected income pledgeable by a working risky borrower is $0.7(1.20) = 0.84$ without and $0.7(1.40) = 0.98$ with the dynamic incentive.

Answer (f). $r_c \leq 20\%$ without and $r_c \leq 40\%$ with the dynamic incentive; at 30% the IC fails only without it (cell C). The dynamic incentive raises pledgeable income by $0.7 \times 0.20 = 0.14$.

Intuition. The offer rate works *before* the borrower is locked in, so it can only change who walks in; the contract rate and the future-loan promise work *after*, so they can only change what borrowers do. A randomised surprise between the two stages lets the economist hold the pool fixed while moving incentives, and an independent future-rate lottery separates a second incentive channel. In this example most of the naive rate–default correlation is moral hazard, and it disappears when future credit is made valuable.

Common mistakes.

- Comparing A with C and calling it adverse selection: that difference mixes the pool and the repayment burden.
- Letting borrowers anticipate the surprise: take-up uses $r_c = r_o$ and $C_b = 0.25$; the lower r_c and the dynamic incentive are news at stage 2.
- Assuming the riskier type always exerts less effort. Here R works harder because it has more at stake in success ($Y(R)$ is larger); the identification does not depend on the effort ranking across types.

Ref: Prof. Ricca, Topic 4B slides 45–58 (Aula/Topic4_B.pdf); Karlan and Zinman (2009), *Econometrica* 77(6); Tirole §3.2.2, pp. 116–117 (IC$_b$, pledgeable income). No book exercise exists.

19. Reading a three-rate experiment: selection, incentives and credit rationing — Original exercise

Statement.

Origin: original exercise on separating adverse selection from moral hazard (Karlan–Zinman 2009) and its implications for credit rationing, built on Prof. Ricca's Topic 4B slides 45–58 (Aula/Topic4_B.pdf), Topic 4A (Aula/Topic4_A.pdf) and Topic 3 slides 4–19 and 47–55 (Aula/Topic_3.pdf) · Answer: in the solutions file

A consumer lender runs a three-rate field experiment like Karlan and Zinman's. **The data below are hypothetical, invented for this exercise; they are not Karlan and Zinman's estimates.** Offer rates r_o and contract rates r_c take two values, $r_L = 30\%$ and $r_H = 40\%$ per loan (four-month loans). Group A was offered and paid r_L; group B was offered r_H, took the loan, and was then surprised with $r_c = r_L$; group C was offered and paid r_H. Independently of the rates, a random half of each group received the dynamic repayment incentive: future loans at a lower rate if this loan is repaid ($C_i = 1$; $C_i = 0$ otherwise). A loan is in default if it is not repaid; assume defaulted loans repay nothing.

Group	r_o	r_c	Dynamic incentive	Loans n	Defaults
A	30%	30%	no	600	69
A	30%	30%	yes	600	51

Group	r_o	r_c	Dynamic incentive	Loans n	Defaults
B	40%	30%	no	750	132
B	40%	30%	yes	750	108
C	40%	40%	no	750	141
C	40%	40%	yes	750	114

(a) Compute the default rate in each cell and in each group (A, B, C, pooling the two halves), and the pooled default rates with and without the dynamic incentive. For each of the three effects — selection (hidden information), repayment burden (hidden action effect 1) and dynamic incentive (hidden action effect 2) — state which comparison identifies it and why, and compute it in percentage points. Also compute the dynamic-incentive effect within each group.

(b) For each of the three effects, compute the standard error of the difference of two proportions, $\text{SE} = \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}$, the z-statistic and the 95% confidence interval ($\pm 1.96\,\text{SE}$). Which effects are significant at 5%?

(c) Slide 58 estimates the linear probability model $Y_i = \alpha + \beta_o r_o + \beta_c r_c + \beta_b C_i + \varepsilon_i$, where Y_i is the default indicator in percentage points (0 or 100) and the rates are in percentage points. Explain why, in this design, the OLS estimates satisfy $\beta_o(r_H - r_L) = \bar{d}_B - \bar{d}_A$, $\beta_c(r_H - r_L) = \bar{d}_C - \bar{d}_B$ and $\beta_b =$ the pooled dynamic-incentive effect ($\bar{d}_g$ is the default rate of group g). Compute β_o, β_c, β_b and α, and state which coefficient signals hidden information and which signal hidden action.

(d) A loan officer proposes raising the rate on new loans from 30% to 40%. Using groups A and C, compute the expected gross repayment per unit lent, $(1-\bar{d})(1+r)$, at each rate. Decompose the change into a rate effect, a selection effect and a repayment-burden effect. What does the result imply for the shape of the lender's expected return as a function of the rate, and why may the lender ration credit rather than raise the rate? Relate the selection part to the lemons problem of Topic 3.

(e) Link the hidden-action results to the Holmström–Tirole fixed-investment model. A borrower with assets A needs $I = 1$; the project yields $R = 1.5$ with probability $p_H = 0.9$ if she behaves and $p_L = 0.5$ if she misbehaves, in which case she enjoys a private benefit $B = 0.2$. Lenders are competitive and risk-neutral with a zero riskless rate. She receives R_b on success and 0 on failure; in addition, a dynamic incentive makes her lose $C_b \geq 0$ (the value of cheaper future loans) if the project fails. (i) Write the incentive constraint and show that it becomes $(\Delta p)(R_b + C_b) \geq B$. (ii) Compute the pledgeable income and the minimum assets $\bar{A}$ needed for financing for $C_b = 0$ and $C_b = 0.2$. (iii) Compute the borrower's net utility when financed with $C_b = 0.2$, counting the loss C_b as a deadweight cost. (iv) Use (i)–(iii) to interpret the repayment-burden and dynamic-incentive effects of part (a).

(f) What can this design not tell you about the source of the selection effect? Give one reason why a significant selection effect does not by itself prove adverse selection on risk type.

Context needed to solve it The design and the regression are those of Prof. Ricca's Topic 4B slides 45–58 (Aula/Topic4_B.pdf), based on Karlan and Zinman (2009), *Econometrica* 77(6). The data are hypothetical.

From the slides (45–47): A positive correlation between interest rates and default can arise due to adverse selection (those with a relatively high probability of default are more likely to accept a high rate) or moral hazard (a high interest rate can undermine incentives and increase the default probability). Karlan and Zinman randomize the offer rate, the contract rate, and the interest rate on future loans conditional on not defaulting (the dynamic repayment incentive). The design compares borrowers who select in at different rates and then face identical repayment incentives (adverse selection) with borrowers who select at identical rates and then face different repayment incentives (moral hazard). It separates hidden information — ex ante selection effects, which include both selection on unobserved risk and selection on unobserved anticipated effort — from hidden action, ex post incentive effects. The design cannot disentangle the two sources of hidden information.

From the slides (52, 55–56): Hidden action effect 1: effort is decreasing in r_c. Hidden action effect 2: effort increases as the cost of default increases, so repayment increases when r_f decreases. Hidden information effect: a higher offer rate leads to a riskier pool of clients and thus a higher default rate. Experiment: $r_o \in \{r_L, r_H\}$; of those who take up $r_o = r_H$, some are randomly surprised with $r_c = r_L$, while the remainder keep $r_o = r_c = r_H$. Hidden information: compare $r_o = r_c = r_L$ with $r_o = r_H$, $r_c = r_L$. Hidden action: compare $r_o = r_c = r_H$ with $r_o = r_H$, $r_c = r_L$.

From the slides (58): Empirical model $Y_i = \alpha + \beta_o r_o + \beta_c r_c + \beta_b C + X_i + \varepsilon$, where Y_i is a measure of default. $\beta_c > 0$ identifies hidden action effect 1 (effect of a higher repayment on effort); $\beta_b < 0$ identifies

hidden action effect 2 (effect of higher dynamic incentives on effort); $\beta_o > 0$ identifies adverse selection (hidden information effect, effect of a higher offer rate on default).

From Tirole §3.2.2, pp. 116–117 (Topic 4A): the borrower behaves if $p_H R_b \geq p_L R_b + B$, or $(\Delta p)R_b \geq B$ (IC_b); the highest income in the case of success that can be pledged to lenders is $R - B/\Delta p$; the pledgeable income is $\mathcal{P} = p_H(R - B/\Delta p)$; the project is financed if and only if $\mathcal{P} \geq I - A$, i.e. $A \geq \bar{A} \equiv I - \mathcal{P}$ (credit rationing below $\bar{A}$).

From Topic 3 (Aula/Topic_3.pdf; summary in Map/lectures-index.md, blocks 3.2–3.3 and 3.6): in the lemons model of credit, lenders who cannot tell good from bad borrowers offer one pooling contract, good types cross-subsidise bad types, and the market breaks down when the share of good types is too low. In Stiglitz–Weiss (1981, slides 47–55), borrowers' projects are mean-preserving spreads of each other; above a threshold rate the safe types leave the pool, the bank's expected return is non-monotonic in the rate, the loan supply curve bends backward, and equilibrium credit rationing can occur.

Data (full statement: list, exercise 19; *hypothetical* numbers, not Karlan and Zinman's estimates). $r_L = 30\%$, $r_H = 40\%$ per loan. Loans and defaults: A (offered and paid 30%): 600/69 without and 600/51 with the dynamic incentive; B (offered 40%, paid 30%): 750/132 and 750/108; C (offered and paid 40%): 750/141 and 750/114. Defaulted loans repay nothing. Part (e): $I = 1$, $R = 1.5$, $p_H = 0.9$, $p_L = 0.5$, $B = 0.2$, $C_b \in \{0, 0.2\}$.

What is asked. (a) Turn the counts into default rates and read off the three effects from the right comparisons. (b) Decide which effects are statistically distinguishable from zero. (c) Map the table into the regression on slide 58. (d) Use the table to evaluate a rate increase from the lender's side and explain why it can backfire. (e) Show in the Holmström–Tirole model how a dynamic incentive relaxes the incentive constraint and loosens rationing. (f) State the limit of the design on the selection side.

Approach. The identification logic is that of slides 46 and 56: B vs A holds the contract fixed and varies the pool; C vs B holds the pool fixed and varies the repayment; the dynamic-incentive lottery is independent of everything else, so its effect can be read within each group or pooled. Because each comparison is between randomised groups, a two-sample test of proportions is valid. Part (d) applies Stiglitz–Weiss logic (Topic 3) to the measured effects, and part (e) applies the Holmström–Tirole IC (Topic 4A).

Notation.

- $\bar{d}_g$: default rate of group $g \in \{A, B, C\}$; $\bar{d}^0$, $\bar{d}^1$: pooled default rates without / with the dynamic incentive.
- $C_i \in \{0, 1\}$: dynamic-incentive dummy; $Y_i \in \{0, 100\}$: default indicator in percentage points.
- SE: standard error of a difference of proportions; z: difference over SE.
- $\mathcal{R}(r) = (1 - \bar{d})(1 + r)$: lender's expected gross repayment per unit lent.
- Part (e): R_b: borrower's pay on success; C_b: her loss on failure from forgone cheap future credit; $\Delta p = p_H - p_L = 0.4$; $\mathcal{P}$: pledgeable income; $\bar{A}$: minimum assets for financing.

Step 1 — (a) Default rates. Each rate is defaults over loans. Cells: A $69/600 = 11.5\%$ and $51/600 = 8.5\%$; B $132/750 = 17.6\%$ and $108/750 = 14.4\%$; C $141/750 = 18.8\%$ and $114/750 = 15.2\%$. Groups:

$$\bar{d}_A = \frac{69 + 51}{1200} = \frac{120}{1200} = 10.0\%, \qquad \bar{d}_B = \frac{132 + 108}{1500} = \frac{240}{1500} = 16.0\%,$$
$$\bar{d}_C = \frac{141 + 114}{1500} = \frac{255}{1500} = 17.0\%.$$

Pooled by dynamic incentive:

$$\bar{d}^0 = \frac{69 + 132 + 141}{2100} = \frac{342}{2100} = 16.2857\%, \qquad \bar{d}^1 = \frac{51 + 108 + 114}{2100} = \frac{273}{2100} = 13.0\%.$$

Step 2 — (a) The three effects. Each comparison isolates one channel.

$$\text{Selection (B vs A, same } r_c\text{):} \quad \bar{d}_B - \bar{d}_A = 16.0 - 10.0 = 6.0 \text{ pp}, \tag{137}$$

$$\text{Repayment burden (C vs B, same pool):} \quad \bar{d}_C - \bar{d}_B = 17.0 - 16.0 = 1.0 \text{ pp}, \tag{138}$$

$$\text{Dynamic incentive (yes vs no):} \quad \bar{d}^1 - \bar{d}^0 = 13.0 - 16.2857 = -3.2857 \text{ pp}. \tag{139}$$

Within groups the dynamic effect is A $8.5 - 11.5 = -3.0$, B $14.4 - 17.6 = -3.2$, C $15.2 - 18.8 = -3.6$ pp. Pooling is valid because the lottery gives exactly half of every group the incentive, so the pooled "yes" and "no" samples have the same mix of A, B and C.

Answer (a). Selection +6.0 pp (B vs A); repayment burden +1.0 pp (C vs B); dynamic incentive −3.29 pp (pooled; −3.0, −3.2, −3.6 pp within A, B, C).

Step 3 — (b) Standard errors and tests. For each difference of two independent proportions,

$$\text{SE} = \sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}.$$

$$\begin{aligned}
\text{Selection:}\quad \text{SE} &= \sqrt{\frac{0.16(0.84)}{1500} + \frac{0.10(0.90)}{1200}} = \sqrt{0.0000896 + 0.0000750} \\
&= \sqrt{0.0001646} = 0.012830, \\
z &= \frac{0.06}{0.012830} = 4.677, \qquad \text{CI} = 0.06 \pm 1.96(0.012830) = [0.034854,\ 0.085146]. \\
\text{Burden:}\quad \text{SE} &= \sqrt{\frac{0.17(0.83)}{1500} + \frac{0.16(0.84)}{1500}} = \sqrt{0.0000941 + 0.0000896} \\
&= \sqrt{0.0001837} = 0.013552, \\
z &= \frac{0.01}{0.013552} = 0.738, \qquad \text{CI} = [-0.016563,\ 0.036563]. \\
\text{Dynamic:}\quad \text{SE} &= \sqrt{\frac{0.13(0.87)}{2100} + \frac{0.162857(0.837143)}{2100}} = \sqrt{0.0000539 + 0.0000649} \\
&= \sqrt{0.0001188} = 0.010899, \\
z &= \frac{-0.032857}{0.010899} = -3.015, \qquad \text{CI} = [-0.054218,\ -0.011496].
\end{aligned}$$

Answer (b). Selection ($z = 4.68$, CI 3.5 to 8.5 pp) and dynamic incentive ($z = -3.01$, CI −5.4 to −1.1 pp) are significant at 5% ($|z| > 1.96$); the repayment burden ($z = 0.74$, CI −1.7 to 3.7 pp) is not.

Step 4 — (c) The slide-58 regression. The design has three rate groups, and the constant, r_o and r_c give exactly three free parameters, so OLS fits the three group means exactly. C_i is balanced (one half in every group), so it is orthogonal to the group structure. By the Frisch–Waugh–Lovell theorem, β_b is the regression of Y on $C_i - \frac{1}{2}$, which is the n-weighted average of the within-group differences:

$$\beta_b = \frac{1200(-3.0) + 1500(-3.2) + 1500(-3.6)}{4200} = \frac{-3600 - 4800 - 5400}{4200} = \frac{-13800}{4200} = -3.285714,$$

the pooled difference (139). Each group's fitted value, averaged over its two halves, equals its mean: $\alpha + \beta_o r_o + \beta_c r_c + \frac{1}{2}\beta_b = \bar{d}_g$. Subtracting the equations of A and B (they differ only in r_o) and of B and C (only in r_c):

$$\begin{aligned}
\beta_o(40 - 30) = \bar{d}_B - \bar{d}_A = 6.0 &\implies \beta_o = 0.6, \\
\beta_c(40 - 30) = \bar{d}_C - \bar{d}_B = 1.0 &\implies \beta_c = 0.1, \\
\alpha = \bar{d}_A - \tfrac{1}{2}\beta_b - 30(\beta_o + \beta_c) &= 10 + 1.642857 - 21 = -9.357143.
\end{aligned}$$

(Solving the 4×4 normal equations directly gives the same four numbers; the negative intercept is an extrapolation to zero rates, outside the data.)

Answer (c). $\beta_o = 0.6$ pp of default per pp of offer rate (hidden information, adverse selection); $\beta_c = 0.1$ (hidden action 1, not significant); $\beta_b = -3.29$ pp (hidden action 2); $\alpha = -9.36$.

Step 5 — (d) The lender's return. With defaulted loans repaying nothing,

$$\mathcal{R}(30\%) = (1 - 0.10)(1.30) = 1.170, \qquad \mathcal{R}(40\%) = (1 - 0.17)(1.40) = 1.162. \tag{140}$$

Decompose the change by adding and subtracting $(1 - \bar{d}_A)(1 + r_H)$:

$$\begin{aligned} \mathcal{R}(40\%) - \mathcal{R}(30\%) &= (1 - \bar{d}_A)(r_H - r_L) - (\bar{d}_C - \bar{d}_A)(1 + r_H) \\ &= \underbrace{0.90(0.10)}_{\text{rate}=+0.090} - \underbrace{0.06(1.40)}_{\text{selection}=0.084} - \underbrace{0.01(1.40)}_{\text{burden}=0.014} = -0.008. \end{aligned}$$

Had the pool and behaviour stayed as in A, the higher rate would have yielded $0.90(1.40) = 1.26$.

Answer (d). 1.170 at 30% vs 1.162 at 40%: the rate increase lowers the expected return by 0.008 ($+0.090$ rate, -0.084 selection, -0.014 burden). The return is non-monotonic in the rate, so the loan supply curve bends backward: a lender facing excess demand at 30% does better to ration than to raise the rate.

Step 6 — (e) Holmström–Tirole with a dynamic incentive. (i) The borrower gets R_b on success and loses C_b on failure. Behaving beats misbehaving iff

$$\begin{aligned} p_H R_b - (1 - p_H)C_b &\geq p_L R_b - (1 - p_L)C_b + B \\ (p_H - p_L)R_b + [(1 - p_L) - (1 - p_H)]C_b &\geq B \\ (\Delta p)(R_b + C_b) &\geq B. \end{aligned} \tag{141}$$

(ii) The minimum stake is $R_b = B/\Delta p - C_b$ (non-negative for these values), so lenders can be promised $R - B/\Delta p + C_b$ on success:

$$\mathcal{P}(C_b) = p_H\Big(R - \frac{B}{\Delta p} + C_b\Big), \qquad \frac{B}{\Delta p} = \frac{0.2}{0.4} = 0.5.$$

For $C_b = 0$: $\mathcal{P} = 0.9(1.5 - 0.5) = 0.9$ and $\bar{A} = I - \mathcal{P} = 1 - 0.9 = 0.1$. For $C_b = 0.2$: $R_b \geq 0.5 - 0.2 = 0.3$, $\mathcal{P} = 0.9(1.5 - 0.5 + 0.2) = 0.9(1.2) = 1.08$, and $\bar{A} = 1 - 1.08 = -0.08 < 0$: every borrower, even one with $A = 0$, is financed. Behaving is worth inducing: the NPV is $p_H R - I = 1.35 - 1 = 0.35 > 0$, while misbehaving gives $p_L R + B - I = 0.75 + 0.2 - 1 = -0.05 < 0$. (iii) With competitive lenders the borrower gets the NPV less the expected deadweight loss:

$$U_b = 0.35 - (1 - p_H)C_b = 0.35 - 0.1(0.2) = 0.35 - 0.02 = 0.33.$$

(iv) The repayment-burden effect is the IC (141) at work: a higher contract rate raises the lenders' claim and lowers R_b, and borrowers whose IC was nearly binding switch to misbehaving. That is why lenders cannot raise repayments beyond $R - B/\Delta p$ and why rationing starts below $\bar{A}$. Its small, insignificant size here says that at these rates few borrowers sit at the margin. The dynamic-incentive effect is C_b in (141): future cheap credit is a punishment for default that costs the lender nothing today, raises pledgeable income, and lowers $\bar{A}$. The significant -3.3 pp says this channel is strong, and that bankruptcy rules or renegotiation that soften the punishment (slides 37–40) cut pledgeable income.

Answer (e). IC: $(\Delta p)(R_b + C_b) \geq B$. $\mathcal{P} = 0.9$, $\bar{A} = 0.1$ without; $\mathcal{P} = 1.08$, $\bar{A} = -0.08$ (no rationing) with $C_b = 0.2$; borrower's utility 0.33.

Step 7 — (f) What the design cannot say. The selection effect compares pools formed at different offer rates, and people differ in two hidden ways when they select: in risk type and in the

effort they *plan* to exert. Both are hidden information at take-up, and the design cannot separate them (slide 47). A borrower who accepts 40% because she already plans to put in little effort produces the same +6 pp as a borrower of a riskier type.

Answer (f). It cannot separate selection on unobserved risk from selection on unobserved anticipated effort, so a significant β_o shows hidden information, not necessarily adverse selection on risk type.

Intuition. Three randomisations give three clean contrasts. In these invented data the pool deteriorates sharply when the offer rate rises, which is the lemons and Stiglitz–Weiss mechanism of Topic 3: price cannot clear the market because it changes the quality of the borrowers. The incentive side shows that the repayment burden matters little at these rates, but future credit is a powerful carrot, the Holmström–Tirole IC in action. A lender that learns this should keep the rate low, ration, and lean on dynamic incentives.

Common mistakes.

- Testing the selection effect with C vs A: that difference (7 pp) also contains the repayment burden.
- Concluding "no moral hazard" from the insignificant burden effect: the dynamic-incentive effect is hidden action too, and it is significant.
- Reading $\mathcal{R}$ as profit: it is gross repayment per unit lent; what matters for rationing is that it *falls* with the rate.

Ref: Prof. Ricca, Topic 4B slides 45–58 (Aula/Topic4_B.pdf); Topic 3 slides 4–19 and 47–55 (lemons, Stiglitz–Weiss; Aula/Topic_3.pdf); Tirole §3.2.2, pp. 116–117. Data hypothetical. No book exercise exists.